

M O D U L A R S Y S T E M

QUADRATICS

Equations, Inequalities, and Functions

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<http://book.zambak.com>

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PREFACE

To the Teacher

This book is about quadratic equations, quadratic inequalities, quadratic functions, and their applications. The book is divided into three chapters, each of which is subdivided into sections. The first chapter, Quadratic Equations, deals with equations involving quadratic polynomials and their solution methods. The second chapter, Quadratic Inequalities, introduces the solution of linear and quadratic inequalities. The third chapter, Quadratic Functions, covers graphs and properties of quadratic functions.

Each chapter section is followed by plenty of exercises. More difficult exercise problems are denoted by a single or double star, where the former indicates problems for upper-intermediate level students and the latter indicates problems at advanced level. Many of the problems reflect skills or problem-solving techniques encountered in the section. Every exercise set also contains problems whose solution method is not covered in an example. In these problems, students may be required to work a little beyond the material discussed in the text, or to use the concepts in ways not illustrated in the examples. All of these problems can be solved using skills the student should already have mastered.

The book follows a linear approach, with material in the latter sections building on concepts and math covered previously in the text. For this reason, there are several self-test 'Check Yourself' sections that check students' understanding of the material at key points. 'Check Yourself' sections include a rapid answer key that allows students to measure their own performance and understanding. Successful completion of each self-test section allows students to advance to the next topic.

Each chapter ends with a brief summary of the main points covered in the chapter, followed by a concept check, and chapter review tests.

To the Student

Your education is one of the most valuable activities of your life. An education gives you the skills to learn, again and again.

You might be studying now because you want to go university. Or, you might be studying because you want a particular job. In either case, you should know exactly why you are taking this course and why it is important for you to learn and succeed. If your goal and purpose are not clear, you may find it difficult to make the daily choices needed to study, learn, and succeed.

To support you in your learning, this study module on Quadratics presents the concepts, problem-solving skills, and applications that are essential for an intermediate algebra course.

Acknowledgements

Many friends and colleagues were of great help in writing this textbook. We would like to thank the following people for their advice and suggestions in correspondence and conversation:

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Finally, we wish to express our thanks to the many other people whose names we have not been able to mention. Any errors that may appear in the text are the responsibility of the authors. We would appreciate having these brought to our attention.

The authors

Using This Book

This book is designed so that you can use it effectively. Each chapter has its own special color that you can see at the bottom of the page.

Chapter 1

Quadratic Equations

Chapter 2

Quadratic Inequalities

Chapter 3

Quadratic Functions

Different pieces of information in this book are useful in different ways. Look at the types of information, and how they appear in the book:

Note

We can apply Vieta's theorem for a c
I and be the root of the

Notes help you focus on important details. When you see a note, read it twice! Make sure you understand it.

Definition boxes give a formal description of a new concept. Notation boxes explain the mathematical way of expressing concepts. Theorem boxes include propositions that can be proved. The information in these boxes is very important for further understanding and for solving examples.

Theorem

The inf

Property

Zero product prop
If $A \cdot B = 0$, then

Definition

sequence

A function which

EXAMPLE

4

Solve the inequality $|2x - 1|$

Solution

$$|2x - 1| < 5$$

$$-5 < 2x - 1 < 5$$

Examples include problems related to the topic and their solution, with explanations. The examples are numbered, so you can find them easily in the book.

Check Yourself sections help you check your understanding of what you have just studied. Solve the problems alone and then check your answers against the answer key provided. If your answers are correct, you can move on to the next section. If you have a wrong answer, go through your working again, and check back through the examples in the section.

Check Yourself 6

Graph the following functions.

1. $y = |x^2 - 6x + 5|$

2. $y = -x|x| +$

Answers

1.

2.

A small notebook in the left margin of a page reminds you of material that is related to the topic you are studying. Notebook text helps you to remember the math you need to understand the material. It might help you to see your mistakes, too! Notebooks are the same color as the section you are studying.

Numbers whose square roots are integers or quotients of integers are perfect squares.

FIND THE MISTAKE!

$$a > 4$$

FIND THE MISTAKE!

Let a and b be two arbitrary numbers such that $a \neq b$.

COMPLETED SQUARE FORM

The completed square form of a quadratic function is $y = a(x - h)^2 + k$.
The vertex of a parabola of a function in complete square form is (h, k) .

Special windows highlight important new information. Windows can contain formulas, properties, or solution procedures. They are the same color as the color of the section.

Exercises at the end of each section cover the material in the whole section. You should be able to solve all the problems without any special symbol. ★ next to a question means the question is a bit harder. ★★ next to a question means the question is for students who are looking for a challenge! The answers to the exercises are at the back of the book.

EXERCISES 3

1. Solve the inequality systems.

a.
$$\begin{cases} x + 2 > 0 \\ 2x - 3 < 0 \end{cases}$$

b.
$$\begin{cases} x - 1 > 0 \\ x^2 - 2x - 3 < 0 \end{cases}$$

CHAPTER SUMMARY

- A quadratic function is described by $y = ax^2 + bx + c$.

Concept Check

- What is the standard form of a quadratic equation?

CHAPTER REVIEW TEST 2B

1. What is the solution of the inequality

$$x + 1 \leq x - 1 > x^2?$$

The Chapter Summary at the end of each chapter summarizes all the important material that has been covered in the chapter. The Concept Check section contains oral questions. In order to answer them you don't need paper or pen. If you answer Concept Check questions correctly, it means you know that topic! The answers to these questions are in the material you studied. Go back over the material if you are not sure about an answer to a Concept Check question. Finally, chapter review tests include questions in increasing order of difficulty and contain multiple choice questions. The answer key for these tests is at the back of the book.

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Equations, Inequalities, and Functions

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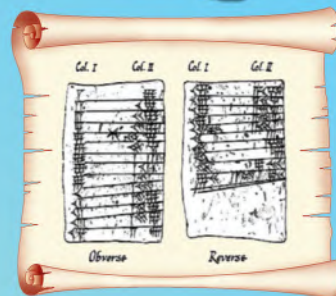
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INTRODUCTION

The study of quadratics started around three thousand years ago with the Babylonians. The Babylonians were one of the world's first civilisations, and developed advanced ideas such as organized agriculture, irrigation, and writing. They plotted the paths of the Sun, the Moon, and the planets, and recorded them on clay tablets. The Babylonians invented the modern ideas of angle, including the way that the circle is divided up into 360 degrees. They also invented the job of taxman! And this was one of the reasons that the Babylonians needed to solve quadratic equations.



Let's suppose that you are a Babylonian farmer. Somewhere on your farm you have a square field on which you grow a crop. What amount of your crop can you grow on the field? Double the length of each side of the field and you find that you can grow four times as much of the crop as before. The reason for this is that the amount of the crop that you can grow is proportional to the area of the field, which is in turn proportional to the square of the length of the side. In mathematical terms, if x is the length of the side of the field, m is the amount of crop you can grow on a square field of side length 1, and c is the amount of crop that you can grow, then

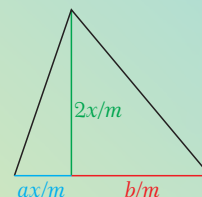
$$c = mx^2.$$

This is our first quadratic equation. Quadratic equations and areas are linked together like brothers and sisters in the same family. However, at the moment we don't have to solve the equation, until the taxman arrives, that is! The taxman says, 'I want you to give me c crops to pay for the taxes on your farm.' You now have a dilemma: how big a field do you need to grow that amount of crop? We can answer this question easily, in fact:

$$x = \sqrt{\frac{c}{m}}.$$

Finding square roots by using a calculator is easy for us, but it was more of a problem for the Babylonians. To help themselves, they developed a method of successive approximation to the answer. This method is identical to the algorithm used by modern computers to solve problems that are much harder than quadratic equations.

Now, not all fields are square. Let's now suppose that you have a more oddly shaped field with two triangular sections as shown on the right.



For appropriate values of a and b the amount of crop that you can grow in this field is given by $c = ax^2 + bx$.

This looks a lot more like the standard form of a quadratic equation that mathematicians today are used to, and even under the evil eye of the tax man, it's a lot harder to solve. Yet the Babylonians found the answer to this problem again. First we divide by a to give

$$x^2 + \frac{b}{a}x = \frac{c}{a}.$$

Now we complete the square by using the fact that

$$\left(x + \frac{b}{2a}\right)^2 = x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}.$$

Combining this with the original equation we have

$$\left(x + \frac{b}{2a}\right)^2 = \frac{c}{a} + \frac{b^2}{4a^2}.$$

This is now an equation that we can solve by taking square roots. The result is the famous ' $-b$ formula':

$$x = -\frac{b}{2a} \pm \sqrt{\frac{c}{a} + \frac{b^2}{4a^2}},$$

which we can rewrite as

$$x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}.$$

(The formula usually has ' $-4ac$ ' because the quadratic equation is more usually written in the form $ax^2 + bx + c = 0$.)

The fact that taking a square root can give a positive or a negative answer leads to the remarkable result that a quadratic equation has two solutions.

The Babylonians, therefore, developed an algorithmic approach to solving certain problems. These problems involved what we now call quadratic equations. The Babylonian method is essentially one of completing the square. However, all such Babylonian problems had answers which were positive (more accurately unsigned) quantities since the usual answer was a length.

In about 300 BC Euclid developed a different geometrical approach which, although it was later used to solve quadratic equations, amounted to finding a length which in our notation was the root of a quadratic equation. Euclid had no notion of the concept of equation or coefficient, but worked with purely geometrical quantities.

Hindu mathematicians took the Babylonian methods further. The Hindu mathematician Brahmagupta (598-665 AD) gives an almost modern method of solving quadratic problems, which admits negative quantities. He also used abbreviations for the unknown terms. Usually the initial letter of a colour was used, and sometimes several different unknowns occurred in a single problem.



Euclid

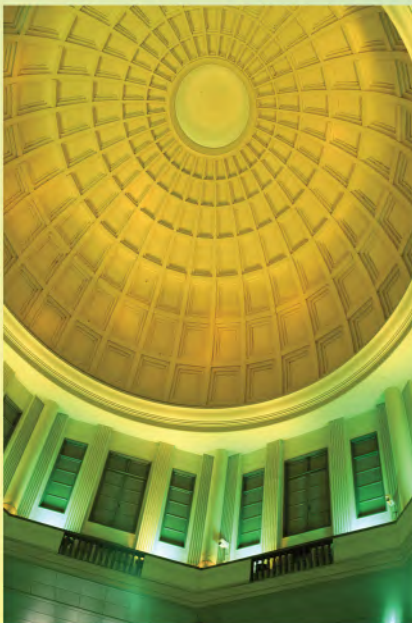
The Arabs did not know about the advances of the Hindus so they had neither negative quantities nor abbreviations for their unknowns. However al-Khwarizmi (around 800 AD) gave a classification of different types of quadratics (although it included only numerical examples of each). The different types arose since al-Khwarizmi had no zero or negatives. He wrote six chapters each devoted to a different type of equation, defined in terms of roots, squares of roots, and numbers. Al-Khwarizmi's classification was:

1. Squares equal to roots.
2. Squares equal to numbers.
3. Roots equal to numbers.
4. Squares and roots equal to numbers, e.g. $x^2 + 10x = 39$.
5. Squares and numbers equal to roots, e.g. $x^2 + 21 = 10x$.
6. Roots and numbers equal to squares, e.g. $3x + 4 = x^2$.

Al-Khwarizmi gives the rule for solving each type of equation, and then a proof for each example.



al-Khwarizmi



Abraham bar Hiyya Ha-Nasi, often known by the Latin name Savasorda, is famous for his book *Liber embadorum*, published in 1145, which was the first book published in Europe to give the complete solution of a quadratic equation.





Chapter 1

QUADRATIC EQUATIONS



SOLVING QUADRATIC EQUATIONS

Definition

An equation that can be written in the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

is called a **quadratic equation**.

I'm sick of being an unknown



In the equation, a , b , and c are **real number coefficients** and x is a **variable**. A quadratic equation written in the form $ax^2 + bx + c = 0$ is said to be in **standard form**. Sometimes, a quadratic equation is also called a **second degree equation**.

For example,

$$x^2 + 3x - 5 = 0, \quad 2x^2 - x - 1 = 0 \quad \text{and} \quad \sqrt{2}x^2 - \frac{1}{2}x + 3 = 0$$

are all quadratic equations. By the definition of a quadratic equation, a cannot be zero. However b or c or both may be zero. For instance,

$$3x^2 + 5x = 0, \quad 2x^2 = 0 \quad \text{and} \quad x^2 - 9 = 0$$

are also quadratic equations.

We can see that quadratic equations are formed by second-degree polynomials. Polynomials of a different degree do not form quadratic equations.

Let us look at the coefficients a , b , and c of some quadratic equations.

Equation	a	b	c
$3x^2 + 5x - 9 = 0$	3	5	-9
$1 - x + 3x^2 = 0$	3	-1	1
$\sqrt{2}x^2 + 5x = 0$	$\sqrt{2}$	5	0
$-x^2 + \frac{1}{2} = 0$	-1	0	$\frac{1}{2}$
$1 - x^2 = 0$	-1	0	1
$(\sqrt{3} + 1)x^2 = 0$	$\sqrt{3} + 1$	0	0
$\frac{x^2}{2} - \frac{x}{3} + \frac{1}{4} = 0$	$\frac{1}{2}$	$-\frac{1}{3}$	$\frac{1}{4}$

Example

1 Determine whether the following equations are quadratic or not.

a. $x^2 + 1 = 0$

b. $\frac{1}{2}x^2 - \sqrt{2}x + 5 = 0$

c. $2x^2 - 3x = 5$

d. $x^2 - 2x^{-1} + 3 = 0$

e. $(x - 1)(x + 2) = 0$

f. $(x - 2)x^2 = 0$

Solution



To the best of our knowledge, the origin of the term 'quadratic' is Latin. It is derived from *quadratus* which is the past participle of *quadrare* which means 'to make square'. From this it is clear that part of the word is connected to the Latin word for 'four': it refers to squaring, and a square is a regular four-sided figure.

a , b , c , and e are quadratic equations. Equation d is not quadratic, since the power of x is -1 , which does not meet the requirements for a quadratic. Equation f is a third degree equation, so it is not quadratic.

To solve a quadratic equation we must find the values of the unknown x which make the left-hand and right-hand sides equal. Such values are called the **solutions** or **roots** of the quadratic equation. A number of techniques are available to help us obtain a solution to any quadratic equation.

A. SOLVING EQUATIONS OF THE FORM $ax^2 = 0$

We have seen that the solutions (or roots) of a quadratic equation are the values of x that make the two sides of the equation equal. Let us find the roots of the simple quadratic $ax^2 = 0$.

$$ax^2 = 0 \quad (a \neq 0)$$

$$x^2 = 0$$

$$x \cdot x = 0$$

$$x = 0 \text{ or } x = 0$$

$$x_1 = x_2 = 0$$



If $A \cdot B = 0$, then
 $A = 0$ or $B = 0$.

We can see that this equation has two equal roots. When the roots of a quadratic equation are the same, we say that the equation has a **double root**.

EXAMPLE

2

Solve the equation $-\frac{3}{2}x^2 = 0$.

Solution

$$-\frac{3}{2}x^2 = 0$$

$$x^2 = 0$$

$$x_1 = x_2 = 0$$

B. SOLVING EQUATIONS OF THE FORM $ax^2 + bx = 0$

Let us look at the solution of this more complex quadratic.

$$ax^2 + bx = 0$$

$$x(ax + b) = 0$$

$$x = 0 \text{ or } ax + b = 0, \text{ so}$$

$$x_1 = 0 \text{ and } x_2 = -\frac{b}{a}.$$

This kind of quadratic equation has two roots and one of them is always zero.

Example

3 Solve the equations.

a. $x^2 + x = 0$

b. $-4x^2 + 5x = 0$

c. $\sqrt{3}x^2 - 2x = 0$

Solution

a. $x^2 + x = 0$

$$x(x + 1) = 0$$

$$x = 0 \text{ or } x + 1 = 0$$

$$x_1 = 0 \text{ or } x_2 = -1$$

b. $-4x^2 + 5x = 0$

$$x(-4x + 5) = 0$$

$$x = 0 \text{ or } -4x + 5 = 0$$

$$x_1 = 0 \text{ or } x_2 = \frac{5}{4}$$

c. $\sqrt{3}x^2 - 2x = 0$

$$x(\sqrt{3}x - 2) = 0$$

$$x = 0 \text{ or } \sqrt{3}x - 2 = 0$$

$$x_1 = 0 \text{ or } x_2 = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$$

C. SOLVING EQUATIONS OF THE FORM $ax^2 + c = 0$

Look at the calculation.

$$ax^2 + c = 0, \quad c \neq 0$$

$$ax^2 = -c$$

$$x^2 = -\frac{c}{a}$$

$$x = \pm \sqrt{-\frac{c}{a}}$$

Here the sign of $\frac{c}{a}$ is important.

If $\frac{c}{a} > 0$, the equation has no real solution, because we cannot find the square root of a negative number.

If $\frac{c}{a} < 0$, the equation has two real solutions. These roots are symmetric, i.e. they are the same numeral with opposite signs.

Note

All positive real numbers have two square roots. One root is the positive square root and the other root is the negative square root, i.e. if $a^2 = b$ and a is a positive real number, then $a = \pm\sqrt{b}$.

Example**4** Solve the equations.

a. $3x^2 - 27 = 0$

b. $2x^2 + 6 = 0$

c. $7 - 4x^2 = 2$

Solution a. $3x^2 - 27 = 0$

$3x^2 = 27$

$x^2 = 9$

$x = \pm 3$

b. $2x^2 + 6 = 0$

$2x^2 = -6$

$x^2 = -3$

no real solution

 $(x^2 \text{ cannot be negative})$

c. $7 - 4x^2 = 2$

$4x^2 = 5$

$x^2 = \frac{5}{4}$

$x = \pm\sqrt{\frac{5}{4}} = \pm\frac{\sqrt{5}}{2}$

Check Yourself 1

Solve the equations.

1. $-3x^2 = 0$

2. $5x^2 - 20x = 0$

3. $7x^2 + 35 = 0$

4. $2x^2 - 8 = 0$

Answers**1.** 0 **2.** 0, 4 **3.** no real solution **4.** ± 2 **FIND THE MISTAKE!**Let a and b be two arbitrary numbers such that $a \neq b$. Then

$$(a - b)^2 = a^2 - 2ab + b^2 = b^2 - 2ab + a^2$$

$$(a - b)^2 = (b - a)^2$$

$$a - b = b - a$$

$$2a = 2b$$

$$a = b.$$

Can you find the mistake in this working?

D. SOLVING EQUATIONS OF THE FORM $ax^2 + bx + c = 0$ There are three basic methods for solving a quadratic equation of the form $ax^2 + bx + c = 0$:

1. factoring,
2. completing the square, and
3. using the quadratic formula.

1. FactoringIf we can write $ax^2 + bx + c = 0$ as the product of two linear factors, then we can easily solve the equation.

To solve a quadratic equation by factoring, follow the steps.

1. Write the equation in standard form, $ax^2 + bx + c = 0$, $a \neq 0$.
2. Factor the left side of the equation.

3. Apply the zero product property, that is, set each factor equal to zero.
4. Solve each equation to obtain the roots.

Example

5 Solve by factoring.

a. $x^2 + 3x + 2 = 0$

b. $6x^2 - 19x - 7 = 0$

c. $2x^2 = x + 3$

Solution

a. $x^2 + 3x + 2 = 0$

$$(x + 2)(x + 1) = 0$$

$$x + 2 = 0 \text{ or } x + 1 = 0$$

$$x_1 = -2, x_2 = -1$$

c. $2x^2 = x + 3$

$$2x^2 - x - 3 = 0$$

$$(x + 1)(2x - 3) = 0$$

$$x + 1 = 0 \text{ or } 2x - 3 = 0$$

$$x_1 = -1, x_2 = \frac{3}{2}$$

b. $6x^2 - 19x - 7 = 0$

$$(2x - 7)(3x + 1) = 0$$

$$2x - 7 = 0 \text{ or } 3x + 1 = 0$$

$$x_1 = \frac{7}{2}, x_2 = -\frac{1}{3}$$

Note

When you are solving an equation, do not divide both sides by an expression containing the variable for which you are solving. You may be dividing by zero. For example, to solve $x^2 - 2x = 0$, do not divide both sides by x , because x may be zero and you will also lose one of the solutions.

Check Yourself 2

Solve the following equations.

1. $3x^2 = 5x + 2$

2. $(5x - 1)(x + 2) = x + 2$

3. $4x(x + 1) = 3$

Answers

1. $-\frac{1}{3}, 2$ 2. $-2, \frac{2}{5}$ 3. $-\frac{3}{2}, \frac{1}{2}$

2. Completing the Square

The idea behind this method is to adjust the left side of the equation $ax^2 + bx + c = 0$ so that it becomes a perfect square, that is, the square of a first-degree polynomial. Expressions in the form $x^2 + 2xy + y^2$ and $x^2 - 2xy + y^2$ are perfect square polynomials.

For example, $x^2 + 6x + 9$ and $x^2 - 4x + 4$ are perfect squares, because $x^2 + 6x + 9 = (x + 3)^2$ and $x^2 - 4x + 4 = (x - 2)^2$.

To write a quadratic equation as a perfect square, follow the steps,

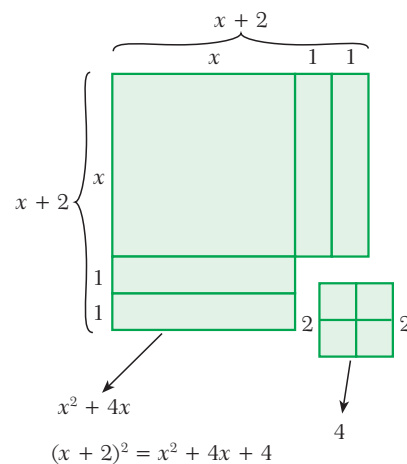


Numbers whose square roots are integers or quotients of integers are perfect squares.



1. Make sure $a = 1$ in the quadratic. If it isn't 1, divide each term by a .
2. Rewrite the equation so that the constant term is alone on one side of the equation.
3. Take half of the coefficient of the x term and square it.
4. Add this number to both sides of the equation.
5. Factor the left-hand side into a perfect square.
6. Solve for x by using the square root property.

Let us look at some examples of completing the square.



Start	Add	Result
$x^2 + 4x$	4	$x^2 + 4x + 4 = (x + 2)^2$
$x^2 + 12x$	36	$x^2 + 12x + 36 = (x + 6)^2$
$x^2 - 6x$	9	$x^2 - 6x + 9 = (x - 3)^2$
$x^2 + x$	$\frac{1}{4}$	$x^2 + x + \frac{1}{4} = (x + \frac{1}{2})^2$

Note

The expression $x^2 + 2bx$ is equivalent to $(x + b)^2 - b^2$.

Example

6 Solve by completing the square.

a. $x^2 + 6x - 7 = 0$

Solution a. $x^2 + 6x - 7 = 0$

$$x^2 + 6x = 7$$

$$x^2 + 6x + 9 = 7 + 9$$

$$(x + 3)^2 = 16$$

$$x + 3 = \pm\sqrt{16}$$

$$x_1 = -7, \quad x_2 = 1$$

b. $2x^2 - 4x + 1 = 0$

b. $2x^2 - 4x + 1 = 0$

$$x^2 - 2x + \frac{1}{2} = 0$$

$$x^2 - 2x = -\frac{1}{2}$$

$$x^2 - 2x + 1 = -\frac{1}{2} + 1$$

$$(x - 1)^2 = \frac{1}{2}$$

$$x - 1 = \pm\sqrt{\frac{1}{2}}$$

$$x_1 = 1 - \frac{\sqrt{2}}{2}, \quad x_2 = 1 + \frac{\sqrt{2}}{2}$$

Check Yourself 3

Solve the equations.

1. $x^2 + 8x - 3 = 0$ 2. $2x^2 - 5x - 3 = 0$ 3. $2x^2 - 2 = 4x$

Answers

1. $-4 \pm \sqrt{19}$ 2. $3, -\frac{1}{2}$ 3. $1 \pm \sqrt{2}$

3. The Quadratic Formula

The final method will work on any quadratic equation. Therefore, we can use it when the other easier methods fail or are not easy to apply. Look at the derivation of the quadratic formula.

$$ax^2 + bx + c = 0 \quad (a \neq 0, a, b, c \in \mathbb{R})$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad (\text{divide both sides by } a)$$

$$x^2 + \frac{b}{a}x = -\frac{c}{a} \quad (\text{try to get a perfect square})$$

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2 \quad (\text{add } \left(\frac{b}{2a}\right)^2 \text{ to both sides})$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{-4ac + b^2}{4a^2}$$

$$x + \frac{b}{2a} = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

QUADRATIC FORMULA

$$\text{If } ax^2 + bx + c = 0, a \neq 0, \text{ then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example

7 Solve $x^2 + 2x - 8 = 0$ using the quadratic formula.

Solution

Make sure that you write a quadratic equation in standard form before you identify the values a , b , and c .

First we identify the coefficients a , b , and c . For this equation $a = 1$, $b = 2$, and $c = -8$. Let us substitute the values of a , b , and c into the quadratic formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{2^2 - 4 \cdot 1 \cdot (-8)}}{2 \cdot 1} = \frac{-2 \pm \sqrt{36}}{2}$$

$$x_1 = \frac{-2 + 6}{2} = \frac{4}{2} = 2, \quad x_2 = \frac{-2 - 6}{2} = \frac{-8}{2} = -4$$

Example

Solve $3x^2 + 2x - 4 = 0$.

Solution

For this equation $a = 3$, $b = 2$, and $c = -4$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{2^2 - 4 \cdot 3 \cdot (-4)}}{2 \cdot 3} = \frac{-2 \pm \sqrt{52}}{6}$$

$$x_1 = \frac{-2 + \sqrt{52}}{6} = \frac{-2 + 2\sqrt{13}}{6} = \frac{-1 + \sqrt{13}}{3}$$

$$x_2 = \frac{-2 - \sqrt{52}}{6} = \frac{-2 - 2\sqrt{13}}{6} = \frac{-1 - \sqrt{13}}{3}$$

EXAMPLE

Find the real solutions of the equation $9 + \frac{3}{x} - \frac{2}{x^2} = 0$, $x \neq 0$.

Solution

In its present form, the equation $9 + \frac{3}{x} - \frac{2}{x^2} = 0$, is not a quadratic equation. However, we can make it quadratic by multiplying each side by x^2 , since $x \neq 0$. The result is $9x^2 + 3x - 2 = 0$, $x \neq 0$. Now, $a = 9$, $b = 3$, and $c = -2$.

$$b^2 - 4ac = 3^2 - 4 \cdot 9 \cdot (-2) = 81$$

$$x_1 = \frac{-3 - \sqrt{81}}{2 \cdot 9} = \frac{-12}{18} = -\frac{2}{3}$$

$$x_2 = \frac{-3 + \sqrt{81}}{18} = \frac{6}{18} = \frac{1}{3}$$

Check Yourself 4

Solve the equations.

1. $4x^2 + 3x - 1 = 0$

2. $x^2 - \frac{5}{2} = -3x$

3. $2x^2 - 4x = 5$

Answers

1. $-1, \frac{1}{4}$ 2. $\frac{-3 \pm \sqrt{19}}{2}$ 3. $\frac{2 \pm \sqrt{14}}{2}$

4. Discriminant of a Quadratic Equation

Definition

discriminant of a quadratic equation

The quantity $b^2 - 4ac$ is called the **discriminant** of a quadratic equation.

The discriminant tells us whether the equation has real solutions, and also tells us how many roots of an equation exist. The discriminant is denoted by Δ (delta).

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}, \quad \Delta = b^2 - 4ac$$

For a quadratic equation $ax^2 + bx + c = 0$, the value of Δ determines the number of real roots.

1. If $\Delta > 0$, there are two distinct real roots.
2. If $\Delta = 0$, there is one real root (a double root).
3. If $\Delta < 0$, there is no real root.

Use the discriminant to check the number of roots before you solve a quadratic equation.

EXAMPLE

10

Solve $x^2 + 6x + 7 = 0$.

Solution First, find Δ .

$$\Delta = b^2 - 4ac = 36 - 4 \cdot 7 \cdot 1 = 8$$

So Δ is positive.

$$x_1 = \frac{-6 + \sqrt{8}}{2} = -3 + \sqrt{2}, \quad x_2 = \frac{-6 - \sqrt{8}}{2} = -3 - \sqrt{2}$$

We can see that $\Delta > 0$ and there are two real roots.

EXAMPLE

11

Solve $x^2 - 4x + 4 = 0$.

Solution First, check Δ .

$$\Delta = b^2 - 4ac = 16 - 4 \cdot 4 \cdot 1 = 0$$

$$\text{So } \Delta = 0 \text{ and } x_1 = \frac{4 - \sqrt{0}}{2}, \quad x_2 = \frac{4 + \sqrt{0}}{2}.$$

$$\text{Hence } x_1 = x_2 = 2.$$

We can see that $\Delta = 0$ and there is only one real root (a double root).

EXAMPLE

12

Solve $x^2 - 2x + 5 = 0$.

$$\text{Solution } \Delta = b^2 - 4ac = 4 - 4 \cdot 1 \cdot 5 = -16$$

Δ is negative, so there is no real root.

EXAMPLE 13 For which values of k does the equation $3x^2 - 4x + k = 0$ have no real solution?

Solution If there is no real solution, the discriminant must be negative. Therefore,

$$\Delta = (-4)^2 - 4 \cdot 3 \cdot k = 16 - 12k < 0.$$

$$16 - 12k < 0, \quad 16 < 12k,$$

$$\frac{16}{12} < k, \quad k > \frac{4}{3}.$$

So the equation has no real solution for $k > \frac{4}{3}$.

EXAMPLE 14 For what values of m does the equation $x^2 + 3mx - 5m - 1 = 0$ have a double root?

Solution A quadratic equation has a double root if its discriminant is 0. Therefore,

$$\Delta = (3m)^2 - 4 \cdot 1 \cdot (-5m - 1) = 0.$$

$$9m^2 + 20m + 4 = 0$$

$$(9m + 2)(m + 2) = 0$$

$$9m + 2 = 0 \quad \text{or} \quad m + 2 = 0.$$

$$\text{So } m_1 = -\frac{2}{9} \quad \text{and} \quad m_2 = -2.$$

EXAMPLE 15 The equation $mx^2 + (2m + 1)x + m - 1 = 0$ has two real roots. Find m .

Solution If the quadratic equation has two real roots, then its discriminant is positive.

$$\Delta = b^2 - 4ac = (2m + 1)^2 - 4m(m - 1)$$

$$= 4m^2 + 4m + 1 - 4m^2 + 4m$$

$$= 8m + 1$$

$$\Delta = 8m + 1 > 0$$

$$8m > -1$$

$$m > -\frac{1}{8}$$

Therefore, the equation has two real roots if $m > -\frac{1}{8}$.

EXAMPLE**16**

Prove that $(a^2 + b^2)x^2 + 2(a + b)x + 2 = 0$ has no real root if a and b are unequal.

Solution

$$\begin{aligned}
 \Delta &= 4(a + b)^2 - 4(a^2 + b^2) \cdot 2 \\
 &= 4(a^2 + 2ab + b^2) - 8(a^2 + b^2) \\
 &= 4a^2 + 8ab + 4b^2 - 8a^2 - 8b^2 \\
 &= -4a^2 + 8ab - 4b^2 \\
 &= -4(a - b)^2
 \end{aligned}$$

But $(a - b)^2$ is always non-negative, and it is also non-zero, since $a \neq b$. So $\Delta < 0$.

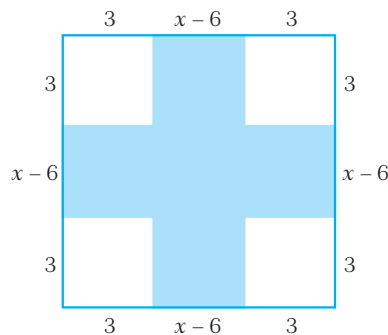
Thus the equation $(a^2 + b^2)x^2 + 2(a + b)x + 2 = 0$ has no real roots if a and b are unequal.

EXAMPLE**17**

From each corner of a square piece of sheet metal, a man removes a square of side 3 cm. He turns up the edges to form an open box. If the box holds 48 cm^3 , what are the dimensions of the piece of sheet metal?

Solution

Let x be the length of a side of the square.



Since the volume of the box is 48 cm^3 , we have

$$\begin{aligned}
 3(x - 6)^2 &= 48 \\
 (x - 6)^2 &= 16 \\
 x - 6 &= \pm 4 \\
 x_1 &= 10, \quad x_2 = 2.
 \end{aligned}$$

We discard the solution $x = 2$ since length cannot be negative. So the sheet of metal is 10 cm by 10 cm.

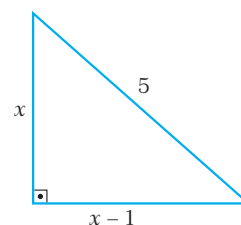
EXAMPLE**18**

A ladder 5 m long is leaning against a house. The distance from the bottom of the ladder to the house is 1 m less than the distance from the top of the ladder to the ground. How far is the bottom of the ladder from the house?

**Solution**

Let us use x to represent the distance from the top of the ladder to the ground. The ladder then forms the right triangle shown in the diagram. By using the Pythagorean Theorem, we get the equation

$$\begin{aligned}x^2 + (x - 1)^2 &= 5^2 \\x^2 + x^2 - 2x + 1 &= 25 \\2x^2 - 2x - 24 &= 0 \\x^2 - x - 12 &= 0 \\(x + 3)(x - 4) &= 0 \\x = -3 \text{ or } x &= 4.\end{aligned}$$



Since the length cannot be negative, $x = 4$. So the distance from the bottom of the ladder to the house is 3 m.

EXAMPLE**19**

A motorboat heads upstream a distance of 48 km on a river whose current is running at 3 km per hour (km/h). Then the motorboat returns. The trip upstream and back takes 12 hours. Assuming that the motorboat maintained a constant speed relative to the water, what was its speed?

Solution

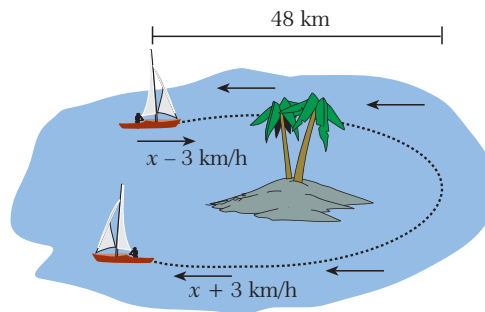
We use x to represent the constant speed of the motorboat relative to the water. Then the true speed going upstream is $x - 3$ km/h, and the true speed going downstream is $x + 3$ km/h. Since Distance = Velocity $\cdot$ Time, we can write Time = Distance / Velocity.

Therefore, the boat takes $\frac{48}{x-3}$ hours to travel upstream and $\frac{48}{x+3}$ hours to travel downstream.

Since the total time is 12 hours,

$$\frac{48}{x-3} + \frac{48}{x+3} = 12; \quad x \neq \pm 3.$$

Multiply both sides by $(x - 3)(x + 3)$:



$$48(x - 3) + 48(x + 3) = 12(x - 3)(x + 3)$$

$$x^2 - 8x - 9 = 0$$

$$(x - 9)(x + 1) = 0$$

$$x = 9 \text{ or } x = -1.$$

Since the speed cannot be negative, the speed of the boat is 9 km/h.

Check Yourself 5

1. Evaluate the discriminant of the equation $2x^2 - x - 3 = 0$, and describe the roots.
2. The sum of two numbers is 10 and sum of their squares is 68. Find the numbers.
3. For which value(s) of m does the equation $mx^2 + mx - 1 = 0$ have a double root?

Answers

1. $\Delta = 25$, two real roots
2. 2, 8
3. -4

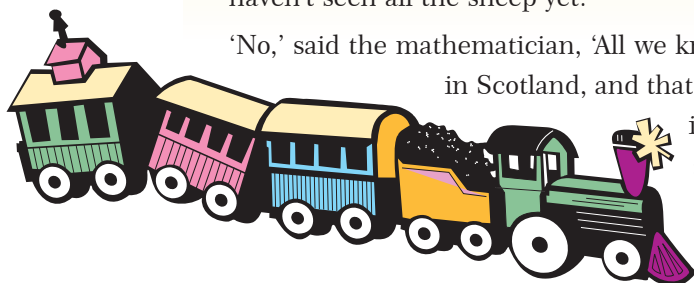
MATH FUN

A mathematician, a physicist, and an engineer were traveling through Scotland when they saw a black sheep through the window of the train.

‘Aha,’ said the engineer, ‘I see that Scottish sheep are black.’

‘Hmm,’ said the physicist, ‘You mean that some Scottish sheep are black. We haven’t seen all the sheep yet.’

‘No,’ said the mathematician, ‘All we know is that there is at least one sheep in Scotland, and that at least one side of that one sheep is black! We haven’t seen the other side of the sheep yet.’



THE GOLDEN RATIO

The Golden Ratio appears again and again in art, architecture, music and nature. Its origins go back to the days of the ancient Greeks, who thought that a rectangle with sides in the Golden Ratio, called a **golden rectangle**, exhibited the most aesthetically pleasing proportion. The use of the Golden Ratio has been of interest to artists and architects since before the building of the Parthenon in Greece in the fifth century B.C. The rectangle drawn around the Parthenon with its upper triangular structure intact, as shown in Figure 1, is a golden rectangle.



Figure 1

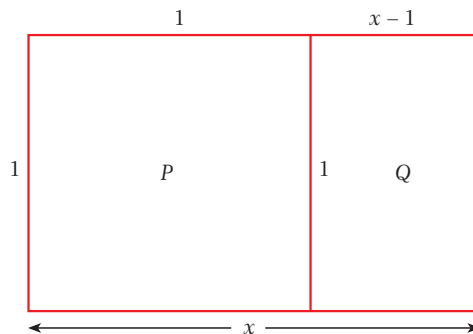


Figure 2

Figure 2 shows the dimensions of a golden rectangle. The ratio of length to width in the rectangle is x to 1 . A square P with sides of 1 unit has been marked, leaving a smaller rectangle Q . For the smaller rectangle Q , the ratio of length to width is 1 to $x - 1$. In order for the larger rectangle to be a golden rectangle, the two ratios need to be equal, creating a proportion:

$$\frac{x}{1} = \frac{1}{x-1}$$

When we apply cross multiplication to this proportion, we obtain a quadratic equation:

$$\frac{x}{1} = \frac{1}{x-1}; \quad x(x-1) = 1; \quad x^2 - x = 1; \quad x^2 - x - 1 = 0.$$

Since we cannot factor this equation, we apply the quadratic formula.

The solutions are $x_1 = \frac{1+\sqrt{5}}{2}$ and $x_2 = \frac{1-\sqrt{5}}{2}$.

Because x represents the length of a rectangle, the negative solution is discarded and the positive solution is $\frac{1+\sqrt{5}}{2} \approx 1.618034$. This is the **Golden Ratio**.

The Golden Ratio occurs in nature as well as in art. For example, in sunflowers, the ratio of the number of clockwise spirals to the number of counterclockwise spirals approximates the Golden Ratio.



EXERCISES 1.1

A. Solving Equations of the Form $ax^2 = 0$

1. Solve the equations.

a. $\sqrt{2}x^2 = 0$ b. $(\sqrt{3} - 2)x^2 = 0$
c. $-\frac{1}{5}x^2 = 0$ d. $0.07x^2 = 0$

B. Solving Equations of the Form

$$ax^2 + bx = 0$$

2. Solve the equations.

a. $2x^2 + 5x = 0$ b. $-7x^2 + 3x = 0$
c. $3x^2 - 8x = 0$ d. $\frac{2}{3}x - 6x^2 = 0$

3. Solve $(x + 2)^2 = 2(x + 2)$.

4. Solve $(2x - 1)(x + 3) = -3$.

5. Solve $(x + 1)^2 - 3(x + 1) = 0$.

C. Solving Equations of the Form

$$ax^2 + c = 0$$

6. Solve the equations.

a. $x^2 - 16 = 0$ b. $7x^2 + 3 = 0$
c. $9x^2 - 25 = 0$ d. $0.6x^2 - 15 = 0$
e. $-5x^2 + 4 = 0$ f. $\frac{2}{25} - \frac{x^2}{8} = 0$

7. Solve the equations.

a. $(x - 2)^2 - 9 = 0$ b. $(x + 3)^2 + 7 = 0$
c. $(x + 1)^2 = 5$ d. $(1 - x)^2 - 1 = 8$

8. Solve $(4x + 1)(x - 1) = (x - 1)(x + 1) - 3(x - 3)$.

D. Solving Equations of the Form

$$ax^2 + bx + c = 0$$

9. Solve by factoring.

a. $x^2 - x = 0$ b. $-3x^2 + x = 0$
c. $x^2 - 49 = 0$ d. $x^2 - 25 = 0$
e. $x^2 - x - 2 = 0$ f. $x^2 + 3x + 2 = 0$
g. $x^2 - 2x + 1 = 0$ h. $6x^2 + x - 15 = 0$
i. $10x^2 - 19x + 6 = 0$ j. $12x^2 - 5x - 2 = 0$

10. Solve by factoring.

a. $x^2 + (a + 1)x + a = 0$
b. $(x + 1)^2 - 2(x + 1)(x - 3) + (x - 3)^2 = 0$
c. $2ax^2 + (5b - 2a)x - 5b = 0$

11. Solve by completing the square.

a. $x^2 - 4x - 1 = 0$ b. $x^2 + 4x = 3$
c. $x^2 - 6x - 13 = 0$ d. $3x^2 - 2x + 4 = 0$
e. $4x^2 + 8x + 15 = 0$ f. $2x^2 + 7x + 11 = 0$
g. $\frac{1}{2}x^2 - 3x + \frac{1}{2} = 0$ h. $3x^2 + x = \frac{1}{2}$

12. Solve by using the quadratic formula.

a. $x^2 - 4x + 2 = 0$ b. $2x^2 + x - 1 = 0$
c. $4x^2 + 12x + 9 = 0$ d. $3x^2 - 5x + 1 = 0$
e. $5x^2 + 4x + 7 = 0$ f. $3x^2 - 7 = 2 - 2x$
g. $x^2 + x + 1 = 0$
h. $25x^2 + 40x + 16 = 0$
i. $(2x - 3)^2 = 11x - 19$
j. $\frac{3}{4}x^2 - \frac{1}{4}x - \frac{1}{2} = 0$
k. $\frac{4x^2 - 1}{3} = x(10x - 9)$
l. $\frac{x^2 - 1}{2} = 11(x + 1)$

13. Solve by using the quadratic formula.

a. $4 - \frac{1}{x} - \frac{2}{x^2} = 0$

b. $\frac{1}{x^2 - 1} + \frac{1}{x - 1} = \frac{2}{3}$

c. $\frac{(x-3)^2}{16} - \frac{(x-2)^2}{4} = \frac{1-x}{2}$

d. $\frac{1}{x-1} + \frac{1}{x-4} = \frac{5}{4}$

14. Use the discriminant to determine whether each quadratic equation has two real solutions, a double root, or no real solution, without solving the equation.

a. $2x^2 + 3x + 1 = 0$ b. $x^2 + 5x - 6 = 0$

c. $4x^2 + 12x + 9 = 0$ d. $25x^2 - 20x + 4 = 0$

e. $x^2 + 4x + 7 = 0$ f. $2x^2 - x + 2 = 0$

15. Consider the equation $ax^2 + 3x + 10 = 0$. For which values of a does the equation have

a. two distinct real roots?

b. one double root?

c. no solution?

16. For which values of m does the equation $2x^2 + 2x + m + 4 = 0$ have

a. two real solutions?

b. one solution?

c. no real solutions?

17. For which values of k does the equation

$$x^2 - (4k + 2)x + 7k + 2 = 0$$
 form a perfect square?

18. The longest side of a right triangle is 6 cm less than twice the length of the medium side. The shortest side is 6 cm. Find the length of the two other sides.

19. A wire that is 32 cm long is cut into two pieces, and each piece is bent to form a square. The total area enclosed by the two squares is 34 cm^2 . Find the length of each piece of wire.

Mixed Problems

20. a , b , c , and d are four consecutive even natural numbers. The sum of a and c is one fifth of the product of b and d . Find a , b , c , and d .

21. Two squares have sides $(x + 6)$ cm and $(2x + 1)$ cm respectively. The sum of their areas is 697 cm^2 . Find the areas of the squares.

22. A year ago, a father was eight times as old as his son. Now his age is the square of his son's age. How old are they now?



**François Viète
(or Vieta)**

1540-1603, French mathematician. Vieta was a founder of modern algebra, who introduced the use of letters as algebraic symbols and correlated algebra with geometry and trigonometry.

Vieta presented methods for solving equations of second, third and fourth degree. He knew the connection between the positive roots of equations and the coefficients of the different powers of the unknown quantity. The word 'coefficient' is actually due to Vieta. When Vieta applied numerical methods to solve equations, he used methods which were similar to those used by earlier Arabic mathematicians.

We have seen that the roots of an equation depend on its coefficients. Therefore, there exist certain relations between the coefficients and the roots of an equation.

In this section we will consider the relations between the roots and the coefficients a , b and c of a quadratic equation.

We know that the roots of the quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$ are

$$x_1 = \frac{-b - \sqrt{\Delta}}{2a} \quad \text{and} \quad x_2 = \frac{-b + \sqrt{\Delta}}{2a}.$$

Let us use these formulas above to find the sum of the roots of a quadratic equation.

$$\begin{aligned} x_1 + x_2 &= \frac{-b - \sqrt{\Delta}}{2a} + \frac{-b + \sqrt{\Delta}}{2a} \\ &= \frac{-b - \sqrt{\Delta} + (-b + \sqrt{\Delta})}{2a} \\ &= \frac{-2b}{2a} = -\frac{b}{a} \end{aligned}$$

Therefore, $x_1 + x_2 = -\frac{b}{a}.$

We can use the same expressions for x_1 and x_2 to find the product of the roots of a quadratic equation.

$$\begin{aligned} x_1 \cdot x_2 &= \left(\frac{-b - \sqrt{\Delta}}{2a} \right) \cdot \left(\frac{-b + \sqrt{\Delta}}{2a} \right) = \frac{(-b - \sqrt{\Delta}) \cdot (-b + \sqrt{\Delta})}{4a^2} \\ &= \frac{b^2 - (b^2 - 4ac)}{4a^2} = \frac{4ac}{4a^2} = \frac{c}{a} \end{aligned}$$

Therefore, $x_1 \cdot x_2 = \frac{c}{a}.$

These relations were discovered by **François Vieta**, a French mathematician, and so they are together called **Vieta's theorem**.

Theorem**Vieta's theorem**

Let x_1 and x_2 be the roots of the quadratic equation of the form $ax^2 + bx + c = 0$, $a \neq 0$. Then

$$x_1 + x_2 = -\frac{b}{a}, \quad x_1 \cdot x_2 = \frac{c}{a}.$$

By using Vieta's theorem we can now find the sum and product of the roots of a quadratic equation without calculating the roots.

Note

By using Vieta's theorem we can also see the following.

1. $\frac{1}{x_1} + \frac{1}{x_2} = \frac{x_1 + x_2}{x_1 x_2} = \frac{-b/a}{c/a} = \frac{-b}{c}$ (sum of the reciprocals of the roots)
2. $x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1 x_2 = \left(\frac{-b}{a}\right)^2 - 2 \cdot \frac{c}{a} = \frac{b^2 - 2ac}{a^2}$ (sum of the squares of the roots)
3. $x_1^3 + x_2^3 = (x_1 + x_2)^3 - 3x_1 x_2 (x_1 + x_2) = \frac{3abc - b^3}{a^3}$ (sum of the cubes of the roots)

EXAMPLE**20**

Derive a formula for the difference of the roots of a quadratic equation in standard form.

Solution

Let us consider the two differences. First,

$$x_1 - x_2 = \frac{-b - \sqrt{\Delta}}{2a} - \frac{-b + \sqrt{\Delta}}{2a} = \frac{-b - \sqrt{\Delta} + b - \sqrt{\Delta}}{2a} = \frac{-2\sqrt{\Delta}}{2a} = \frac{-\sqrt{\Delta}}{a}.$$

$$\text{Similarly, } x_2 - x_1 = \frac{\sqrt{\Delta}}{a}.$$

$$\text{So we can say that } |x_1 - x_2| = \frac{\sqrt{\Delta}}{|a|}.$$

EXAMPLE**21**

Find the sum and product of the roots of the given equations, without solving the equations.

a. $2x^2 + 6x + 5 = 0$

b. $x^2 - 3x - 5 = 0$

Solution

a. $a = 2$, $b = 6$ and $c = 5$

b. $a = 1$, $b = -3$ and $c = -5$

$$x_1 + x_2 = -\frac{b}{a} = \frac{-6}{2} = -3$$

$$x_1 + x_2 = -\frac{b}{a} = \frac{3}{1} = 3$$

$$x_1 \cdot x_2 = \frac{c}{a} = \frac{5}{2}$$

$$x_1 \cdot x_2 = \frac{c}{a} = \frac{-5}{1} = -5$$

Note

The quadratic equation $2x^2 + 6x + 5 = 0$ has no real root. However, by Vieta's theorem, the sum of the roots is -3 and the product of the roots is $\frac{5}{2}$, which are real numbers. Can you say why?

Check Yourself 6

1. Find the sum and the product of the roots of the following equations, using Vieta's theorem.

a. $3x^2 + 5x - 1 = 0$

b. $x^2 - 4x + \frac{7}{2} = 0$

c. $-x^2 + 7x - 1 = 0$

2. For the previous equations, find

a. $\frac{1}{x_1} + \frac{1}{x_2}$.

b. $x_1^2 + x_2^2$.

c. $x_1^3 + x_2^3$.

Answers

1. a. $-\frac{5}{3}, -\frac{1}{3}$ b. $4, \frac{7}{2}$ c. 7, 1

2. a. 5 b. 9 c. 322

EXAMPLE

22

x_1 and x_2 are the non-negative roots of the equation $3x^2 + 2mx + 1 = 0$. Given $x_1 = 3x_2$, find x_1 , x_2 , and m .

Solution

$$x_1 = 3x_2$$

$$x_1 \cdot x_2 = \frac{c}{a} = \frac{1}{3}; \quad 3x_2 \cdot x_2 = \frac{1}{3}; \quad x_2^2 = \frac{1}{9}; \quad x_2 = \pm \frac{1}{3} \text{ since } x_2 > 0; \quad x_2 = \frac{1}{3}$$

$$x_1 = 3x_2 \text{ then } x_1 = 3 \cdot \frac{1}{3} = 1$$

$$x_1 + x_2 = -\frac{2m}{3}; \quad 1 + \frac{1}{3} = -\frac{2m}{3}; \quad \frac{4}{3} = -\frac{2m}{3}$$

$$m = -2$$

EXAMPLE

23

x_1 and x_2 are the roots of the equation $x^2 - 3x + 1 = 0$. Find the value of $\frac{x_1}{\sqrt{x_2}} + \frac{x_2}{\sqrt{x_1}}$.

Solution

Since $x_1 + x_2 = -\frac{(-3)}{1} = 3 > 0$ and $x_1 \cdot x_2 = \frac{1}{1} = 1 > 0$, both x_1 and x_2 are positive.

$$\frac{x_1}{\sqrt{x_2}} + \frac{x_2}{\sqrt{x_1}} = k$$

$$k^2 = \frac{x_1^2}{x_2} + \frac{2x_1x_2}{\sqrt{x_1x_2}} + \frac{x_2^2}{x_1}$$

$$k^2 = \frac{x_1^3 + x_2^3}{x_1x_2} + 2\sqrt{x_1x_2}$$

$$k^2 = \frac{-9 + 27}{1} + 2\sqrt{1} = 20$$

$$k_1 = 2\sqrt{5}, k_2 = -2\sqrt{5}$$

$$k = 2\sqrt{5} \text{ (why?)}$$

EXAMPLE**24**

$x^2 + x - 6m = 0$ and $x^2 - 2mx + 3 = 0$ have a common root. Find m .

Solution Let x_1 and x_2 be the roots of $x^2 + x - 6m = 0$ and
 x_1 and x_3 be the roots of $x^2 - 2mx + 3 = 0$, then

$$(1) \begin{cases} x_1 + x_2 = -1 \\ x_1 + x_3 = 2m \end{cases} \quad \text{and} \quad (2) \begin{cases} x_1 x_2 = -6m \\ x_1 x_3 = 3 \end{cases}.$$

From (1) we get $x_2 - x_3 = -1 - 2m$.

From (2) we get $\frac{x_2}{x_3} = -2m$; $x_2 = -2mx_3$.

From (1) and (2),

$$-2mx_3 - x_3 = -1 - 2m$$

$$x_3(2m + 1) = 1 + 2m$$

$$x_3 = \frac{1 + 2m}{1 + 2m} = 1.$$

If we substitute 1 for x in the second equation, we have

$$4 - 2m = 0, \text{ so } m = 2.$$

Note

We can apply Vieta's theorem to a cubic polynomial equation.

Let x_1 , x_2 and x_3 be the roots of the equation $ax^3 + bx^2 + cx + d = 0$. Then,

$$1. \ x_1 + x_2 + x_3 = -\frac{b}{a}. \quad 2. \ x_1 x_2 + x_1 x_3 + x_2 x_3 = \frac{c}{a}. \quad 3. \ x_1 x_2 x_3 = -\frac{d}{a}.$$

Check Yourself 7

- One of the roots of the equation $2x^2 - mx + 8 = 0$ is 3 more than the other root. Find m .
- x_1 and x_2 are the roots of the equation $x^2 + x - m + 1 = 0$. $x_1 - x_2 = 5$ is given. Find m , x_1 , and x_2 .

Answers

1. ± 10 2. 7, 2, -3

Mathematics is one component of any plan for liberal education. Mother of all the sciences, it is a builder of the imagination, a weaver of patterns of sheer thought, an intuitive dreamer, a poet. The study of mathematics cannot be replaced by any other activity.



EXERCISES 1.2

1. Find the sum and the product of the roots of each equation, without solving it.
 - a. $10x^2 - 11x - 12 = 0$
 - b. $\frac{7}{8}x^2 + \frac{9}{7}x - \frac{5}{3} = 0$
 - c. $(x - 2)(3x - 4) = 13$
 - d. $x + 7 = (2x - 1)(3x - 2)$
 - e. $(4x + 3)^2 = (3x + 1)^2$
 - f. $x^2 = \sqrt{2}(3x - \sqrt{2}x)$
2. The sum of the roots of the equation $(5k + 2)x^2 + 7kx - 8k = 0$ is 3. Find the product of the roots.
3. The product of the roots of the equation $(4m^2 - 1)x^2 + (2m + 1)x + 2m - 1 = 0$ is 5. Find m .
4. Find the sum of the squares of the roots of the following quadratic equations.
 - a. $2x^2 + 5x + 1 = 0$
 - b. $-x^2 + 7x + 2 = 0$
 - c. $3(x + 2)(x - 1) = 4(x - 2) - 1$
 - d. $4x(3 - 4x) = x - 1$
5. One of the roots of the equation $20x^2 + (2m^2 + 2m + 1)x + 20 = 0$ is $-\frac{4}{5}$. Find m and the other root of the equation.
6. One of the roots of the equation $6x^2 + 13x + n^2 + 2n - 2 = 0$ is $-\frac{3}{2}$. Find n and the other root of the equation.
7. Let x_1 and x_2 be the roots of the equation $4x^2 + 5x = 0$. Find $x_1^2x_2 + x_1x_2^2$.
8. Consider the equation $(m + 2n)x^2 - (m + 2n)x + m - n = 0$. If the arithmetic and the geometric means of the roots of the equation are equal, find the relation between m and n .
9. Let x_1 and x_2 be the roots of the equation $x^2 - 8x + 5k = 0$. $x_1 = 2x_2 - 1$ is given. Find k .
10. Let x_1 and x_2 be the roots of the equation $x^2 + (m + 1)x + m + 2 = 0$. $2x_1 + 3x_2 = -13$ is given. Find m .
11. Let x_1 and x_2 be the roots of the equation $(k + 1)x^2 - kx + k - 4 = 0$. $\frac{6}{x_1} + \frac{6}{x_2} = 30$ is given. Find k .

12. Consider the equation $x^2 - (3k + 1)x + 8 = 0$.
 $x_1^2 + x_2^2 = 20$ is given. Find k .

13. Consider the equation $x^2 + (m + 1)x - m = 0$.
 $x_1^2 + x_2^2 = 13$ is given. Find m .

14. Let x_1 and x_2 be the roots of the equation
 $kx^2 + (k - 1)x - 2 + k = 0$.
 $\frac{2}{x_1 + 3} + \frac{2}{x_2 + 3} = \frac{3}{2}$ is given. Find k .

15. Consider the equation $x^2 + (k - 2)x + k - 6 = 0$.
If the equation has two negative roots, find all the possible values of k .

16. Find two positive consecutive numbers such that the sum of their squares is 85.

17. Find the product of the roots of the equation
 $\frac{1}{a} + \frac{1}{b} = \frac{1}{a + b + x} - \frac{1}{x}$, $a + b \neq 0$.

18. Let p and q be the roots of the equation
 $2x^2 - 5x + p^2 + q^2 = 0$. Find the discriminant of the equation.

19. Find $x_1^2 + \frac{1}{x_1^2}$, if x_1 is the root of the equation
 $\left(x + \frac{1}{x}\right)^2 - 6\left(x + \frac{1}{x}\right) + 9 = 0$.

20. Consider the equation $x^3 + cx + 1 = 0$.
 $x_1 = \frac{1}{x_2} + \frac{1}{x_3}$ is given. Find the roots of the equation and c .

21. The roots of the equation $x^2 - 5x + p = 0$ are also the roots of the equation $x^3 + qx + 30 = 0$. Find $p + q$.

22. The sum of the two roots of the equation
 $2x^3 - x^2 - 7x - 3 = 0$ is 1. Find the roots of the equation.

23. From each corner of a square piece of sheet metal, a man removes a square of side 2 cm. He turns up the edges to form an open box which holds 24 cm^3 . What are the dimensions of the piece of sheet metal?



DERIVING QUADRATIC EQUATIONS

Let x_1 and x_2 be the roots of the equation $ax^2 + bx + c = 0$.

We can write the equation $ax^2 + bx + c = 0$ as $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$.

$$\text{We know that } \begin{cases} x_1 + x_2 = -\frac{b}{a}, \text{ and} \\ x_1 \cdot x_2 = \frac{c}{a}. \end{cases}$$

So we can write the equation again as $x^2 - (x_1 + x_2)x + (x_1 \cdot x_2) = 0$.

In other words, if we know the sum and product of the roots of a quadratic equation then we can write the equation as $x^2 - Sx + P = 0$ where $S = x_1 + x_2$ and $P = x_1 \cdot x_2$. This means that we can derive (find) a quadratic equation if we know its roots.

Note

If the roots of a quadratic equation are x_1 and x_2 , then $(x - x_1) \cdot (x - x_2) = 0$.

EXAMPLE

25

Find a quadratic equation whose roots are -1 and 5 .

Solution

$$x_1 = -1 \text{ and } x_2 = 5$$

$$S = x_1 + x_2 = -1 + 5 = 4$$

$$P = x_1 \cdot x_2 = (-1) \cdot 5 = -5$$

Hence, the equation is

$$x^2 - Sx + P = 0$$

$$x^2 - 4x - 5 = 0.$$

EXAMPLE

26

Find a quadratic equation whose only root is $\frac{1}{3}$.

Solution

$$x_1 = x_2 = \frac{1}{3}$$

$$S = x_1 + x_2 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

$$P = x_1 \cdot x_2 = \frac{1}{9}.$$

Hence, the equation is $x^2 - \frac{2}{3}x + \frac{1}{9} = 0$, or $9x^2 - 6x + 1 = 0$.

EXAMPLE 27 Find the equation whose roots are $2 + \sqrt{3}$ and $2 - \sqrt{3}$.

Solution $x_1 = 2 + \sqrt{3}$, $x_2 = 2 - \sqrt{3}$
 $S = x_1 + x_2 = 4$
 $P = x_1 \cdot x_2 = (2 + \sqrt{3}) \cdot (2 - \sqrt{3}) = 1$
Hence, the equation is $x^2 - 4x + 1 = 0$.

EXAMPLE 28 Find the equation whose roots are 1 more than the roots of $x^2 - 3x - 4 = 0$.

Solution Let the roots of $x^2 - 3x - 4 = 0$ be x_1 and x_2 . Let the roots of the equation we are looking for be x_3 and x_4 .
 $x_1 + x_2 = 3$, $x_1 \cdot x_2 = -4$
 $x_3 = x_1 + 1$, $x_4 = x_2 + 1$
 $S = x_3 + x_4 = (x_1 + 1) + (x_2 + 1) = x_1 + x_2 + 2 = 3 + 2 = 5$
 $P = x_3 \cdot x_4 = (x_1 + 1) \cdot (x_2 + 1) = x_1 \cdot x_2 + x_1 + x_2 + 1 = (-4) + (3) + 1 = 0$
Hence the equation is $x^2 - 5x = 0$.

EXAMPLE 29 The equation $x^2 + (k + 4)x + k - 4 = 0$ has symmetric roots. Find them.

Solution $x_1 = t$, $x_2 = -t$ (symmetric roots)
$$x_1 + x_2 = -\frac{b}{a} \qquad x_1 \cdot x_2 = \frac{c}{a}$$
$$t + (-t) = \frac{-(k+4)}{1} \qquad t \cdot (-t) = \frac{k-4}{1}$$
$$0 = -k - 4 \qquad -t^2 = -8$$
$$k = -4 \qquad t^2 = 8$$
$$\qquad \qquad \qquad t = \pm 2\sqrt{2}$$

Therefore, $x_1 = 2\sqrt{2}$, $x_2 = -2\sqrt{2}$.

Check Yourself 8

- Find a quadratic equation whose roots are -2 and 3 .
- The roots of the equation $x^2 - x - 2 = 0$ are x_1 and x_2 . Find the equation whose roots are x_3 and x_4 , where $x_3 = 2x_1 + 1$ and $x_4 = 2x_2 + 1$.

Answers

1. $x^2 - x - 6 = 0$ 2. $x^2 - 4x - 5 = 0$

Mathematics has beauty but not everyone sees it.

EXERCISES 1.3

- Find the equation with the given roots.
 - $-1, 1$
 - $\frac{1}{2}, \frac{3}{2}$
 - $0, 4$
 - $2 + \sqrt{2}, 2 - \sqrt{2}$
 - $\sqrt{3} - \sqrt{2}, \sqrt{3} + \sqrt{2}$
 - $\frac{1}{p^2}, \frac{1}{q^2}$
- Find the quadratic equation whose sum of the roots is -2 and product of the roots is 5 .
- The roots of the equation $x^2 - 4x - 3 = 0$ are x_1 and x_2 . Find the equation whose roots are x_3 and x_4 , such that $x_3 = 2x_1$ and $x_4 = 2x_2$.
- Find the equation whose roots are 2 less than the roots of the equation $2x^2 - 6x + 58 = 0$.
- The roots of each given equation are x_1 and x_2 . Write a new equation with roots x_3 and x_4 .
 - $x^2 - 2x - 3 = 0$
 $x_3 = x_1 - 1$
 $x_4 = x_2 - 1$
 - $x^2 + 5x + 4 = 0$
 $x_3 = 2x_1 + 1$
 $x_4 = 2x_2 + 1$
 - $6x^2 - 9x - 6 = 0$
 $x_3 = x_1 - x_2$
 $x_4 = x_2 - x_1$
 - $x^2 - 2x + 1 = 0$
 $x_3 = \frac{x_1^2}{x_2}$
 $x_4 = \frac{x_2^2}{x_1}$
 - $x^2 - 2mx + 3m - 2 = 0$
 $x_3 = x_1 - 2x_2$
 $x_4 = x_2 - 2x_1$
 - $mx^2 - 2mx - 1 = 0$
 $x_3 = 2x_1 - x_2$
 $x_4 = 2x_2 - x_1$
- The roots of the equation $x^2 - kx - 3k + 1 = 0$ are two more than the roots of the equation $x^2 - (k - 4)x - 4k = 0$. Find k .
- Find three consecutive integers a, b, c , such that $a^2 + b^2 + c^2 = 110$.
- Find the number that is 3 more than twice its square root.
- A swimming pool can be filled by two pipes together in six hours. If the larger pipe alone takes five hours less than the smaller pipe alone to fill the pool, find the time in which each pipe alone would fill the pool.
- Alex can do a job in one hour less than Jane. If Alex and Jane work together the job takes $\frac{6}{5}$ hours. How long would it take each person working alone?
- A man completed a job for \$156. It took him seven hours longer than he expected and so he earned \$14 an hour less than he anticipated. How long did the man expect the job to take?
- The sum of the numerator and denominator of a certain positive fraction is 11. If 1 is added to both the numerator and the denominator, the fraction is increased by $\frac{3}{56}$. Find the fraction.
- The area of an isosceles right triangle is 81 m^2 . Find the perimeter of the triangle.

14. The distance between two cities A and B is 140 km. A car driving from A to B left at the same time as a car driving from B to A . The cars met after one hour, then the first car reached city B 35 minutes later than the second car reached city A . Find the speed of each car.
15. An aeroplane traveled a distance of 400 km at an average speed of x km/h. Write down an expression for the time taken. On the return journey, the speed increased by 40 km/h. Write down an expression for the time for the return journey. If the return journey took 30 minutes less than the outward journey, write down an equation in x and solve it.
16. A car drove from one city to another and returned by a different route. The outward journey was 48 km and the return journey was 8 km shorter. The speed of the car on the return journey increased by 4 km/h. The return journey took one hour less time. Find the speed of the car on the outward journey.
17. A group of women plan to share equally in the \$14,000 cost of a boat. At the last minute three of the women decide not to pay. This raises the share for each of the remaining women by \$1500. How many women were in the original group?
18. A car drove from a city A to a city B . The distance between the two cities is 350 km. After 200 km, the speed of the car decreased by 20 km/h. The total trip took 5 hours. Find the speed of the car in the first part of the journey.
19. Two painters working together can paint the front of a house in 16 hours. One of the painters alone can finish this job in 24 hours less time than the other painter alone. How much time does each worker need to do this job alone?
20. A motorcyclist traveled at a constant speed for 60 km. If he had gone 10 km/h faster, he would have shortened his traveling time by one hour. Find the speed of the motorcyclist.
21. The area of a triangle is 36 m^2 . The length of the base is twice the height. Find the length of the base and the height of the triangle.
22. A jeweler wishes to mix an alloy of 25% silver with another alloy of 40% silver. How much of each should he use to produce 60 kg of an alloy which is 30% silver?
23. One of two pipes can fill a pool 24 hours faster than the other one. The slower pipe filled the pool for eight hours, then the other pipe was opened. The pipes filled together for twenty hours and filled $\frac{2}{3}$ of the pool. Find the time that each pipe requires to fill the pool alone.
24. A worker can clean a pool in four hours less time than it takes another worker. If the men work together the job takes $\frac{8}{3}$ hours. How long would it take each man working alone?
25. The length of a rectangle is 2 m more than its width and the area is 48 m^2 . Find the length and width of the rectangle.

A. WRITING EQUATIONS IN QUADRATIC FORM

Definition

standard form of an equation

An equation is in **standard form** if the only term on the right-hand side of the equation is zero.

For example, the equations $6x^2 + 2x - 3 = 0$ and $x^4 - 5 = 0$ are both in standard form. The equation $6x^2 + 2x = 3$ and $x^4 = 5$ are not in standard form.

Certain equations that are not quadratic can be expressed in quadratic form using substitutions. These equations can be recognized because when they are written in standard form, the exponent of the variable in one term is half the exponent of variable in the other term.

For example, we can write standard form equations such as

$$x^4 + 17x^2 + 72 = 0$$

$$2x^8 + 4x^4 = 0$$

$$x - \sqrt{x} - 12 = 0$$

as quadratic equations, because the exponent of the first variable is twice the exponent of the second variable.

Look at the steps to write an equation as a quadratic.

1. Let t be a variable term with the half exponent.
2. Substitute t in all the terms with the variable.
3. Solve for t .
4. Back substitute for the original variable.

EXAMPLE

30

Solve $x^4 - 13x^2 + 36 = 0$.

Solution 1

The equation $x^4 - 13x^2 + 36 = 0$ is not a quadratic equation but we can write it as $(x^2)^2 - 13x^2 + 36 = 0$. For this reason, it is a quadratic in x^2 . Let $x^2 = t$.

First we solve for t , then solve the resulting equations for x .

$$(x^2)^2 - 13x^2 + 36 = 0$$

$$x^2 = t, \text{ so } t^2 - 13t + 36 = 0. \text{ By factoring,}$$

$$(t - 4)(t - 9) = 0$$

$$t = 4 \quad \text{or} \quad t = 9.$$

$$\text{Since } t = x^2$$

$$x^2 = 4 \quad \text{or} \quad x^2 = 9$$

$$x = \pm 2 \quad \text{or} \quad x = \pm 3.$$

Solution 2 Alternatively, we can directly factorize the equation.

$$x^4 - 13x^2 + 36 = 0$$

$$(x^2 - 4)(x^2 - 9) = 0$$

$$x^2 = 4 \quad \text{or} \quad x^2 = 9$$

$$x = \pm 2 \quad \text{or} \quad x = \pm 3$$

In both solutions, the roots of the given equation are $-3, -2, 2, 3$.

EXAMPLE 31 Solve $(5x - 1)^2 + 4(5x - 1) - 5 = 0$.

Solution For the equation $(5x - 1)^2 + 4(5x - 1) - 5 = 0$, we let $t = 5x - 1$ so that $t^2 = (5x - 1)^2$. Then the original equation becomes $t^2 + 4t - 5 = 0$. First solve for t :

$$(t + 5)(t - 1) = 0, \text{ so}$$

$$t = -5 \text{ or } t = 1.$$

Now, solve for x .

$$5x - 1 = -5 \quad \text{or} \quad 5x - 1 = 1$$

$$5x = -4 \quad \text{or} \quad 5x = 2$$

$$x_1 = \frac{-4}{5} \quad \text{or} \quad x_2 = \frac{2}{5}$$

Hence, the roots of the equation are $\frac{-4}{5}$ and $\frac{2}{5}$.

EXAMPLE 32 Solve $(a^2 - a)^2 - 2(a^2 - a) = 0$.

Solution Let $t = a^2 - a$.

The equation becomes

$$t^2 - 2t = 0$$

$$t(t - 2) = 0$$

$$t = 0 \text{ or } t = 2.$$

Now, solve for a .

1. $a^2 - a = 0$

$$a(a - 1) = 0$$

$$a = 0 \text{ or } a = 1$$

2. $a^2 - a = 2$

$$a^2 - a - 2 = 0$$

$$(a - 2)(a + 1) = 0$$

$$a = 2 \text{ or } a = -1$$

Hence, the roots of the equation are $-1, 0, 1, 2$.

EXAMPLE**33**Solve $\left(x - \frac{1}{x}\right)^4 + 5 = 2x^2 + \frac{2}{x^2}$.**Solution** First $x^2 \neq 0$, so $x \neq 0$.Let $x - \frac{1}{x} = t$.How can we write the right-hand side in terms of t ?Let's take the square of t :

$$t^2 = \left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2, \text{ so}$$

 $x^2 + \frac{1}{x^2} = t^2 + 2$. Now, let us multiply both sides of the equation by 2:

$$2x^2 + \frac{2}{x^2} = 2t^2 + 4.$$

Then we have the equation

$$t^4 + 5 = 2t^2 + 4$$

$$t^4 - 2t^2 + 1 = 0$$

$$(t^2 - 1)^2 = 0$$

$$t^2 - 1 = 0$$

$$t = 1 \text{ or } t = -1.$$

$$\text{Since } t = x - \frac{1}{x},$$

$$x - \frac{1}{x} = 1 \text{ or } x - \frac{1}{x} = -1.$$

Let us multiply the equations by x :

$$x^2 - x - 1 = 0 \text{ or } x^2 + x - 1 = 0.$$

By using the quadratic formula,

$$x_1 = \frac{1 - \sqrt{5}}{2}, \quad x_2 = \frac{1 + \sqrt{5}}{2}, \quad x_3 = \frac{-1 + \sqrt{5}}{2}, \quad x_4 = \frac{-1 - \sqrt{5}}{2}.$$

Check Yourself 9

Solve the equations.

1. $(x + 15)^2 - 3(x + 15) - 18 = 0$.

2. $(x + 5)^4 - 6(x + 5)^2 - 7 = 0$.

3. $x - 9\sqrt{x} + 14 = 0$.

Answers

1. $-18, -9$ 2. $-5 \pm \sqrt{7}$ 3. $4, 49$

B. EQUATIONS INVOLVING PRODUCTS AND QUOTIENTS

If the product of two or more numbers is zero, then at least one of the factors must be zero.
If the quotient of a division is zero, then the dividend must be zero.

Note

1. $P \cdot Q = 0$ if and only if $P = 0$ or $Q = 0$, where $P = P(x)$ and $Q = Q(x)$.
2. $\frac{P}{Q} = 0$ if and only if $P = 0$ and $Q \neq 0$.

EXAMPLE 34 Solve $(x^2 - 1)(x^2 - 2x - 8) = 0$.

Solution Try to factorize each part if possible:

$$(x - 1)(x + 1)(x - 4)(x + 2) = 0.$$

If the product is zero then at least one of the factors is zero.

$$x - 1 = 0 ; \quad x = 1$$

$$x + 1 = 0 ; \quad x = -1$$

$$x - 4 = 0 ; \quad x = 4$$

$$x + 2 = 0 ; \quad x = -2$$

Thus, the roots of the equation are $-2, -1, 1, 4$.

EXAMPLE 35 Solve $(x^2 - 4x + 10)(x^2 - 5x + 2) = 0$.

Solution We cannot factorize the parts.

Let us try to solve $x^2 - 4x + 10 = 0$ and $x^2 - 5x + 2 = 0$ as two different quadratic equations.

$$x^2 - 4x + 10 = 0; \quad \Delta = 16 - 4 \cdot 1 \cdot 10 = -24 < 0.$$

Therefore, this equation has no real root.

$$x^2 - 5x + 2 = 0 ; \quad \Delta = 25 - 4 \cdot 1 \cdot 2 = 17$$

This time, by the quadratic formula, the roots are

$$x_1 = \frac{5 - \sqrt{17}}{2}, \quad x_2 = \frac{5 + \sqrt{17}}{2}.$$

EXAMPLE**36**Solve $x^3 - x^2 - 4x + 4 = 0$.**Solution** First try to factorize the expression.

$$x^3 - x^2 - 4x + 4 = x^2(x - 1) - 4(x - 1) = (x - 1)(x^2 - 4) = (x - 1)(x - 2)(x + 2).$$

Now, the question becomes:

$$\text{solve } (x - 1)(x - 2)(x + 2) = 0.$$

So the solution is

$$x - 1 = 0; \quad x_1 = 1$$

$$x - 2 = 0; \quad x_2 = 2$$

$$x + 2 = 0; \quad x_3 = -2.$$

EXAMPLE**37**Solve $\frac{x^2 - 5x - 6}{x + 2} = 0$.**Solution** The denominator of a fraction cannot be equal to zero, so $x + 2 \neq 0$, $x \neq -2$.We need to solve $x^2 - 5x - 6 = 0$.

$$x^2 - 5x - 6 = (x - 6)(x + 1)$$

$$(x - 6)(x + 1) = 0$$

$$x_1 = -1 \text{ or } x_2 = 6$$

Since these roots are not equal to -2 , $x_1 = -1$ and $x_2 = 6$ are both solutions to the equation.**EXAMPLE****38**Solve $\frac{x^2 + 7x - 8}{x^2 - 3x + 2} = 0$.**Solution** First solve $x^2 - 3x + 2 \neq 0$: $x \neq 2$, $x \neq 1$.

$$x^2 + 7x - 8 = (x - 1) \cdot (x + 8) = 0$$

So $x_1 = 1$ and $x_2 = -8$ are the roots of the numerator.We know that $x \neq 1$, so the only solution is -8 .**Note**

It is very important to check the roots of the numerator to see whether they make the denominator zero or not. We can do this either by substituting the roots of the numerator in the denominator, or by finding the roots of the denominator directly and checking whether they are common or not.

EXAMPLE**39**

Solve $\frac{x-7}{x-4} - \frac{x+3}{x+4} = 1$.

Solution $x - 4 \neq 0$, and $x + 4 \neq 0$, so $x \neq 4$, and $x \neq -4$ (since the denominators cannot be zero).

Now we have $\frac{x-7}{x-4} - \frac{x+3}{x+4} - 1 = 0$.

Let us make the denominators common:

$$\frac{(x-7)(x+4) - (x+3)(x-4) - (x^2-16)}{(x+4)(x-1)} = 0.$$

$$\frac{-x^2 - 2x}{(x-4)(x+4)} = 0$$

$$-x^2 - 2x = 0; \quad -x(x+2) = 0; \quad x_1 = 0 \text{ and } x_2 = -2.$$

EXAMPLE**40**

Solve $\frac{4}{x+1} - \frac{x+1}{x-1} = \frac{x^2-5}{x^2-1}$.

Solution In its present form, the equation $\frac{4}{x+1} - \frac{x+1}{x-1} = \frac{x^2-5}{x^2-1}$ is not a quadratic equation. However, we can make it quadratic by multiplying each side by $x^2 - 1$ since $(x^2 - 1 \neq 0; x \neq \pm 1)$. The result is

$$4(x-1) - (x+1)^2 = x^2 - 5$$

$$4x - 4 - x^2 - 2x - 1 = x^2 - 5$$

$$-2x^2 + 2x = 0$$

$$-2x(x-1) = 0$$

$$x_1 = 0 \text{ or } x_2 = 1, \text{ but } x \neq \pm 1, \text{ so } x = 0 \text{ is the only possible solution.}$$

EXAMPLE**41**

Solve $\left(\frac{x}{x+2}\right)^2 - \frac{5x}{x+2} + 6 = 0$.

Solution $x + 2 \neq 0; x \neq -2$ Let $t = \frac{x}{x+2}$, then the equation becomes

$$t^2 - 5t + 6 = 0$$

$$(t-3)(t-2) = 0$$

$$t_1 = 2 \text{ or } t_2 = 3.$$

Now, solve for x :

$$\begin{array}{ll}\frac{x}{x+2} = 2 & \text{or} \quad \frac{x}{x+2} = 3 \\ x = 2x + 4 & x = 3x + 6 \\ x_2 = -4 & x_1 = -3.\end{array}$$

So the roots are -4 and -3 .

Check Yourself 10

1. Solve $\frac{9}{4x+3} + 2x = 4$.
2. Solve $\frac{x+1}{x-2} - \frac{x-3}{x+2} = \frac{12}{x^2-4}$.

Answers

1. $-\frac{1}{4}; \frac{3}{2}$
2. no real solution

C. EQUATIONS INVOLVING RADICALS

When the variable in an equation occurs in a square root, cube root, and so on, that is, when it occurs in a radical, the equation is called a **radical equation**. For example, the equations $\sqrt{x+1} = 2$ and $\sqrt[3]{x-1} - 5 = 0$ are radical equations. Sometimes a suitable operation will change a radical equation to an equation that is linear or quadratic.

To solve a radical equation, follow the procedure.

1. Isolate the radicals

Isolating a radical means putting the radical on one side of the equation and everything else on the other side, using inverse operations. If there are two radicals in the equation, isolate one of the radicals.

2. Get rid of the radical sign

Raise both sides of the equation to a power equal to the index of the isolated radical.

3. If there is still a radical sign left, repeat steps 1 and 2.

4. Solve the remaining equation

5. Check for extraneous solutions

When you solve a radical equation, extra solutions may come up when you raise both sides to an even power. These extra solutions are called **extraneous solutions**. In radical equations you check for extraneous solutions by putting the values you found into the original equation. If the left side of the equation does not equal the right side then you have an extraneous solution.

Note

1. If a value is an extraneous solution, it is not a solution to the original problem.
2. It is very important to check your results in the original equation. In many equations, one of the results may not satisfy the original equation. However, sometimes it is possible that all results that you have found will be acceptable.

EXAMPLE

42

Solve the radical equation $\sqrt{2x+5} = 7$.

Solution

Here the radicand is already alone; we do not need to isolate it. So take the square of both sides:

$$2x + 5 = 49 ; \quad 2x = 44 ; \quad x = 22.$$

Now let us check to see if $x = 22$ is an extraneous solution:

$$\sqrt{2x+5} = 7$$

$$\sqrt{2 \cdot 22 + 5} = 7$$

$$\sqrt{49} = 7$$

$$7 = 7.$$

Since the last statement is true, $x = 22$ is not an extraneous solution. Therefore, there is one solution to this radical equation, $x = 22$.

EXAMPLE

43

Solve $\sqrt{2x-5} + x = 4$.

Solution

First we isolate the radical:

$$\sqrt{2x-5} = 4 - x.$$

Now take the squares of both sides to eliminate the square root:

$$2x - 5 = 16 - 8x + x^2.$$

The new equation is $x^2 - 10x + 21 = 0$

$$(x - 3) \cdot (x - 7) = 0$$

$$x = 3 \text{ or } x = 7.$$

Let us check the results in the original equation:

$$x = 7 ; \sqrt{2 \cdot 7 - 5} + 7 = 4$$

$$\sqrt{9} + 7 = 4$$

$$3 + 7 = 4$$

$$10 = 4$$

$$x = 3 ; \sqrt{2 \cdot 3 - 5} + 3 = 4$$

$$\sqrt{1} + 3 = 4$$

$$4 = 4$$

This is true, so $x = 3$ is a solution.

This is false, so 7 is an extraneous solution.

Hence, the only solution to the equation is $x = 3$.

EXAMPLE

44

Solve $\sqrt{10x+56} - \sqrt{2x+8} = 4$.

Solution

In this question there are two radical expressions. We can isolate only one expression, so it is better to isolate the more complex one. So we have

$$\sqrt{10x+56} = \sqrt{2x+8} + 4.$$

Take the squares of both sides:

$$10x+56 = (2x+8) + 2 \cdot 4 \cdot \sqrt{2x+8} + 16.$$

This is a new equation involving radical expressions. Follow the same steps again to isolate the second radical.

$$8x+32 = 8 \cdot \sqrt{2x+8}$$

$$x+4 = \sqrt{2x+8}$$

$$x^2+8x+16 = 2x+8$$

$$x^2+6x+8 = 0$$

$$(x+4) \cdot (x+2) = 0$$

$$x = -4, \quad x = -2$$

Now, check these results in the original equation.

$$x = -4 ; \sqrt{2 \cdot (-4) + 8} - \sqrt{10 \cdot (-4) + 56} = -4$$

$$0 - \sqrt{16} = -4$$

$$-4 = -4$$

This is true!

$$x = -2 ; \sqrt{2 \cdot (-2) + 8} - \sqrt{10 \cdot (-2) + 56} = -4$$

$$\sqrt{4} - \sqrt{36} = -4$$

$$2 - 6 = -4$$

$$-4 = -4$$

This is true!

Hence, both -4 and -2 are solutions to the equation.

EXAMPLE**45**Solve the equation $5 + \sqrt[3]{x+3} = 3$.

Solution $5 + \sqrt[3]{x+3} = 3$

$$\sqrt[3]{x+3} = -2 \quad (\text{by taking the cube of both sides})$$

$$x+3 = -8$$

$x = -11$. We do not need to check for extraneous solutions because this is an odd power.

Therefore, -11 is the only solution to the equation.

EXAMPLE**46**Solve the equation $\sqrt{4x+1} + \sqrt{x+2} = \sqrt{10x+5}$.**Solution**

$$(\sqrt{4x+1} + \sqrt{x+2})^2 = (\sqrt{10x+5})^2$$

$$4x+1 + 2\sqrt{(4x+1)(x+2)} + x+2 = 10x+5$$

$$2\sqrt{(4x+1)(x+2)} = 5x+2$$

$$(2\sqrt{(4x+1)(x+2)})^2 = (5x+2)^2$$

$$4(4x+1)(x+2) = 25x^2 + 20x + 4$$

$$16x^2 + 36x + 8 = 25x^2 + 20x + 4$$

$$9x^2 - 16x - 4 = 0$$

$$(9x+2)(x-2) = 0$$

$$x = -\frac{2}{9} \quad \text{or} \quad x = 2$$

Now, check these results in the original equation.

$$x = -\frac{2}{9}; \quad \sqrt{-\frac{8}{9}+1} + \sqrt{-\frac{2}{9}+2} = \sqrt{-\frac{20}{9}+5} = \sqrt{\frac{1}{9}} + \sqrt{\frac{16}{9}} = \frac{5}{3}; \quad \frac{5}{3} = \frac{5}{3}$$

$$x = 2; \quad \sqrt{8+1} + \sqrt{2+2} = \sqrt{20+5}; \quad 5 = 5$$

Both statements are true, so both $x = -\frac{2}{9}$ and $x = 2$ are solutions.

Check Yourself 11

Solve the equations.

1. $\sqrt{5x+3} = 4$

2. $\sqrt{2x-4} = x-2$

3. $\sqrt{x+2} - \sqrt{4x+8} = -3$

4. $\sqrt{3x^2-2x+15} + \sqrt{3x^2-2x+8} = 7$

Answers

1. $\frac{13}{5}$ 2. 2, 4 3. 7 4. $-\frac{1}{3}, 1$

D. EQUATIONS INVOLVING AN ABSOLUTE VALUE

On the real number line, the **absolute value** of x is the distance from the origin to the point x . For example, there are two points whose distance from the origin is three units, -3 and 3 . So the equation $|x| = 3$ has two solutions, 3 and -3 . Let us first remember the mathematical definition of absolute value.

Definition

absolute value of a function

For all real numbers x ,

$$|f(x)| = \begin{cases} f(x), & \text{for } f(x) \geq 0 \\ -f(x), & \text{for } f(x) < 0. \end{cases}$$

We can use this information to begin solving equations involving one or more absolute values.

EXAMPLE

47

Solve the equation $|x-2| = 5$.

Solution

Case 1

$$x-2 \geq 0; \quad x \geq 2$$

$$x-2 = 5$$

$$x = 7$$

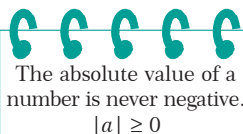
So the solutions are -3 and 7 .

Case 2

$$x-2 < 0; \quad x < 2$$

$$-(x-2) = 5$$

$$x = -3$$



EXAMPLE

48

Solve $|2x-3| = x+1$.

Solution 1

Case 1

$$2x-3 \geq 0; \quad x \geq \frac{3}{2}$$

$$2x-3 = x+1$$

$$x = 4$$

since $4 > \frac{3}{2}$, $x = 4$ is a solution.

So the solutions are 4 and $\frac{2}{3}$.

Case 2

$$2x-3 < 0; \quad x < \frac{3}{2}$$

$$-(2x-3) = x+1$$

$$x = \frac{2}{3} \quad \text{since } \frac{2}{3} < \frac{3}{2},$$

$$x = \frac{2}{3} \text{ is a solution.}$$

Solution 2 Let us take the square of both sides:



$$|f(x)|^2 = f^2(x) = f(x) \cdot f(x)$$

$$4x^2 - 12x + 9 = x^2 + 2x + 1$$

$$3x^2 - 14x + 8 = 0$$

$$(x - 4) \cdot (3x - 2) = 0.$$

So the solutions are 4 and $\frac{2}{3}$.

EXAMPLE

49

Solve $x^2 - 2|x + 2| + 1 = 0$.

Solution

Case 1

$$x + 2 \geq 0; \quad x \geq -2$$

$$x^2 - 2(x + 2) + 1 = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x - 3)(x + 1) = 0$$

$$x = 3 \text{ or } x = -1$$

Both 3 and -1 are greater than -2, so the solutions to the equation are -1 and 3.

Case 2

$$x + 2 < 0; \quad x < -2$$

$$x^2 - 2(-(x + 2)) + 1 = 0$$

$$x^2 + 2x + 5 = 0$$

$$\Delta = -16$$

Since $\Delta < 0$, there are no real roots.

EXAMPLE

50

Solve the equation $|x^2 - 5x + 3| = 3$.

Solution

Case 1

$$x^2 - 5x + 3 \geq 0$$

$$x^2 - 5x + 3 = 3$$

$$x^2 - 5x = 0$$

$$x(x - 5) = 0$$

$$x = 0 \text{ or } x = 5$$

These values satisfy the inequality, so they are solutions.

Hence, the solutions to the equation are 0, 2, 3, 5.

Case 2

$$x^2 - 5x + 3 < 0$$

$$-(x^2 - 5x + 3) = 3$$

$$x^2 - 5x + 6 = 0$$

$$(x - 3)(x - 2) = 0$$

$$x = 3 \text{ or } x = 2$$

These values also satisfy the inequality, so they are also solutions.

EXAMPLE

51

Solve the equation $x^2 - 3x + 1 = \sqrt{x^2 - 4x + 4}$.

Solution

Remember that $x^2 - 4x + 4 = (x - 2)^2$. Hence, the expression becomes



For any real number x ,

$$\sqrt{x^2} = |x|.$$

$$x^2 - 3x + 1 = \sqrt{(x - 2)^2} = |x - 2|$$

$$x^2 - 3x + 1 = |x - 2|.$$

Case 1

$$x - 2 \geq 0 ; \quad x \geq 2$$

$$x^2 - 3x + 1 = x - 2$$

$$x^2 - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$x = 3 \text{ or } x = 1$$

Since $1 \leq 2$,

$x = 3$ is the only solution.

Hence, the solutions to the equation are $(1 - \sqrt{2})$ and 3.

Case 2

$$x - 2 < 0 ; \quad x < 2$$

$$-(x^2 - 3x + 1) = x - 2$$

$$x^2 - 2x - 1 = 0$$

$$x_1 = 1 - \sqrt{2}, \quad x_2 = 1 + \sqrt{2}$$

Since $(1 + \sqrt{2})$ is greater than 2,

$x = (1 - \sqrt{2})$ is the only solution.

EXAMPLE

52

Solve the equation $|2x - 3| = |x - 7|$.

Solution 1 Let's take the square of both sides. Then,

$$4x^2 - 12x + 9 = x^2 - 14x + 49$$

$$3x^2 + 2x - 40 = 0$$

$$(3x - 10)(x + 4) = 0$$

$$x_1 = -4 \text{ or } x_2 = \frac{10}{3}.$$

Therefore, both -4 and $\frac{10}{3}$ are solutions to the equation.

Solution 2 $2x - 3 = x - 7$ or $2x - 3 = -(x - 7)$

$$x = -4 \qquad x = \frac{10}{3}$$

So both -4 and $\frac{10}{3}$ are solutions to the equation.



If $|f(x)| = |g(x)|$
then $f(x) = \pm g(x)$.

Check Yourself 12

Solve the equations.

1. $|2 - x| = 2x + 1$

2. $x|x + 1| - 2 = 2x$

Answers

1. $\frac{1}{3}$ 2. $-2, -1, 2$

Algebraic symbols are used when you do not know what you are talking about.



PUZZLE

The following problem is posed and solved in the Chiu Chang Suan-Shu, a Chinese mathematical treatise which is over two thousand years old.

A tree of height 20 m has a circumference of 3 m. An arrowroot vine winds seven times around the tree before it reaches the top. What is the length of the vine?

Can you solve this puzzle?



EXERCISES 1.4

A. Writing Equations in Quadratic Form

1. Solve the equations.

a. $x^4 - 13x^2 + 36 = 0$

b. $3x^4 - 8x^2 - 3 = 0$

c. $(x^2 - 9)^2 - 4(x^2 - 9) + 3 = 0$

d. $(x + 5)^4 - 6(x + 5)^2 - 7 = 0$

e. $\left(\frac{3-4x}{5}\right)^2 - 2(3-4x) + 25 = 0$

f. $\sqrt{x+2} - \sqrt[4]{x+2} = 6$

g. $x^3 - 7x^{\frac{3}{2}} - 8 = 0$

h. $\left(x + \frac{1}{x}\right)^2 - \frac{7}{2}\left(x + \frac{1}{x}\right) = 2$

i. $\frac{x^2 + 2}{x} + \frac{8x}{x^2 + 2} = 6$

2. Solve the equations.

a. $3\left(x^2 + \frac{1}{x^2}\right) - 7\left(x + \frac{1}{x}\right) = 0$

b. $(x^2 - 2x - 5)^2 - 2(x^2 - 2x - 3) - 4 = 0$

c. $\frac{3}{1+x+x^2} = 3 - x - x^2$

d. $\frac{1}{x^2 - 3x + 3} + \frac{2}{x^2 - 3x + 4} = \frac{6}{x^2 - 3x + 5}$

e. $(x + 1)(x + 2)(x + 3)(x + 4) = 120$

B. Equations Involving Products and Quotients

3. Solve the equations.

a. $(16x^3 - x)(x^2 - 6x + 5) = 0$

b. $(x^2 + 8x)(x^2 + 8x - 6) = 280$

c. $x^3 + 4x^2 - 24 = 0$

d. $x^3 - 5x^2 + 9x - 45 = 0$

e. $\frac{2x+3}{4x-1} = \frac{3x-2}{3x+2}$

f. $(x^2 - 5x + 6)^2 - (2x^2 - 5x + 1)^2 = 0$

g. $(6x^2 - 5x - 4)^2 + (10x^2 - 29x + 21)^2 = 0$

h. $\frac{3x^2 - 9x}{2} - \frac{12}{x^2 - 3x} = 3$

i. $\frac{3x+4}{x-2} - \frac{6(x-2)}{3x+4} = 1$

j. $\frac{4}{x^2} + \frac{x^2}{9} + 4\left(\frac{x}{3} - \frac{2}{x}\right) + \frac{8}{3} = 0$

C. Equations Involving Radicals

4. Solve the equations.

a. $\sqrt{x-1} - 1 = 0$

b. $\sqrt{x} + x = 5$

c. $\sqrt[3]{x-1} = -2$

d. $\sqrt{x-1} = x-1$

e. $\sqrt{x-3} + \sqrt{x-8} = 5$

f. $3 - \sqrt{x+1} = \sqrt{x-2}$

g. $\sqrt[3]{x^2 - x + 6} - 2 = 0$

h. $\sqrt{\sqrt{x+16} - \sqrt{x}} = 2$

i. $x^2 - 6x - \sqrt{x^2 - 6x - 3} = 5$

j. $\sqrt{4x-1} + \sqrt{2x+3} = 1$

5. Solve the equations.



a. $\sqrt{x+1} - \sqrt{9-x} = \sqrt{2x-12}$

b. $\sqrt{2x+5} + \sqrt{5x+6} = \sqrt{12x+25}$

c. $\sqrt{x} - \sqrt{x+1} + \sqrt{x+9} - \sqrt{x+4} = 0$

d. $\sqrt{2x + \sqrt{6x^2 + 1}} = x + 1$

e. $\sqrt{1 - \sqrt{x^2 - 3x}} = \sqrt{x+1}$

f. $\sqrt{x^2 + 1} + \sqrt{x^2 - 8} = 1$

g. $\sqrt{3x^2 + 5x + 8} - \sqrt{3x^2 + 5x + 1} = 1$

h. $1 + \sqrt{x+2} = \sqrt{2\sqrt{x+2} + 5}$

D. Equations Involving an Absolute Value

6. Solve the equations.

a. $|x| = x + 2$

b. $|2x - 5| = x - 2$

c. $x \cdot |x - 1| = 2$

d. $|x^2 - 4x| = 5$

e. $|x^2 - 2x + 3| = 6$

f. $x^2 + |x - 1| + 1 = 0$

g. $x^2 + |2x - 1| + 3 = 4x + 2$

h. $\sqrt{x^2 - 2x + 1} = 2x - 3$

i. $x^2 + \sqrt{x^2 - 6x + 9} = 0$

7. Solve the equations.



a. $|x| + x^2 = 0$

b. $(|x| + 1)(|x| - 1) = -\frac{1}{2}$

c. $(x + 1)^2 - 2|x + 1| + 1 = 0$

d. $|x - x^2 - 1| = |2x - 3 - x^2|$

e. $\frac{|x^2 - 4x| + 3}{x^2 + |x - 5|} = 1$

f. $|x + |3 - 2x|| = 3 - x$

Mixed Problems

8. Solve the equations.



a. $\sqrt{x^2 - 2x + 1} = 1 - x$

b. $x^2 + \sqrt{x^2 - 6x + 9} = 2x + 3$

c. $x + |x + 1| = 2\sqrt{x^2 - 4x + 4}$

d. $x^2 - |x + 2| = \sqrt{x^2 + 4x + 4} + x$



SYSTEMS OF QUADRATIC EQUATIONS

Sometimes we need to solve two or more equations simultaneously. A set of equations like this is called a **system** of equations. There are no concrete rules that we can follow to solve systems of equations, but let us look at some general strategies.

EXAMPLE 53 Solve the system of equations.

$$\begin{cases} x + y = 5 \\ xy = 6 \end{cases}$$

Solution We can write $y = 5 - x$. Now substitute this value of y in the second equation.

$$x(5 - x) = 6$$

$$x^2 - 5x + 6 = 0$$

$$(x - 2)(x - 3) = 0$$

$$x = 2 \text{ or } x = 3$$

If $x = 2$ then $y = 3$.

If $x = 3$ then $y = 2$.

Therefore, the solutions of the system are $(2, 3)$ and $(3, 2)$.

EXAMPLE 54 Solve the system.

$$\begin{cases} x^2 + y^2 = 65 \\ x \cdot y = 28 \end{cases}$$

Solution 1 Multiply the second equation by 2 and then add and subtract the resulting equations.

$$\begin{array}{r} x^2 + y^2 = 65 \\ + \quad 2x \cdot y = 56 \\ \hline x^2 + 2xy + y^2 = 121 \\ (x + y)^2 = 121 \\ x + y = \pm 11 \end{array}$$

$$\begin{array}{r} x^2 + y^2 = 65 \\ - \quad 2x \cdot y = 56 \\ \hline x^2 - 2xy + y^2 = 9 \\ (x - y)^2 = 9 \\ x - y = \pm 3 \end{array}$$

Therefore, we have four cases.

Case 1	Case 2	Case 3	Case 4
$x + y = 11$	$x + y = 11$	$x + y = -11$	$x + y = -11$
$+ x - y = 3$	$+ x - y = -3$	$+ x - y = 3$	$+ x - y = -3$
$2x = 14$	$2x = 8$	$2x = -8$	$2x = -14$
$x = 7$	$x = 4$	$x = -4$	$x = -7$
$y = 4$	$y = 7$	$y = -7$	$y = -4$

The solutions of the system are therefore $(7, 4)$, $(4, 7)$, $(-7, -4)$, $(-4, -7)$.

Solution 2 $\begin{cases} x^2 + y^2 = 65 \\ x \cdot y = 28 \end{cases}$

$y = \frac{28}{x}$, $x \neq 0$, substitute y in the first equation.

$$x^2 + \left(\frac{28}{x}\right)^2 = 65$$

$$x^2 + \frac{28^2}{x^2} = 65$$

$$x^4 - 65x^2 + 784 = 0$$

$$(x^2 - 49)(x^2 - 16) = 0$$

$$x^2 = 49 \text{ or } x^2 = 16$$

$$x = \pm 7 \text{ or } x = \pm 4$$

By substituting these x values in $y = \frac{28}{x}$, we get the y values.

The solutions of the system are $(7, 4)$, $(-7, -4)$, $(4, 7)$, $(-4, -7)$.

EXAMPLE

55

Solve the system.

$$\begin{cases} x^2 = 13x + 4y \\ y^2 = 4x + 13y \end{cases}$$

Solution Let us subtract the two equations side by side.

$$x^2 - y^2 = 9x - 9y$$

$$(x - y)(x + y) = 9(x - y)$$

$$x - y = 0 \text{ or } x + y = 9$$

Case 1

$$x - y = 0 ; x = y$$

Let us substitute x in the equation $y^2 = 4x + 13y$.

$$y^2 = 4y + 13y$$

$$y^2 - 17y = 0$$

$$y_1 = 0 \text{ or } y_2 = 17$$

$$x_1 = 0 \text{ or } x_2 = 17$$

Therefore we have two solutions, $(0, 0)$, $(17, 17)$.

Case 2

$$x + y = 9 ; x = 9 - y$$

Let us substitute x in the equation $y^2 = 4x + 13y$.

$$y^2 = 4(9 - y) + 13y$$

$$y^2 - 9y - 36 = 0$$

$$(y - 12)(y + 3) = 0$$

$$y_3 = 12 \text{ or } y_4 = -3$$

$$x_3 = -3 \text{ or } x_4 = 12$$

Therefore we have two more solutions, $(-3, 12)$, $(12, -3)$.

Hence, the solutions of the system are $(0, 0)$, $(17, 17)$, $(-3, 12)$, $(12, -3)$.

EXAMPLE

56

Solve the system.

$$\begin{cases} \frac{x+y}{x-y} + \frac{x-y}{x+y} = \frac{5}{2} \\ x^2 + y^2 = 20 \end{cases}$$

Solution $x - y \neq 0 ; x \neq y$ and $x + y \neq 0 ; x \neq -y$.

Let us multiply both sides of the equation by $2(x - y)(x + y)$.

$$2(x + y)^2 + 2(x - y)^2 = 5(x - y)(x + y)$$

$$2x^2 + 4xy + 2y^2 + 2x^2 - 4xy + 2y^2 = 5x^2 - 5y^2$$

$$-x^2 + 9y^2 = 0.$$

Now, we have the system
$$\begin{cases} -x^2 + 9y^2 = 0 \\ x^2 + y^2 = 20 \end{cases}.$$

Let us add these two equations side by side.

$$10y^2 = 20$$

$$y^2 = 2; y = \pm\sqrt{2}$$

$$x^2 = 18; x = \pm 3\sqrt{2}$$

Hence, the solutions of the system are $(3\sqrt{2}, \sqrt{2})$, $(-3\sqrt{2}, \sqrt{2})$, $(3\sqrt{2}, -\sqrt{2})$, $(-3\sqrt{2}, -\sqrt{2})$.

Check Yourself 13

Solve the systems of equations.

1.
$$\begin{cases} x^2 - y = 2 \\ x + y = 4 \end{cases}$$

2.
$$\begin{cases} x^2 + 3xy - 10y^2 = 0 \\ x^2 + 2xy - y^2 = 28 \end{cases}$$

Answers

1. $(2, 2)$, $(-3, 7)$

2. $(4, 2)$, $(-4, -2)$, $(5\sqrt{2}, -\sqrt{2})$, $(-5\sqrt{2}, \sqrt{2})$

One person's constant is another person's variable.



EXERCISES 1.5

1. Solve each system of equations.

a.
$$\begin{cases} x - y = 12 \\ x \cdot y = 108 \end{cases}$$

b.
$$\begin{cases} 9x^2 - y^2 = 44 \\ 3x - y = 11 \end{cases}$$

c.
$$\begin{cases} x = 3 - y \\ y^2 - x = 39 \end{cases}$$

d.
$$\begin{cases} \frac{1}{x} + \frac{1}{y} = \frac{1}{6} \\ x + y = 25 \end{cases}$$

e.
$$\begin{cases} x^2 - 3y^2 = 13 \\ xy = -4 \end{cases}$$

f.
$$\begin{cases} x^2 + y^2 = 18 \\ xy = 9 \end{cases}$$

g.
$$\begin{cases} x + y + xy = 5 \\ xy + x - y = 13 \end{cases}$$

h.
$$\begin{cases} x^2 + 3x - 4y = 20 \\ x^2 - 2x + y = -5 \end{cases}$$

2. Solve each system of equations.

a.
$$\begin{cases} x^2 + y^2 + 6x + 2y = 0 \\ x + y + 8 = 0 \end{cases}$$

b.
$$\begin{cases} 3x + xy - y^2 = 0 \\ 2x^2 - 3xy + y^2 = 0 \end{cases}$$

c.
$$\begin{cases} 6x^2 + xy - y^2 = 0 \\ 3x^2 - xy - y^2 = 0 \end{cases}$$

d.
$$\begin{cases} \frac{1}{x+y} + \frac{1}{x-y} = 2 \\ \frac{3}{x+y} + \frac{4}{x-y} = 7 \end{cases}$$

e.
$$\begin{cases} x + y - xy = -2 \\ xy(x + y) = 48 \end{cases}$$

f.
$$\begin{cases} x^3 - y^3 = 8 \\ x - y = 2 \end{cases}$$

g.
$$\begin{cases} \frac{x+y}{x-y} - \frac{2(x-y)}{x+y} = 1 \\ x^2 - 5xy + 2y^2 = 4 \end{cases}$$

h.
$$\begin{cases} 3x^2 - 2xy - y^2 = 7 \\ x^2 + xy + 8y^2 = 14 \end{cases}$$

CHAPTER 1 SUMMARY

- A quadratic equation in standard form is an equation that is in the form

$$ax^2 + bx + c = 0, \quad a \neq 0.$$

In this standard form, x is a variable and a , b , and c are real number coefficients.

Methods of solving quadratic equations include

- Factoring and using the zero product property:

$$a \cdot b = 0 \text{ if and only if } a = 0 \text{ or } b = 0.$$

- Using the square root property:

$$\text{If } a^2 = b, \text{ then } a = \pm\sqrt{b}.$$

- Completing the square:

$$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2$$

- Using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- For a quadratic equation $ax^2 + bx + c = 0$, $\Delta = b^2 - 4ac$,
 - if $\Delta > 0$, there are two distinct real roots.
 - if $\Delta = 0$, there is one real root (a double root).
 - if $\Delta < 0$, there is no real solution.

Δ is called the discriminant of the quadratic.

- Vieta's Theorem

For a quadratic equation $ax^2 + bx + c = 0$

$$x_1 + x_2 = -\frac{b}{a} \text{ and } x_1 \cdot x_2 = \frac{c}{a}.$$

- By using Vieta's Theorem, we can also conclude.

$$1. \quad \frac{1}{x_1} + \frac{1}{x_2} = -\frac{b}{c}.$$

$$2. \quad x_1^2 + x_2^2 = \frac{b^2 - 2ac}{a^2}.$$

$$3. \quad x_1^3 + x_2^3 = \frac{3abc - b^3}{a^3}.$$

- Let x_1 , x_2 , and x_3 be the roots of the equation

$$ax^3 + bx^2 + cx + d = 0. \text{ Then,}$$

$$1. \quad x_1 + x_2 + x_3 = -\frac{b}{a}.$$

$$2. \quad x_1x_2 + x_1x_3 + x_2x_3 = \frac{c}{a}.$$

$$3. \quad x_1x_2x_3 = -\frac{d}{a}.$$

- We can write a quadratic equation with roots x_1 and x_2 as $x^2 - (x_1 + x_2)x + (x_1 \cdot x_2) = 0$.

- To solve a radical equation, follow the procedure.

- Isolate the radicals.
- Get rid of the radical sign.
- If there is still a radical sign left, repeat steps 1 and 2.
- Solve the remaining equation.
- Check for extraneous solutions.

When solving radical equations, extraneous solutions are extra solutions that may come up when you raise both sides to an even power. If a solution is extraneous, it is not a solution to the given problem.

- For all real numbers x ,

$$|f(x)| = \begin{cases} f(x), & \text{for } f(x) \geq 0 \\ -f(x), & \text{for } f(x) < 0 \end{cases}.$$

- If x is a real number, then $\sqrt{x^2} = |x|$.

Concept Check

- What is the standard form of a quadratic equation?
- What do we mean by a root of a quadratic equation?
- Explain the square root property.

- If $a \cdot b = 0$ then what can you conclude about the factors a and b ?
- Can a quadratic equation with rational coefficients have one rational root and one irrational root? Explain why or why not.
- What is the discriminant of a quadratic equation?
- Explain why it is a good idea to find the discriminant of a quadratic equation before you try to solve the equation.
- How can we write a quadratic equation if we know only its roots?
- Explain Vieta's Theorem.
- Form a radical equation that has no solution.
- What do we mean by an extraneous solution?
- Form a radical equation that has an extraneous solution.
- Describe the step in the solution process for radical equations that leads to the possibility of extraneous solutions. Why are there no extraneous solutions for linear and quadratic equations?

CHAPTER REVIEW TEST 1A

1. Find the solutions of the quadratic equation $2x^2 + 5x + 3 = 0$.

A) $\frac{1}{2}$ and 3 B) 3 and $\frac{1}{2}$ C) -1 and $-\frac{3}{2}$
 D) $-\frac{1}{2}$ and -3 E) -1 and $\frac{1}{2}$

2. -1 is a root of $x^2 + mx + n = 0$ and $m + n = 5$. Find m .

A) 1 B) -1 C) 2 D) -2 E) 3

3. Which one of the following is **false** for the equation $x^2 - 7x + 1 = 0$?

A) The roots are not integer.
 B) The roots have the same sign.
 C) The roots are rational.
 D) The roots are both positive.
 E) The sum of the roots is positive.

4. Find the discriminant of $x^2 - 2\sqrt{5}x + 4 = 0$.

A) 2 B) 4 C) 36 D) 16 E) 24

5. The sum and product of the roots of the equation $2x^2 - hx + 2k = 0$ are 4 and -3, respectively.

Find the values of h and k .

A) 8 and -6 B) 4 and -3 C) -3 and 4
 D) -3 and 8 E) 8 and -3

6. What is the sum of the roots of the equation $x^4 - 13x^2 + 36 = 0$?

A) 13 B) 18 C) 1 D) 0 E) -13

7. Which one of the following equations has roots -2 and 3?

A) $2x^2 - x - 6 = 0$ B) $x^2 + x - 6 = 0$
 C) $x^2 - x - 6 = 0$ D) $x^2 + x + 6 = 0$
 E) $2x^2 + x - 6 = 0$

8. The sum of a number and its reciprocal is $\frac{25}{12}$. Find the number.

A) 1 B) -1 C) $\frac{4}{3}$ D) $-\frac{1}{2}$ E) -3

9. What are the roots of the equation

$$\frac{1}{x^2} - \frac{3}{x} + 2 = 0?$$

- A) -1 and $\frac{1}{2}$ B) $\frac{1}{2}$ and 1 C) $-\frac{1}{2}$ and $\frac{4}{3}$
D) $-\frac{1}{2}$ and -1 E) -1 and 1

10. One of the roots of the equation $ax^2 + bx + c = 0$ is twice the other root. What is the relation between the coefficients a , b , and c ?

- A) $4b^2 = 4c$ B) $2b^2 = 9ac$ C) $2b^2 = 9a$
D) $b^2 = 8ac$ E) $9b^2 = 2ac$

11. The diagonal of a rectangle is 5 cm, and the area is 12 cm^2 . Find the perimeter of the rectangle.

- A) 12 B) 24 C) 10 D) 14 E) 15

12. $\sqrt{x-1} - \sqrt{x+1} + 1 = 0$. Find the value of $4x$.

- A) 5 B) 4 C) 0 D) 16 E) -4

13. What is the solution of the equation $x = \sqrt{x} + 6$?

- A) -9 B) 4 C) -4 D) 16 E) 9

14. Which one of the following is **not** a solution of the system

$$\begin{cases} 5x^2 - 6xy + 5y^2 = 29 \\ 7x^2 - 8xy + 7y^2 = 43 \end{cases} ?$$

- A) $(2, 3)$ B) $(-2, -3)$ C) $(-2, 3)$
D) $(3, 2)$ E) $(-3, -2)$

15. Two technicians can complete a mailing job in 12 hours while working together. Alone, the first technician can complete the mailing job seven hours faster than the other technician. How long will it take each technician to complete the mailing alone?

- A) 14 and 21 B) 21 and 28 C) 7 and 14
D) 11 and 18 E) 15 and 22

16. The equations $6x^2 - 18x + 12 = 0$ and $mx^2 - nx + 1 = 0$ have the same solution. Find m and n .

- A) $-\frac{3}{2}$ and $-\frac{1}{2}$ B) $-\frac{3}{2}$ and $\frac{1}{2}$ C) $\frac{3}{2}$ and $\frac{1}{2}$
D) $\frac{1}{2}$ and $\frac{3}{2}$ E) $\frac{3}{2}$ and $-\frac{1}{2}$

CHAPTER REVIEW TEST 1B

- x_1 and x_2 are roots of the equation $x^2 + (2 - x_1)x + 7x_2 = 0$. What is $x_1 + x_2$?
A) -2 B) 5 C) 7 D) 10 E) -14
- Which one of the following is **true** for the equation $|x|^2 + |x| - 6 = 0$?
A) There is only one root.
B) The product of the roots is -3.
C) The product of the roots is 4.
D) The sum of the roots is 0.
E) The sum of the roots is 1.
- The sum of the squares of the roots of the equation $x^2 + 2hx = 3$ is 10. Find the value of h .
A) $\frac{1}{2}$ B) -3 C) $\frac{3}{2}$ D) $-\frac{1}{2}$ E) -1 or 1
- Find the discriminant of $2x^2 - 2x + 5 = 0$.
A) -36 B) 4 C) 25 D) -16 E) 9
- The longest side of a right triangle is 3 cm less than twice the length of the medium side. The shortest side is 3 cm. Find the lengths of the other two sides.
A) 8 and 15 B) 7 and 14 C) 8 and 10
D) 4 and 5 E) 6 and 8
- A triangle has area of 2 cm^2 . The base of the triangle is 3 cm longer than its height. Find the length of the base and the height of the triangle.
A) 5 and 2 B) 4 and 1 C) 6 and 3
D) 7 and 4 E) 8 and 5
- Which one of the following equations has roots $1 - \sqrt{3}$ and $1 + \sqrt{3}$?
A) $x^2 - 2x - 2 = 0$ B) $x^2 + 2x - 2 = 0$
C) $x^2 - x - \sqrt{3} = 0$ D) $x^2 - 2x + 2 = 0$
E) $x^2 - 6 = 0$
- The sum and product of two numbers are 21 and 104, respectively. What is the bigger number?
A) 16 B) 22 C) 13 D) 14 E) 19

9. What are the roots of the equation

$$3\left(\frac{x}{x+1}\right)^2 + 7\left(\frac{x}{x+1}\right) - 6 = 0?$$

- A) $-\frac{3}{4}$ and 2 B) $\frac{1}{2}$ and 2 C) $-\frac{1}{2}$ and $\frac{4}{3}$
D) $-\frac{1}{2}$ and -3 E) -1 and 2

10. Which number is a solution to the equation

$$x^2 + |x - 1| - 3 = 0?$$

- A) 3 B) $-\frac{\sqrt{2}}{2}$ C) $\frac{\sqrt{3}}{2}$ D) 0 E) -1

11. $x - y = 12$ and $x \cdot y = 108$ are given. Find $x + y$.

- A) ± 20 B) ± 24 C) ± 28 D) ± 30 E) ± 32

12. $x - 5 = \sqrt{x - 3}$ is given. Find the value of x .

- A) 5 B) -7 C) 1 D) 7 E) -4

13. What is the solution to the equation $x^{\frac{1}{2}} + x^{\frac{1}{4}} = 2$?

- A) 9 B) 1 C) -4 D) 4 E) 16

14. Which one of the following is **not** a solution of the system

$$\begin{cases} x^2 + y^2 = 26 \\ x^2 - y^2 = 24 \end{cases} ?$$

- A) (1, -5) B) (-5, -1) C) (-5, 1)
D) (5, 1) E) (5, -1)

15. A pipe can fill a pool four hours faster than another pipe. The slower pipe filled the pool for seven hours, then the other pipe was opened. The pipes then filled the pool in two hours, working together. How long would each pipe take to fill the pool alone?

- A) 14 and 18 hours B) 8 and 12 hours
C) 11 and 15 hours D) 9 and 13 hours
E) 10 and 14 hours

16. The roots of the equation $x^3 - 3x + 7 = 0$ are p , q , and r . What is the product $(p + 1)(q + 1)(r + 1)$?

- A) -8 B) -9 C) 7 D) 8 E) 9



Chapter 2

QUADRATIC INEQUALITIES



LINEAR INEQUALITIES

An equation is a statement that says two expressions are equal. For example, $a = b$ is an equation. An **inequality** is a statement that says two expressions may or may not be equal. For example, $a > b$, $a \leq b$, and $a < b$ are all inequalities. An equation usually has a finite number of solutions, but an inequality may have an infinite number of solutions.

We can show the solutions of an inequality as an interval and as a graph. Look at the following examples of intervals and their graphs.

Trichotomy Property:
For any two real numbers a and b , exactly one of the following is true:
 $a < b$, $a = b$ or $a > b$.

Interval	Inequality	Graph
the interval $(-\infty, a)$	$x < a$	
the interval $(-\infty, a]$	$x \leq a$	
the interval (a, ∞)	$x > a$	
the interval $[a, \infty)$	$x \geq a$	
the open interval (a, b)	$a < x < b$	
the closed interval $[a, b]$	$a \leq x \leq b$	
the half-open interval $[a, b)$	$a \leq x < b$	
the half-open interval $(a, b]$	$a < x \leq b$	
the interval $(-\infty, \infty)$	$\square$	

Notice in the table that an open circle on a graph shows an open interval, i.e. a line segment with an open endpoint (a point which is not included in the segment). A filled circle shows a closed endpoint (a point which is included in the segment).

EXAMPLE

1 Write each inequality using interval notation and show it as a graph.

a. $1 \leq x < 5$

b. $x > 7$

c. $-3 < x < 4$

d. $x \leq 9$

Solution

a. The interval notation is $[1, 5)$ and the graph is

b. $(7, \infty)$ c. $(-3, 4)$ d. $(-\infty, 9]$

Definition**linear inequality**

A **linear inequality** is an inequality that can be written in one of the forms

$$\begin{array}{ll} ax + b > 0, & ax + b < 0 \\ ax + b \geq 0, & ax + b \leq 0 \end{array}$$

for the real numbers a and b , $a \neq 0$.

Property

If both sides of an inequality are multiplied or divided by a **negative** number, the direction of the resulting inequality must be reversed.

For example, if we multiply both sides of the inequality $a < b$ by -2 , we obtain $-2a > -2b$. The order of the inequality is reversed.

EXAMPLE**2**

Solve the inequality $2x + 3 \leq 5x$ and graph this solution.

Solution

$$2x + 3 \leq 5x$$

$$-3x \leq -3$$

$$1 \leq x$$

Therefore, $x \in [1, \infty)$, or .

EXAMPLE**3**

Solve the inequality $2 < 3x - 1 \leq 8$ and graph this solution.

Solution

$$2 < 3x - 1 \leq 8$$

$$3 < 3x \leq 9$$

$$1 < x \leq 3$$

Therefore, $x \in (1, 3]$, or .

Property

For $k > 0$, the following statements are true.

1. $|x| \leq k$ means $-k \leq x \leq k$.
2. $|x| > k$ means $x < -k$ or $x > k$.

EXAMPLE**4**

Solve the inequality $|2x - 1| < 5$ and graph this solution.

Solution

$$|2x - 1| < 5$$

$$-5 < 2x - 1 < 5$$

$$-4 < 2x < 6$$

$$-2 < x < 3$$

Therefore, $x \in (-2, 3)$, or .

Check Yourself 1

Solve the inequalities.

1. $-x + 7 > 2x + 1$

2. $-11 \leq \frac{x+3}{4} \leq 17$

3. $|2x - 1| + 2 \leq 7$

Answers

1. $x < 2$ 2. $-47 \leq x \leq 65$ 3. $-2 \leq x \leq 3$

FIND THE MISTAKE!

$$a > 4$$

$$4a > 16$$

$$4a - a^2 > 16 - a^2$$

$$a(4 - a) > (4 - a)(4 + a)$$

$$a > 4 + a$$

$$0 > 4$$

Can you find the mistake in this working?

Sign Chart

In general, to solve a linear inequality such as $ax + b > 0$ or $ax + b \leq 0$ we need to know the sign of the polynomial $ax + b$, $a \neq 0$.

Look at the steps.

First we find the zero of the polynomial:

$$ax + b = 0 ; x = -\frac{b}{a}.$$

Then we construct a **sign chart**.

x	$-\infty$	$x_0 = -\frac{b}{a}$	∞
$ax+b$	$ax+b$ has sign opposite to a		$ax+b$ has the same sign as a

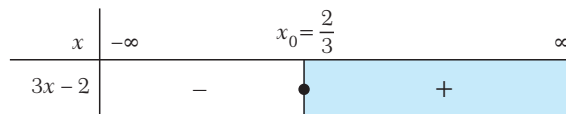
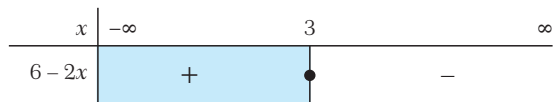
This sign chart shows us:

if $x \in \left(-\infty, -\frac{b}{a}\right)$, the sign of the polynomial is opposite to the sign of a ,

if $x \in \left(-\frac{b}{a}, \infty\right)$, the sign of the polynomial is the same as the sign of a .

EXAMPLE**5** Solve the inequality $3x - 2 \geq 0$.**Solution** Find the zero: $3x - 2 = 0$; $x = \frac{2}{3}$.

Draw the sign chart:

If $x \in \left(-\infty, \frac{2}{3}\right)$, then $3x - 2$ is negative.If $x \in \left(\frac{2}{3}, \infty\right)$, then $3x - 2$ is positive.Since $3x - 2 \geq 0$, x is in the interval $\left[\frac{2}{3}, \infty\right)$, i. e. $x \in \left[\frac{2}{3}, \infty\right)$.**EXAMPLE****6** Solve the inequality $6 - 2x \geq 0$ by using a sign chart.**Solution** $6 - 2x = 0$; $x = 3$ Since $6 - 2x \geq 0$, the sign is positive. So $x \in (-\infty, 3]$.**Check Yourself 2**

Solve the inequalities by using a sign chart.

1. $5 - 2x < 0$ 2. $\frac{x+14}{6} - \frac{x-12}{8} \leq 3$ 3. $6x + (x-2)(x+2) \geq (x+4)^2$

Answers

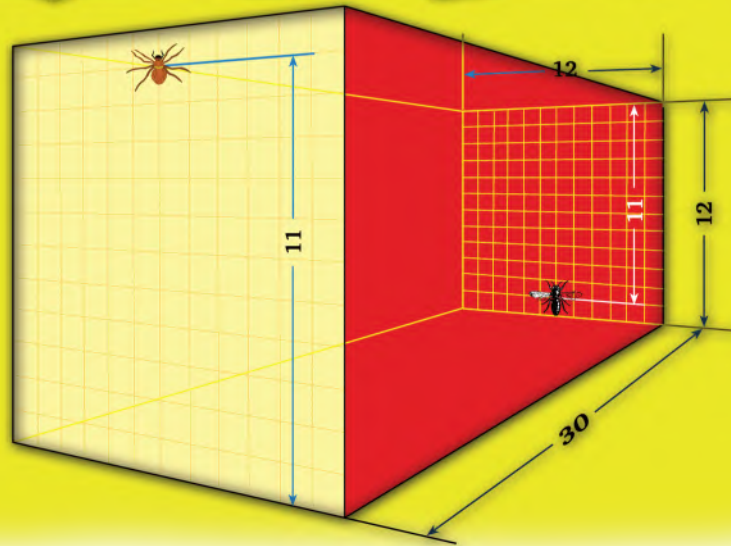
1. $x > \frac{5}{2}$ 2. $x \leq -20$ 3. $x \leq -10$



Mathematics, of course, is not the only cornerstone of opportunity in today's world. Reading is even more fundamental as a basis for learning and for life. What is different today is the great increase in the importance of mathematics to so many areas of education, citizenship, and careers.

Puzzle

A rectangular room has dimensions $30\text{ m} \times 12\text{ m} \times 12\text{ m}$. A spider is in the horizontal center of one end wall, one unit away from the ceiling. A fly is in the horizontal center of the opposite wall, one unit away from the floor. What is the shortest distance the spider needs to travel (without leaving a surface) to get to the fly (which remains stationary)?



EXERCISES 2.1

1. Write each inequality using interval notation.

- | | |
|---------------------------|----------------------------------|
| a. $x > -2$ | b. $x \leq 6$ |
| c. $x < -7$ | d. $x \geq 4$ |
| e. $2 \leq x < 4$ | f. $-1 < x \leq 0$ |
| g. $-5 < x < \frac{1}{2}$ | h. $0,5 \leq x \leq \frac{3}{2}$ |

2. Determine the sign of each polynomial.

- | | |
|---------------------------------|--------------------|
| a. $4x + 1$ | b. $-3x - 5$ |
| c. $-\frac{x}{2} + \frac{7}{3}$ | d. $\sqrt{3}x - 6$ |

3. Solve and graph the inequalities.

- | |
|---|
| a. $-2x + 3 < 4$ |
| b. $\frac{x}{2} + \frac{x}{3} - \frac{x}{4} \geq \frac{x}{5}$ |
| c. $-\frac{3x}{2} + 5 < 1 + \frac{x}{2}$ |
| d. $-\frac{x+3}{4} \leq \frac{x-2}{2} - \frac{2x}{3}$ |
| e. $(3x + 1)^2 - (x + 2)(4x - 1) > 5(x - 1)^2 + 6x$ |
| f. $(a + 3)x - 5 \leq 1 \quad (a > -3)$ |
| g. $(a^4 + 4)x + 3 > 0 \quad (a \in \mathbb{Q})$ |

Definition

quadratic inequality

A **quadratic inequality** is an inequality that can be written in one of the forms

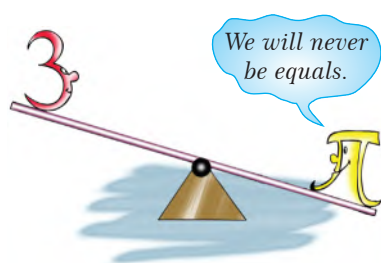
$$ax^2 + bx + c > 0,$$

$$ax^2 + bx + c < 0,$$

$$ax^2 + bx + c \geq 0,$$

$$ax^2 + bx + c \leq 0,$$

for the real numbers a , b , and c , $a \neq 0$.



We have seen how to solve linear inequalities such as

$$2(x + 3) + 3 \leq 5(x - 1).$$

But how do we solve quadratic inequalities such as

$$x^2 - 5x \geq -6?$$

First we write the inequality in standard form, leaving only zero on the right side:

$$x^2 - 5x + 6 \geq 0.$$

In this example, we are looking for values of x that will make the quadratic on the left side greater than or equal to zero.

Note

If $\leq$ and $\geq$ are used in the inequality, then remember that the zeros of the polynomial are included in the solution set.

First, let us find the zeros of the polynomial $x^2 - 5x + 6$:

$$(x - 2)(x - 3) = 0$$

$$x = 2 \text{ or } x = 3.$$

Then we construct a sign chart for each linear factor of the polynomial, and their product.

x	$-\infty$	2	3	∞
$x - 2$	−	○	+	+
$x - 3$	−	−	○	+
$(x - 2)(x - 3)$	+	●	−	+

Since $x^2 - 5x + 6 \geq 0$, we need to take the positive intervals.

Therefore, the solution is the union of intervals $(-\infty, 2]$ and $[3, \infty)$, i.e. $x \in (-\infty, 2] \cup [3, \infty)$.

positive-positive=positive
positive-negative=negative
negative-negative=positive

We can also construct the sign chart in one step. The zeros of the polynomial divide the real number line into three intervals, $(-\infty, 2]$, $[2, 3]$ and $[3, \infty)$. We know that the polynomial has constant sign in each of these three intervals. If we select a test number in each interval and evaluate the polynomial at that number, then the sign of the polynomial at this test number must be the sign for the whole interval.

Let us try testing each interval in our problem. Choose a number from each interval, and substitute for x in the original inequality. For example, we could choose the numbers 1, 2.5, and 4.

Test number	1	2.5	4
Value of the polynomial	2	-0.25	5
Sign of the polynomial	+	-	+

With this information, we can draw the sign chart.



This is the same as the last line of the previous chart.

We can also use the discriminant of a quadratic to help complete the sign chart. We have already seen that the discriminant Δ of a quadratic equation $ax^2 + bx + c = 0$ tells us if the equation has two real roots ($\Delta > 0$), one double root ($\Delta = 0$), or no real solutions ($\Delta < 0$).

If $\Delta < 0$, then the polynomial $ax^2 + bx + c$ always has the same sign as a .

x	$-\infty$	∞
$ax^2 + bx + c$	same sign as a	

If $\Delta = 0$, then of the polynomial $ax^2 + bx + c$ has the same sign as a but we must consider the zero of the polynomial.

x	$-\infty$	$x_1 = x_2 = -\frac{b}{2a}$	∞
$ax^2 + bx + c$	same sign as a		same sign as a

If $\Delta > 0$, the polynomial $ax^2 + bx + c$ has the opposite sign to a between the zeros of the polynomial and the same sign as a in other intervals.

x	$-\infty$	x_1	x_2	∞	
$ax^2 + bx + c$	same sign as a	○	opposite sign to a	○	same sign as a

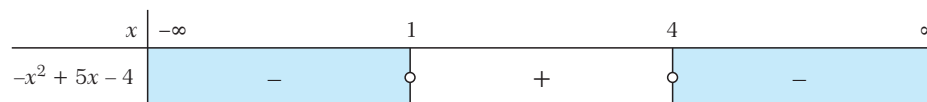
EXAMPLE**7** Solve the inequality $-x^2 - 4 < -5x$.

Solution $-x^2 - 4 < -5x$

$$-x^2 + 5x - 4 < 0$$

$$(-x + 4)(x - 1) = 0$$

$$x = 4 \text{ or } x = 1$$

Therefore, $x \in (-\infty, 1) \cup (4, \infty)$.**EXAMPLE****8** Solve the inequalities.

a. $3x + 4 \geq x^2$

b. $x(x + 2) > 35$

c. $9x^2 - 12x + 4 \leq 0$

d. $x^2 - \frac{2x-1}{4} < 0$

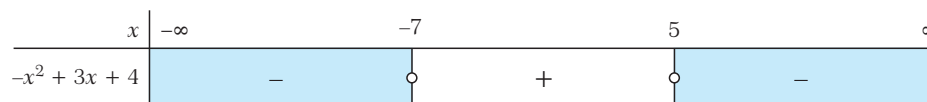
Solution **a.** $3x + 4 \geq x^2$; $-x^2 + 3x + 4 \geq 0$

$$\Delta = 9 + 16 = 25; x_1 = -1, x_2 = 4$$

So $x \in [-1, 4]$.

b. $35 < x(x + 2)$; $-x^2 - 2x + 35 < 0$

$$\Delta = 4 + 140 = 144; x_1 = -7, x_2 = 5$$

So $x \in (-\infty, -7) \cup (5, \infty)$.

c. $9x^2 - 12x + 4 \leq 0$

$$\Delta = 144 - 144 = 0; x_1 = x_2 = \frac{12}{18} = \frac{2}{3}$$

So the only solution is $x = \frac{2}{3}$.

Plot the roots in the sign chart line in ascending order.

c. $x^2 - \frac{2x-1}{4} < 0$; $4x^2 - 2x + 1 < 0$

$\Delta = 4 - 16 = -12 < 0$.

x	$-\infty$					∞
$4x^2 - 2x + 1$		+	+	+	+	+

$4x^2 - 2x + 1$ is always positive, so there is no solution.

EXAMPLE



Consider the equation $x^2 - 2(m+1)x + 1 = 0$. For which values of m does the equation have

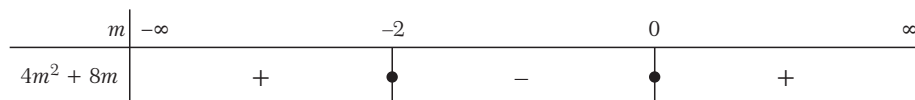
- no real root?
- one double root?
- two distinct real roots?

Solution Let us check the sign of discriminant.

$$\Delta = 4(m+1)^2 - 4 = 4m^2 + 8m + 4 - 4 = 4m^2 + 8m$$

$$4m^2 + 8m = 0$$

$$m = -2 \text{ or } m = 0$$



Now we can answer the question.

- For $\Delta < 0$, there is no real root, and $m \in (-2, 0)$.
- For $\Delta = 0$, there is one double root, and $m = -2$ or $m = 0$.
- For $\Delta > 0$, there are two real roots, and $m \in (-\infty, -2) \cup (0, \infty)$.

EXAMPLE

10

Solve the inequalities.

a. $(3-x)(x^3 - 2x^2 - 8x)(x^2 + 3) < 0$

b. $\frac{(3x+2)(x-5)}{x(x-1)(x^2+x+1)} \geq 0$

c. $\frac{(x^2-x-6)(x-1)^{17}}{x^8(1-x^2)^{55}} \leq 0$

Solution a. First we find all the zeros of the polynomials, then we determine the sign for each polynomial and multiply the signs of each polynomial.

$$(3 - x)(x^3 - 2x^2 - 8x)(x^2 + 3) = (3 - x)x(x^2 - 2x - 8)(x^2 + 3)$$

$$3 - x = 0 ; x = 3$$

$$x = 0$$

$$x^2 - 2x - 8 = 0 ; x = -2 \text{ or } x = 4$$

$$x^2 + 3 = 0 ; \text{ no real solution.}$$

x	$-\infty$	-2	0	3	4	∞
$3 - x$	+	+	+	○	-	-
x	-	-	○	+	+	+
$x^2 - 2x - 8$	+	○	-	-	○	+
$x^2 + 3$	+	+	+	+	+	+
$(3 - x)x(x^2 - 2x - 8)(x^2 + 3)$	-	○	+	○	+	○

We need a value less than zero, so $x \in (-\infty, -2) \cup (0, 3) \cup (4, \infty)$.

b. First we find all the zeros of the polynomials. The equality part of the original inequality is satisfied for these zeros and they must be included in the final solution set. On the other hand, since division by zero is never allowed, the zeros of $x^4 - x$ must not be included in the solution set.

$$\frac{(3x + 2)(x - 5)}{x(x - 1)(x^2 + x + 1)}$$

$$3x + 2 = 0 ; x = -\frac{2}{3}$$

$$x - 5 = 0 ; x = 5$$

$$x = 0$$

$$x - 1 = 0 ; x = 1$$

$$x^2 + x + 1 = 0 ; \text{ no real solution.}$$

x	$-\infty$	$-\frac{2}{3}$	0	1	5	∞
$3x + 2$	-	●	+	+	+	+
$x - 5$	-	-	-	-	●	+
x	-	-	○	+	+	+
$x - 1$	-	-	-	○	+	+
$x^2 + x + 1$	+	+	+	+	+	+
$\frac{(3x + 2)(x - 5)}{x^4 - x}$	+	●	○	○	●	+

We need a value greater than or equal to zero, so $x \in \left[-\frac{2}{3}, 1\right] \cup [5, \infty)$.

$\frac{\text{positive}}{\text{positive}} = \text{positive}$
 $\frac{\text{positive}}{\text{negative}} = \text{negative}$
 $\frac{\text{negative}}{\text{negative}} = \text{positive}$



If the power is an odd number, you can ignore it when you calculate the zero. If the power is an even number, consider it just as 2 when you calculate the zero.

c. $\frac{(x^2 - x - 6)(x - 1)^{17}}{x^8(1 - x^2)^{55}} \leq 0$

$$x^2 - x - 6 = 0 ; x = -2 \text{ or } x = 3$$

$$(x - 1)^{17} = 0 ; x - 1 = 0 ; x = 1$$

$$x^8 = 0 ; x = 0 \text{ (double root)}$$

$$(1 - x^2)^{55} = 0 ; 1 - x^2 = 0 ; x = 1 \text{ or } x = -1 \text{ (1 is also a double root)}$$

x	$-\infty$	-2	-1	0	1	3	∞
$x^2 - x - 6$	+	•	-	-	-	•	+
$(x - 1)^{17}$	-	-	-	-	○	+	+
x^8	+	+	+	○	+	+	+
$(1 - x^2)^{55}$	-	-	○	+	+	○	-
$\frac{(x^2 - x - 6)(x - 1)^{17}}{x^8(1 - x^2)^{55}}$	+	•	-	○	+	○	-

We need a value less than or equal to zero, so $x \in [-2, -1) \cup [3, \infty)$.



The signs of $M \cdot N$ and $\frac{M}{N}$, $N \neq 0$ are the same.

Let us summarize the key steps to solving any inequality.

1. Write the polynomial inequality in standard form.
2. Find all zeros of the polynomial(s).
3. Determine the character of the roots.
4. Determine the sign of the coefficient of leading term of the polynomial(s).
5. Construct a sign chart.
6. In the sign chart, from right to left start with the sign of the coefficient of the leading term a . After each root change the sign, but if there is a double root do not change the sign.

Let us solve Example 10c in another way.

EXAMPLE

11

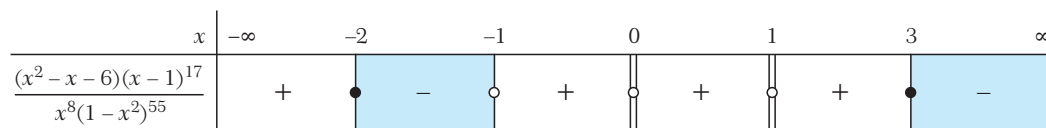
Solve the inequality $\frac{(x^2 - x - 6)(x - 1)^{17}}{x^8(1 - x^2)^{55}} \leq 0$.

Solution $x^2 - x - 6 = 0 ; x = -2 \text{ or } x = 3$

$$(x - 1)^{17} = 0 ; x - 1 = 0 ; x = 1$$

$$x^8 = 0 ; x = 0 \text{ (double root)}$$

$$(1 - x^2)^{55} = 0 ; 1 - x^2 = 0 ; x = 1 \text{ or } x = -1 \text{ (1 is also a double root)}$$



Therefore, $x \in [-2, -1) \cup [3, \infty)$.

EXAMPLE 12 Solve the inequality $(x^2 - 2x - 8)(x^2 + x - 12) \leq 0$.

Solution Find the zeros of the polynomials.

$$x^2 - 2x - 8 = 0; \quad x = 4 \text{ or } x = -2$$

$$x^2 + x - 12 = 0; \quad x = -4 \text{ or } x = 3$$



So, $x \in [-4, -2] \cup [3, 4]$.

EXAMPLE 13 Find the domain of the function $y = \frac{x+2}{\sqrt{3x-12x^2}}$.

Solution $3x - 12x^2 > 0$

$$3x - 12x^2 = 0$$

$$x = 0 \text{ or } x = \frac{1}{4}$$



So $x \in \left(0, \frac{1}{4}\right)$.

EXAMPLE 14 Solve the inequality $(x - 1)^2(x - 2)^3(x - 3)^4(x - 4)^5 \geq 0$.

Solution $(x - 1)^2 = 0; \quad x - 1 = 0; \quad x = 1$ (double root)

$$(x - 2)^3 = 0; \quad x - 2 = 0; \quad x = 2$$

$$(x - 3)^4 = 0; \quad x - 3 = 0; \quad x = 3$$
 (double root)

$$(x - 4)^5 = 0; \quad x - 4 = 0; \quad x = 4$$



So $x \in (-\infty, 2] \cup \{3\} \cup [4, \infty)$.

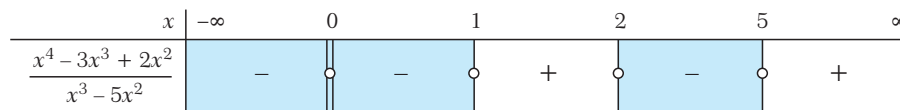
EXAMPLE**15**Solve the inequality $\frac{x^4 - 3x^3 + 2x^2}{x^3 - 5x^2} < 0$.**Solution**

$$\frac{x^4 - 3x^3 + 2x^2}{x^3 - 5x^2} < 0$$

$$\frac{x^2(x-1)(x-2)}{x^2(x-5)} < 0$$

$$x = 0 \quad (\text{double root})$$

$$x = 1, \quad x = 2, \quad x = 5$$

So $x \in (-\infty, 0) \cup (0, 1) \cup (2, 5)$.**EXAMPLE****16**Solve the inequality $\frac{x-1}{x} - \frac{x+1}{x-1} < 2$.**Solution**

$$\frac{x-1}{x} - \frac{x+1}{x-1} - 2 < 0; \quad x \neq 0 \quad \text{and} \quad x \neq 1$$

$$\frac{(x-1)^2 - x(x+1) - 2x(x-1)}{x(x-1)} < 0$$

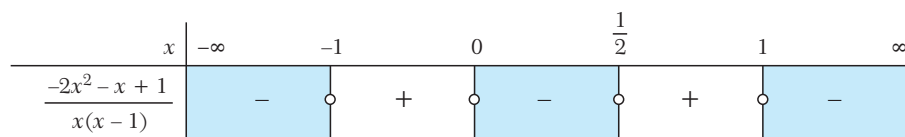
$$\frac{-2x^2 - x + 1}{x(x-1)} < 0$$

$$-2x^2 - x + 1 = 0$$

$$x = -1 \quad \text{or} \quad x = \frac{1}{2}$$

$$x(x-1) = 0$$

$$x = 0, \quad x = 1$$

So $x \in (-\infty, -1) \cup \left(0, \frac{1}{2}\right) \cup (1, \infty)$.

EXAMPLE**17**Solve the inequality $\frac{1}{x+1} - \frac{2}{x^2-x+1} \leq \frac{1-2x}{x^3+1}$.**Solution**

$$\frac{1}{x+1} - \frac{2}{x^2-x+1} - \frac{1-2x}{x^3+1} \leq 0; \quad x \neq -1$$

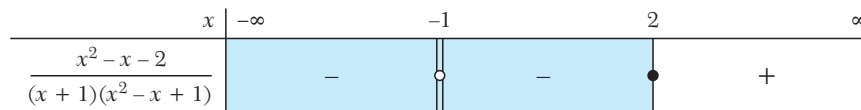
$$\frac{x^2-x+1-2(x+1)-(1-2x)}{(x+1)(x^2-x+1)} \leq 0$$

$$\frac{x^2-x-2}{(x+1)(x^2-x+1)} \leq 0$$

$$x^2-x-2=0; \quad x=-1 \text{ or } x=2$$

$$x+1=0; \quad x=-1 \text{ (double root)}$$

$$x^2-x+1=0; \text{ no real solution}$$

So $x \in (-\infty, -1) \cup (-1, 2]$.**EXAMPLE****18**Solve the inequality $\frac{|2x^2+3x-2| \cdot (1-x^2)}{(x^2+3x) \cdot |x-2|} \leq 0$.**Solution**Since $|2x^2+3x-2|$ and $|x-2|$ are non-negative, just check their roots.

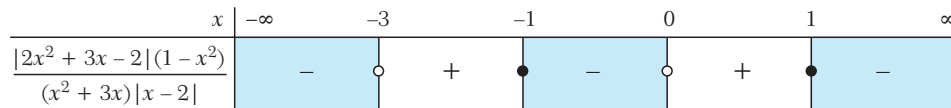
$$2x^2+3x-2=0; \quad x=-2, \quad x=\frac{1}{2}$$

$$x-2=0; \quad x=2$$

-2 and $\frac{1}{2}$ satisfy the inequality, so these values are in the solution set. However, 2 is not in the solution set because it makes the denominator zero.

$$1-x^2=0; \quad x=-1, \quad x=1$$

$$x^2+3x=0; \quad x=0, \quad x=-3$$

So $x \in (-\infty, -3) \cup [-1, 0) \cup \{-2, \frac{1}{2}\} \cup [1, 2) \cup (2, \infty)$.

The absolute value of a number is never negative.
 $|x| \geq 0, \quad x \in \mathbb{R}$

EXAMPLE**19**

Find the domain of the function $y = \sqrt[4]{\frac{x^2 - 6x - 16}{x^2 - 12x + 11}} + \sqrt[7]{3x^4 - 4x^3 + x - 1}$ if y is a real number.

Solution $\sqrt[7]{3x^4 - 4x^3 + x - 1}$ is a real number for all values of x because the index is an odd number.

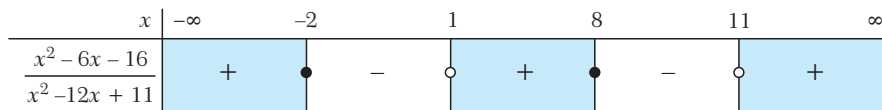
$$\sqrt[4]{\frac{x^2 - 6x - 16}{x^2 - 12x + 11}} \text{ is a real number if } \frac{x^2 - 6x - 16}{x^2 - 12x + 11} \geq 0$$

Therefore, y is a real number if

$$\frac{x^2 - 6x - 16}{x^2 - 12x + 11} \geq 0.$$

$$x^2 - 6x - 16 = 0 ; x = -2, x = 8$$

$$x^2 - 12x + 11 = 0 ; x = 1, x = 11$$



So $x \in (-\infty, -2] \cup (1, 8] \cup (11, \infty)$.

EXAMPLE**20**

Solve the inequality $\sqrt[3]{2^{\frac{3x-1}{x-1}}} < 8^{\frac{x-3}{3x-7}}$.

Solution

$$\sqrt[3]{2^{\frac{3x-1}{x-1}}} < 8^{\frac{x-3}{3x-7}} ; \left(2^{\frac{3x-1}{x-1}}\right)^{\frac{1}{3}} < (2^3)^{\frac{x-3}{3x-7}}$$

$$2^{\frac{3x-1}{3(x-1)}} < 2^{\frac{3(x-3)}{3x-7}}$$

$$\frac{3x-1}{3(x-1)} < \frac{3(x-3)}{3x-7}$$

$$\frac{3x-1}{3x-3} - \frac{3x-9}{3x-7} < 0$$

$$\frac{(3x-1)(3x-7) - (3x-9)(3x-3)}{(3x-3)(3x-7)} < 0$$

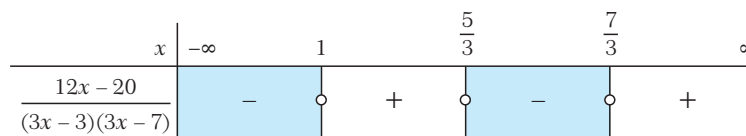
$$\frac{12x-20}{(3x-3)(3x-7)} < 0$$

$$12x-20=0 ; x=\frac{5}{3}$$

$$3x-3=0 ; x=1$$

$$3x-7=0 ; x=\frac{7}{3}$$

If $a^b < a^c$ then $b < c$.



So $x \in (-\infty, 1) \cup \left(\frac{5}{3}, \frac{7}{3}\right)$.

Check Yourself 3

Solve the inequalities.

1. $x^2 + 5x - 6 > 0$

2. $(x + 3)^3(x - 1)^2(x - 4) \geq 0$

3. $\frac{2}{2-x} + \frac{3}{2+x} \geq \frac{4}{4-x^2}$

4. $\frac{(x^2-2)(x^2+4)}{x^2-4} < 0$

5. $\left(\frac{1}{2}\right)^x \leq 4^{x^2-x}$

Answers

1. $(-\infty, -6) \cup (1, \infty)$

2. $(-\infty, -3] \cup \{1\} \cup [4, \infty)$

3. $(-2, 2) \cup [6, \infty)$

4. $(-2, -\sqrt{2}) \cup (\sqrt{2}, 2)$

5. $(-\infty, 0] \cup \left[\frac{1}{2}, \infty\right)$

'Obvious' is the most dangerous word in mathematics.



EXERCISES 2.2

1. Determine the sign of each polynomial.

a. $x^2 - 5x + 4$

b. $2x^2 + x - 6$

c. $2x^2 - 3x + 4$

d. $-16x^2 + 8x - 1$

e. $-4x^2 + 10x - 25$

f. $12x^2 + 4\sqrt{3}x + 1$

2. Solve the inequalities.

a. $x^2 < 9x - 20$

b. $4x - 7x^2 > 0$

c. $x(6x + 7) \geq 0$

d. $2x - 6 < 3x^2$

e. $3x^2 + 8x \leq -\frac{16}{3}$

f. $x(x-1) + 1 < \frac{4}{5}x + \frac{1}{5}$

g. $(2x-1)^2 > (x-5)^2$

3. For which values of k does the equation

$$x^2 + 2(1-k)x + 1 = 0$$

a. no real solution?

b. one double root?

c. two distinct real roots?

4. For which values of b does the equation

$$3x^2 + (b+1)x + 1 = 0$$

have two distinct real roots?

5. For which values of a does the equation

$$ax^2 + (a+1)x + 2 = 0$$

have no real root?

6. The product of a number and four plus the number is less than 15. Find all possible integer values of the number.

7. Solve the inequalities.

a. $x(x-1)^2 > 0$

b. $(2-x)(3x+1)(2x-3) > 0$

c. $(3x-2)(x-3)^3(x+1)^3(x+2)^4 < 0$

d. $\frac{(x-1)(3x-2)}{5-2x} > 0$

e. $\frac{(x+1)(x+2)(x+3)}{(2x-1)(x+4)(3-x)} > 0$

f. $\frac{x^4+x^2+1}{x^2-4x-5} < 0$

g. $\frac{3x-2}{2x-3} < 3$

h. $\frac{2x^2+18x-4}{x^2+9x+8} > 2$

i. $\frac{x+1}{x-2} \geq \frac{3}{x-2} - \frac{1}{2}$

j. $\frac{2-x}{x^3+x^2} \geq \frac{1-2x}{x^3-3x^2}$

k. $\left(x - \frac{1}{x}\right)\left(x^2 - \frac{3}{2}\right) \leq 0$

l. $\frac{1}{3x-2-x^2} > \frac{3}{7x-4-3x^2}$

m. $\frac{(x^2+4x+4)(x-2)^{1001}}{(x-2)^{2004}|x+4|} < 0$

n. $|6x^2-2x+1| \leq 1$

o. $\left|\frac{x+2}{2x-3}\right| < 3$

p. $\left|\frac{x^2-3x+2}{x^2+3x+2}\right| > 1$

q. $\left|\frac{x^2-3x-1}{x^2+x+1}\right| \leq 1$

8. Solve the inequalities.



a. $1 - \frac{1}{x} + \frac{1}{x^2} - \frac{1}{x^3} + \frac{1}{x^4} - \frac{1}{x^5} < 0$

b. $\frac{(a^2+a+1)x^2+4}{a^4x-a^4x^2} > 0$

c. $\frac{3^{-x}(x^2+x+3)}{2-x+\sqrt{2x^2-5x+2}} < 0$

d. $\frac{|x-3|}{x^2-5x+6} \geq 2$

9. Find the domain of each function.

a. $y = \frac{2}{\sqrt[3]{x^2-49}}$

b. $y = \frac{1}{\sqrt{x-4}}$

c. $y = \sqrt{\frac{3x-6}{x+2}}$

d. $\sqrt[4]{(x^4-5x^3+6x^2)(1-x^2)}$

10. Solve the inequality $\frac{12}{x^2+2x} - \frac{3}{x^2+2x-2} > 1$.



11. Solve the inequality $\frac{x^4+3x^3+4x^2-8}{x^2} < 0$.



SYSTEMS OF INEQUALITIES

We saw in Chapter 1 that a set of simultaneous equations to solve is called a system of equations. A system that of equations includes more than one inequality is called an **inequality system**. To find the solution of a system, we solve each inequality separately and then find the **intersection of the solutions**.

EXAMPLE

21

Solve the inequality system $\begin{cases} x^2 - 7x - 8 > 0 \\ x^2 - 4x + 3 > 0 \end{cases}$.

Solution First find the zeros of each polynomial.

(1) $x^2 - 7x - 8 > 0$

$x^2 - 7x - 8 = 0$; $x = -1, x = 8$

(2) $x^2 - 4x + 3 > 0$

$x^2 - 4x + 3 = 0$; $x = 1, x = 3$

We need to find values so that both polynomials are greater than zero. Let us check the chart.

x	$-\infty$	-1	1	3	8	∞	
(1)	+	○	-	-	-	○	+
(2)	+	+	○	-	○	+	+
system		○	○	○	○		

We can see that both polynomials are greater than zero when $x \in (-\infty, -1) \cup (8, \infty)$. This is the intersection of the solutions.

EXAMPLE

22

Solve the inequality system $\begin{cases} \frac{x^2 + x - 4}{x} < 1 \\ x^2 < 64 \end{cases}$.

Solution (1) $\frac{x^2 + x - 4}{x} < 1$; $\frac{x^2 + x - 4 - x}{x} < 0$; $\frac{x^2 - 4}{x} < 0$

$x^2 - 4 = 0$; $x = \pm 2$; $x = 0$

(2) $x^2 < 64$; $x^2 - 64 < 0$

$x^2 - 64 = 0$; $x = \pm 8$

x	$-\infty$	-8	-2	0	2	8	∞
(1)	-	○	-	○	+	○	+
(2)	+	○	-	○	-	○	+
system		○	○	○	○	○	

This time, both polynomials need to be less than zero, so $x \in (-8, -2) \cup (0, 2)$.

EXAMPLE

23

Solve the inequality system $\begin{cases} x^5 \geq 100x^3 \\ \frac{(x+9)(5x-x^2-18)}{x^2-18x+45} \geq 0 \end{cases}$.

Solution

(1) $x^5 \geq 100x^3$

$$x^5 - 100x^3 \geq 0; \quad x^3(x^2 - 100) = 0$$

$$x = 0, \quad x = \pm 10$$

(2) $\frac{(x+9)(5x-x^2-18)}{x^2-18x+45} \geq 0$

$$x + 9 = 0; \quad x = -9$$

$$5x - x^2 - 18 = 0; \quad \text{no solution}$$

$$x^2 - 18x + 45 = 0; \quad x = 3, \quad x = 15$$

x	$-\infty$	-10	-9	0	3	10	15	∞
(1)	-	•	+	+	•	-	+	+
(2)	+	+	•	-	-	+	+	-
system		•	•	○	○	•	○	

So $x \in [-10, -9] \cup [10, 15)$.

EXAMPLE

24

Solve the inequality $\sqrt{2x-1} < x+2$.

Solution

$$\begin{cases} 2x-1 \geq 0 & (1) \\ x+2 > 0 & (2) \\ 2x-1 < (x+2)^2 & (3) \end{cases}$$

(1) $2x - 1 \geq 0$

$$2x - 1 = 0$$

$$x = \frac{1}{2}$$

(2) $x + 2 > 0$

$$x + 2 = 0$$

$$x = -2$$

(3) $2x - 1 < x^2 + 4x + 4$

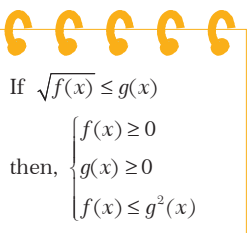
$$x^2 + 2x + 5 > 0$$

$$x^2 + 2x + 5 = 0$$

since $\Delta < 0$, there is no real solution

x	$-\infty$	-2	$\frac{1}{2}$	∞
(1)	-	-	•	+
(2)	-	○	+	+
(3)	+	+	+	+
system			•	

So $x \in [\frac{1}{2}, \infty)$.



EXAMPLE

25

Solve the inequality $\sqrt{x^2 + x - 2} > x$.

Solution

Case 1

$$\begin{cases} x < 0 & (1) \\ x^2 + x - 2 \geq 0 & (2) \end{cases}$$

$$\begin{aligned} (1) \quad & x < 0 \\ (2) \quad & x^2 + x - 2 = 0; \\ & x_1 = -2, \quad x_2 = 1 \end{aligned}$$

Case 2

$$\begin{cases} x \geq 0 & (1) \\ x^2 + x - 2 > x^2 & (2) \end{cases}$$

$$\begin{aligned} (1) \quad & x \geq 0 \\ (2) \quad & x - 2 > 0; \\ & x - 2 = 0, \quad x = 2 \end{aligned}$$



If $\sqrt{f(x)} \geq g(x)$ then

$$\begin{cases} g(x) < 0 \\ f(x) \geq 0 \end{cases}$$

or

$$\begin{cases} g(x) \geq 0 \\ f(x) \geq g^2(x) \end{cases}$$

		x	$-\infty$	-2	0	1	2	∞
Case 1	(1)		-	-	○	+	+	+
	(2)		+	●	-	-	+	+
	system			●				
Case 2	(1)		-	-	●	+	+	+
	(2)		-	-	-	-	○	+
	system						○	

So $x \in (-\infty, -2] \cup (2, \infty)$.

EXAMPLE

26

Solve the inequality $(x^2 + x + 1)^2 - 4(x^2 + x + 1) + 3 < 0$.

Solution

For the inequality, we let $t = x^2 + x + 1$. Then the original inequality becomes $t^2 - 4t + 3 < 0$. First let us solve the inequality for t .

$$t^2 - 4t + 3 = 0; \quad t = 1, \quad t = 3$$

x	$-\infty$	1	3	∞
$t^2 - 4t + 3$	+	○	○	+

So $1 < t < 3$. Now solve for x .

$$1 < x^2 + x + 1 < 3$$

$0 < x^2 + x < 2$, which gives us the system of inequalities

$$\begin{cases} x^2 + x > 0 & (1) \\ x^2 + x - 2 < 0 & (2) \end{cases}$$

$$(1) \quad x^2 + x > 0$$

$$x^2 + x = 0; \quad x = 0, \quad x = -1$$

$$(2) \quad x^2 + x - 2 < 0$$

$$x^2 + x - 2 = 0; \quad x = -2, \quad x = 1$$

x	$-\infty$	-2	-1	0	1	∞	
(1)	+	+	○	−	○	+	+
(2)	+	○	−	−	−	○	+
system		○		○		○	

So $x \in (-2, -1) \cup (0, 1)$.

EXAMPLE

27

Solve the inequality $|x^2 - 3x + 2| \leq 2x - x^2$.

Solution

Case 1

$$\begin{cases} x^2 - 3x + 2 \geq 0 & (1) \\ x^2 - 3x + 2 \leq 2x - x^2 & (2) \end{cases}$$

$$(1) \quad x^2 - 3x + 2 \geq 0$$

$$x^2 - 3x + 2 = 0; \quad x = 1, \quad x = 2$$

$$(2) \quad x^2 - 3x + 2 \leq 2x - x^2; \quad 2x^2 - 5x + 2 \leq 0$$

$$2x^2 - 5x + 2 = 0; \quad x = \frac{1}{2}, \quad x = 2$$

Case 2

$$\begin{cases} x^2 - 3x + 2 < 0 & (1) \\ -(x^2 - 3x + 2) \leq 2x - x^2 & (2) \end{cases}$$

$$(1) \quad x^2 - 3x + 2 < 0$$

$$x^2 - 3x + 2 = 0; \quad x = 1, \quad x = 2$$

$$(2) \quad -(x^2 - 3x + 2) \leq 2x - x^2$$

$$-x^2 + 3x - 2 \leq 2x - x^2$$

$$x - 2 \leq 0$$

$$x - 2 = 0; \quad x = 2$$

x		$-\infty$	$\frac{1}{2}$	1	2	∞
Case 1	(1)	+	+	•	–	+
	(2)	+	•	–	–	+
	system		•	•	•	
Case 2	(1)	+	+	○	–	+
	(2)	–	–	–	•	+
	system		○	○	○	

So $x \in \left[\frac{1}{2}, 1\right] \cup (1, 2) \cup \{2\}$, i.e.

$$x \in \left[\frac{1}{2}, 2\right].$$

Check Yourself 4

Solve the systems.

1.
$$\begin{cases} x^2 - 4x \geq 0 \\ x - 3 \geq 0 \end{cases}$$

2.
$$\begin{cases} x^2 - 4x + 3 > 0 \\ x^2(x - 7)^2 > 0 \end{cases}$$

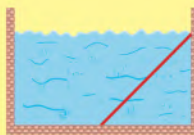
3. $\sqrt{x^2 - x} > 1 + x$

Answers

1. $[4, \infty)$ 2. $x \neq 0, x \neq 7, (-\infty, 1) \cup (3, \infty)$ 3. $(-\infty, -\frac{1}{3})$

Puzzle

In the center of a square pond whose side measures 10 m grows a plant whose top reaches 1 m above the water level. If we pull the plant toward the bank, its top becomes even with the water's surface. What is the depth of the pond and the length of the plant?



Do not worry about your difficulties in mathematics. I can assure you that mine are still greater.

Einstein



EXERCISES 2.3

1. Solve the inequality systems.

$$\text{a. } \begin{cases} x+2 > 0 \\ 2x-3 < 0 \end{cases} \quad \text{b. } \begin{cases} x-1 > 0 \\ x^2-2x-3 < 0 \end{cases}$$

$$\text{c. } \begin{cases} x^2+3x-10 > 0 \\ 2x^2+11x-6 < 0 \end{cases} \quad \text{d. } \begin{cases} \frac{x-2}{x+1} \leq 0 \\ \frac{x^2-1}{x-3} > 0 \end{cases}$$

$$\text{e. } \begin{cases} x^2-4x+3 < 0 \\ 2x-4 < 0 \end{cases} \quad \text{f. } \begin{cases} 2x^2+2 < 5x \\ x^2 \geq x \end{cases}$$

$$\text{g. } \begin{cases} \frac{x+2}{x+1} > 1 \\ \frac{x}{x-2} > 2 \end{cases} \quad \text{h. } \begin{cases} \frac{x^2-1}{3x} < 0 \\ \frac{x-1}{x+1} < 0 \end{cases}$$

2. Solve the inequality systems.

$$\text{a. } \begin{cases} \frac{x+3}{3-x} < 2 \\ x^3 < 16x \\ 4 \geq x^2 \end{cases} \quad \text{b. } 4x-2 < x^2+1 < 4x+6$$

$$\text{c. } \frac{5x-7}{x-5} < 4 - \frac{x}{5-x} + \frac{3x}{x^2-25} < 4$$

3. Solve the inequalities.

$$\text{a. } |4-3x| \geq 2-x$$

$$\text{b. } |x^2-3x-15| < 2x^2-x$$

$$\text{c. } |2x^2+x+11| > x^2-5x+6$$

$$\text{d. } |x-4| + |2x+6| > 10$$

$$\text{e. } \frac{|x-3|}{x^2-5x+6} \geq 2$$

$$\text{f. } |2x+1| - |5x-2| \geq 1$$

$$\text{g. } \frac{x^2-|x|-12}{x-3} \geq 2x$$

4. Solve the inequalities.

$$\text{a. } \sqrt{2x+10} < 3x-5 \quad \text{b. } \sqrt{\frac{x+3}{4-x}} \geq 2$$

$$\text{c. } \sqrt{x^2-x-12} < x \quad \text{d. } \sqrt{\frac{17-15x-2x^3}{x+3}} > 0$$

$$\text{e. } \sqrt{x+1} - \sqrt{x-2} \leq 1 \quad \text{f. } \frac{x-4}{\sqrt{x+2}} < x-8$$

5. Solve the inequalities.

$$\text{a. } \sqrt{x^2-9} < x+2$$

$$\text{b. } \sqrt{-2-3x-x^2} \geq 2x+3$$

$$\text{c. } \sqrt{x^2-x-2} \leq x-1$$

$$\text{d. } \sqrt{4-\sqrt{1-x}} < \sqrt{2-x}$$

$$\text{e. } \sqrt{x-2} + \sqrt{x-3} < 4$$

6. Solve the inequalities.

$$\text{a. } \sqrt{2+x} \geq |x|-1 \quad \text{b. } \sqrt{1+x} \geq |x|-1$$

$$\text{c. } \sqrt{6-x^2} < |x|-1$$

7. Solve the inequality

$$\star \quad \sqrt{x+3-4\sqrt{x-1}} + \sqrt{x+8-6\sqrt{x-1}} > 1.$$

8. For which values of m are the following inequalities satisfied for any real number x ?

$$\text{a. } 2x^2+5x+m > x^2-x-7$$

$$\text{b. } 2x^2+mx^2-2x+m-1 > (1+m)x^2-4x-m$$

$$\text{c. } (m+2)x^2-3x+m-2 < 0$$

$$\text{d. } x^2-6x+4m^2 > 7$$

9. Solve the inequality

$$\star \quad (x^2+6x+14)^2 - 9 \cdot (x^2+6x+15) + 9 < 0.$$

CHAPTER 2 SUMMARY

- To solve a linear inequality, we examine the sign of the polynomial $ax + b$, $a \neq 0$. First, find the zero of the polynomial.

$$ax + b = 0 ; \quad x = -\frac{b}{a}$$

Then, construct a sign chart.

x	$-\infty$	$x_0 = -\frac{b}{a}$	∞
$ax + b$	$ax + b$ has sign opposite to a		$ax + b$ has the same sign as a

If $x \in \left(-\infty, -\frac{b}{a}\right)$, the polynomial has the opposite sign to a .

If $x \in \left(-\frac{b}{a}, \infty\right)$, the polynomial has the same sign as a .

- The equation $ax^2 + bx + c = 0$ has two real roots if $\Delta > 0$, one double root if $\Delta = 0$, and no real solutions if $\Delta < 0$.

If $\Delta < 0$, then the sign of the polynomial $ax^2 + bx + c$ always has the same sign as a .

x	$-\infty$	∞
$ax^2 + bx + c$	same sign as a	

If $\Delta = 0$, then the polynomial $ax^2 + bx + c$ has the same sign as a but we must consider the zero of the polynomial.

x	$-\infty$	$x_1 = x_2 = -\frac{b}{2a}$	∞
$ax^2 + bx + c$	same sign as a		same sign as a

If $\Delta > 0$, then the polynomial $ax^2 + bx + c$ has the opposite sign to a between the zeros of the polynomial and the same sign as a in other intervals.

x	$-\infty$	x_1	x_2	∞
$ax^2 + bx + c$	same sign as a		opposite sign to a	same sign as a

- Key steps to solving any inequality
 - Write the polynomial inequality in standard form.
 - Find all zeros of the polynomial(s).
 - Determine the character of the roots.
 - Determine the sign of the coefficient of leading term of the polynomial(s).
 - Construct a sign chart.
 - In the sign chart, from right to left start with the sign of the coefficient of the leading term a . After each root change the sign, but if there is a double root do not change the sign.

- A system including more than one inequality is called an inequality system.
To find the solution of a system, solve each inequality separately and then find the intersection of the solutions.

Concept Check

- What is an inequality?
- Which symbols do we use to express inequality?
- Show the different interval notation for each type of inequality.
- Write an inequality that has no solution.
- Write an inequality that is always positive.
- Write an inequality that is always negative.
- What are the basic steps to solving an inequality?
- What is the sign of a polynomial?
- Why do we need to determine the sign of a polynomial?
- How do we determine the sign of a polynomial?
- In how many different ways can we write solution to an inequality?
- Why is it a bad idea to multiply each side of an inequality by an expression which includes a variable?
- Why do we need to write an inequality in standard form?
- Can an inequality have exactly one solution? Give an example.

CHAPTER REVIEW TEST 2A

1. What is the solution of the inequality

$$2 - 3(1 - x) < x + 1.8?$$

- A) $x > 1.4$ B) $x < 0.4$ C) $x > 0.7$
 D) $x > -0.7$ E) $x < 1.4$

2. What is the solution of the inequality

$$x^2 + 5x - 14 \leq 0?$$

- A) $[-2, 7]$ B) $[-7, 2]$ C) $(-7, 2)$
 D) $[2, 7]$ E) $(-2, 7]$

3. What is the solution of the inequality $\frac{x^2 - 4}{x^2 - 1} > 0$?

- A) $(-\infty, -2) \cup (1, \infty)$ B) $(-\infty, -2) \cup (2, \infty)$
 C) $(-\infty, -1) \cup (1, \infty)$ D) $\square - \{-1, 1\}$
 E) $(-\infty, -2) \cup (-1, 1) \cup (2, \infty)$

4. What is the solution of the inequality

$$\frac{(2 - 3x)(3x - 3)}{3x + 1} \leq 0?$$

- A) $\left(-\frac{1}{3}, \frac{2}{3}\right] \cup [1, \infty)$ B) $\left(-\infty; -\frac{1}{3}\right) \cup \left[\frac{2}{3}; 1\right]$
 C) $\left\{\frac{2}{3}\right\} \cup (1, +\infty)$ D) $\left(-\frac{1}{3}, \frac{2}{3}\right) \cup (1, +\infty)$
 E) $\left(-\infty; -\frac{1}{3}\right] \cup \left[\frac{2}{3}; 1\right]$

5. What is the solution of the inequality

$$\frac{1}{x+2} < \frac{1}{x-1}?$$

- A) $x < 0$ or $x > 2$ B) $-2 < x < 1$
 C) $x < -2$ or $x > 1$ D) $0 < x < 2$
 E) $-2 < x < 2$

6. What is the greatest integer value of x which satisfies the inequality $x^2 - 13x + 36 < 0$?

- A) 13 B) 7 C) -8 D) 8 E) -13

7. What is the solution of the inequality

$$\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} < 0?$$

- A) $(-\infty, 0)$ B) $(0, \infty)$ C) $\square$ D) $\emptyset$ E) $\{0\}$

8. How many integer values of x are there which satisfy the inequality $|x + 2| \leq \sqrt{2x + 7}$?

- A) 1 B) 3 C) 4 D) 5 E) 7

9. What is the solution of the inequality

$$\frac{(1+x^2)^{2001} \cdot (1-x)^{2002}}{(x+2)^{2003} \cdot (3-x)^{2005}} \geq 0?$$

- A) $x > -2$ B) $x < -2$ C) $x < 3$
D) $x > 3$ E) $-2 < x < 3$

10. The equation $2x(ax - 4) - x^2 + 6 = 0$ has no real roots. What is the smallest possible integer value of a ?

- A) -1 B) 2 C) 3 D) 4 E) 5

11. What is the solution of the inequality

$$\frac{(x-3)(x+2)}{x^2-1} < 1?$$

- A) $(-2, -1) \cup (1, 3)$ B) $(-2, -1)$ C) $(1, 3)$
D) $(-5, -1) \cup (1, \infty)$ E) $(-1, 1) \cup (5, \infty)$

12. What is the sum of the integer values of x which satisfy the inequality $\frac{x^2 - 8x + 7}{(x+2)^2} < 0$?

- A) 32 B) 28 C) 24 D) 20 E) 16

13. What is the domain of the function

$$y = \frac{x-2}{\sqrt{20-8x-x^2}}?$$

- A) $[-1, 7]$ B) $[-10, 2]$ C) $(-10, 2)$
D) $[5, 4)$ E) $(-2, 10]$

14. What is the solution of the inequality system

$$\begin{cases} x^2 - 5x + 6 \leq 0 \\ x^2 + x - 2 \geq 0 \end{cases}?$$

- A) $(2, 3)$ B) $[2, 3]$ C) $(-2, 3)$
D) $[3, 2)$ E) $(-3, -2)$

15. What is the solution of the inequality system

$$\begin{cases} |x| < 2 \\ (x-3)(x-2) \geq 0 \end{cases}?$$

- A) $(-2, 2)$ B) $[2, 3]$ C) $(-2, 3)$
D) $[3, 2)$ E) $(-3, -2)$

16. What is the solution of the inequality

$$3\sqrt{x} - \sqrt{x+3} > 1?$$

- A) $(-2, 2)$ B) $(1, \infty)$ C) $(-2, -3)$
D) $(-\infty, 2)$ E) $(2, \infty)$

CHAPTER REVIEW TEST 2B

1. What is the solution of the inequality

$$\frac{x+1}{2} - \frac{x-1}{3} > x?$$

- A) $x > 1$ B) $x < -3$ C) $x < 3$
D) $x < 1$ E) $x > 6$

2. What is the solution of the inequality

$$(x-6)(x+3) \leq 2-2x?$$

- A) $[-5, 4]$ B) $[-4, 5]$ C) $(-4, 2)$
D) $[-5, 4)$ E) $(-4, 5]$

3. What is the solution of the inequality

$$\frac{x^2+x}{5x-3} < 1?$$

- A) $\left(-\infty, \frac{3}{5}\right) \cup (1, 3)$ B) $\left(\frac{3}{5}, 1\right) \cup (3, \infty)$
C) $(-\infty, 1)$ D) $\square - \{-1, 1\}$
E) $(-1, 1) \cup (3, \infty)$

4. What is the solution of the inequality

$$\frac{1}{2-x} + \frac{5}{2+x} < 1?$$

- A) $[-\infty, -1) \cup (1, 3)$ B) $[3, \infty]$
C) $(-\infty, -2) \cup (2, \infty)$ D) $(1, 3)$
E) $(1, 3)$

5. What is the solution of the inequality

$$\sqrt{x^2} > x+1?$$

- A) $(-2, 4)$ B) $\left[-1, -\frac{1}{2}\right]$ C) $(-\infty, 2)$
D) $[2, \infty)$ E) $\left(-\infty, -\frac{1}{2}\right)$

6. What is the smallest possible integer value of x which satisfies the inequality $(x+7)^2(x-4) \geq 0$?

- A) 3 B) 4 C) -8 D) 8 E) -13

7. What is the solution of the inequality

$$(x^2 + 8x - 9)(x^2 - 4) \geq 0?$$

- A) $(-\infty, -2) \cup [1, \infty)$ B) $(-\infty, -2] \cup (2, \infty)$
C) $(-\infty, -1] \cup [1, \infty)$ D) $\mathbb{R} - \{-1, 1\}$
E) $(-\infty, -9] \cup [-2, 1] \cup [2, \infty)$

8. Which inequality is represented in this sign chart?



- A) $x^2 - 3x + 10 < 0$ B) $x^2 > 3x - 10$
C) $x^2 < 3x + 10$ D) $x^2 - 2x + 5 \geq 0$
E) $x^2 < 2x + 5$

9. What is the solution of the inequality

$$\frac{(x-1)^3(x-2)}{(x-3)^2} \leq 0?$$

- A) $(-\infty, 1)$ B) $[2, \infty]$ C) $[1, 2]$
D) $[-1, 2)$ E) $(2, \infty)$

10. The equation $3x^2 + bx + 12 = 0$ has real roots. What are the possible values of b ?

- A) $[-12, 12]$ B) $(-\infty, -12]$ C) $[12, \infty)$
D) $(-\infty, -12] \cup [12, \infty)$ E) $(-1, 1) \cup (5, \infty)$

11. What is the solution of the inequality $\left| \frac{x}{x+1} \right| \leq 1$?

- A) $\left[-\frac{1}{2}, \infty\right)$ B) $(-2, -1)$ C) $(1, 3)$
D) $\left(\frac{1}{2}, \infty\right)$ E) $(-1, 1)$

12. What is the sum of the integer values of x which satisfy the inequality $\frac{(x-2)(x-4)x^2}{(x+1)^2} \leq 0$?

- A) 8 B) 9 C) 10 D) 11 E) 16

13. What is the domain of the function

$$y = \sqrt{4x - x^2} + \frac{7}{\sqrt{x-2}}?$$

- A) $(-2, 4]$ B) $[-1, 2]$ C) $(-4, 2)$
D) $[2, 4)$ E) $(2, 4]$

14. What is the solution of the inequality system

$$\begin{cases} \frac{x-5}{x+3} > 0 \\ \frac{x+7}{x-3} < 0 \end{cases} ?$$

- A) $(2, 3)$ B) $[-7, 3]$ C) $(-7, -3)$
D) $[-3, 3)$ E) $(7, -2)$

15. How many integer values of x are there which satisfy the inequality $(x+8)(x-1)^2(x-5) < 0$?

- A) 8 B) 9 C) 10 D) 11 E) 16

16. x_1 and x_2 are the roots of $x^2 - 3mx + m - 3 = 0$.

$$\frac{1}{x_1} + \frac{1}{x_2} > 4 \text{ is given.}$$

What are the possible values of m ?

- A) $(-\infty, \infty)$ B) $[2, \infty]$ C) $(-\infty, -3)$
D) $(3, 12)$ E) $(0, 12)$



Chapter 3

QUADRATIC FUNCTIONS



GRAPHS OF QUADRATIC FUNCTIONS

Definition

quadratic function

A function $f(x)$ is a **quadratic function** if

$$y = f(x) = ax^2 + bx + c, \quad a \neq 0$$

where a , b , and c are real numbers.

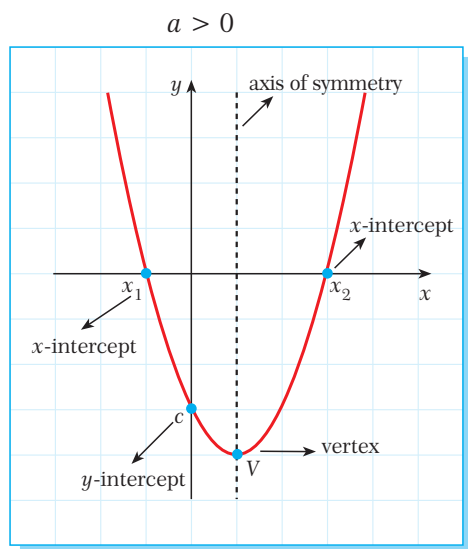
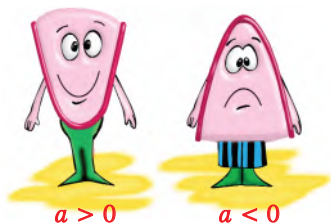
For example, $f(x) = 2x^2 + 3x + 1$, $y = x^2 - 1$, $y = -\sqrt{2}x^2$ and $f(x) = 2x - x^2$ are all quadratic functions. $y = 3x - 4$ is not a quadratic function, because $a = 0$.

The **domain** of a quadratic function is the set of all real numbers.

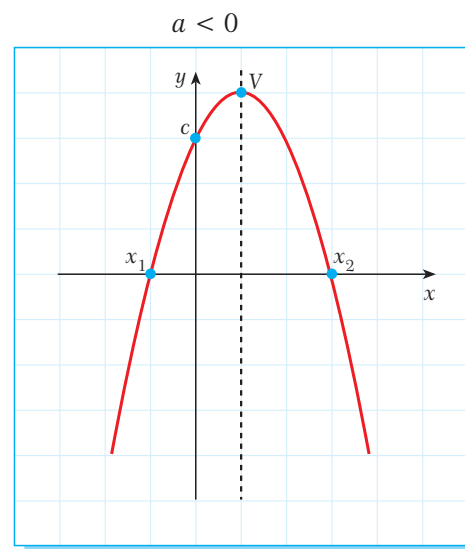
The graph of a quadratic function $f(x) = ax^2 + bx + c$ is called a **parabola**.



The sign of a has no effect on the size or shape of the parabola, the sign of a determines whether the parabola opens upward or downward.



When $a > 0$, the parabola opens upward.



When $a < 0$, the parabola opens downward.

The point V in the parabolas above is called the **vertex** of the parabola. The vertex is the lowest or the highest point of the parabola.

The vertical line drawn through the vertex is called the **axis of symmetry**. It divides the curve into two symmetrical halves.

The points x_1 and x_2 are called the **x-intercepts** of the parabola. They are the zeros of the function.

The point c is called the **y-intercept** of the parabola.

A. GRAPHING $y = ax^2$

How can we find the graph of the quadratic function $y = ax^2$? If we have the function, we can plot the graph by making a table of values. To find the values, we substitute different values of x into the equation to obtain the corresponding y values. These x and y values provide the coordinates for points which we can plot to form the shape of the graph.

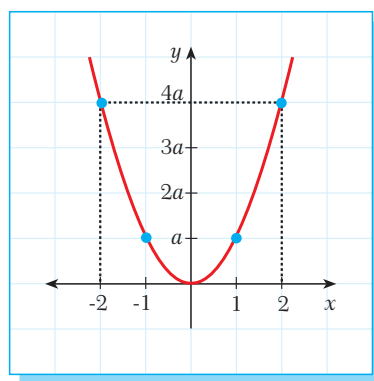
Let us graph the function $y = ax^2$.

If $a > 0$, we get the table of

ordered pairs opposite.

Then we plot the points on a graph and draw a parabola through them.

x	$-\infty$	-2	-1	0	1	2	$+\infty$
y	$+\infty$	$4a$	a	0	a	$4a$	$+\infty$

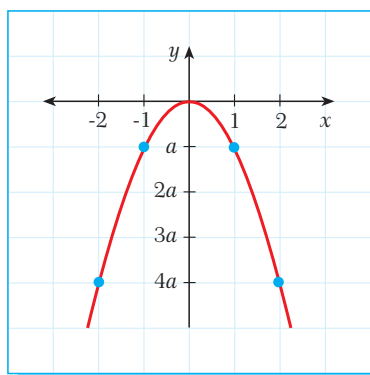


$$y = ax^2, \quad a > 0$$

We can see that the vertex of the parabola is at the origin $(0, 0)$, and the axis of symmetry lies along the y -axis (the line $x = 0$).

If $a < 0$, we get a different set of ordered pairs.

x	$-\infty$	-2	-1	0	1	2	$+\infty$
y	$-\infty$	$4a$	a	0	a	$4a$	$-\infty$



Example

Sketch the graphs of the functions.

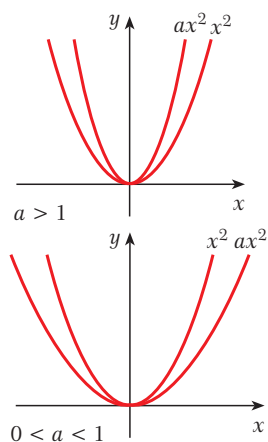
a. $y = x^2$, $y = 2x^2$ and $y = \frac{1}{2}x^2$

b. $y = -x^2$, $y = -2x^2$ and $y = -\frac{1}{2}x^2$

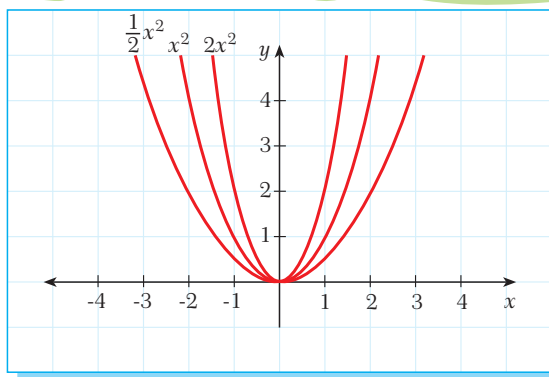
Solution a.



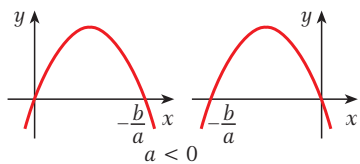
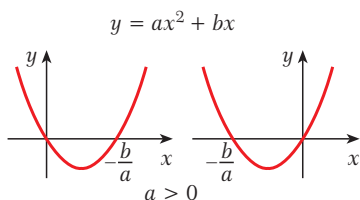
As $|a|$ increases, the parabola becomes narrower. As $|a|$ decreases, the parabola becomes wider.



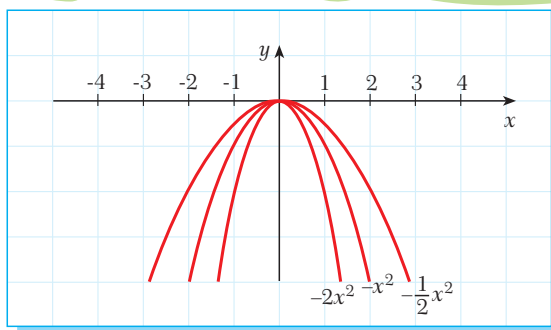
x	$-\infty$	-2	-1	0	1	2	∞
$y = x^2$	∞	4	1	0	1	4	∞
$y = 2x^2$	∞	8	2	0	2	8	∞
$y = \frac{1}{2}x^2$	∞	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2	∞



b.



x	$-\infty$	-2	-1	0	1	2	∞
$y = -x^2$	$-\infty$	-4	-1	0	-1	-4	$-\infty$
$y = -2x^2$	$-\infty$	-8	-2	0	-2	-8	$-\infty$
$y = -\frac{1}{2}x^2$	$-\infty$	-2	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-2	$-\infty$



Check Yourself 1

Graph the functions.

1. $y = 3x^2$

2. $y = \frac{1}{3}x^2$

3. $y = -3x^2$

4. $y = -\frac{1}{3}x^2$

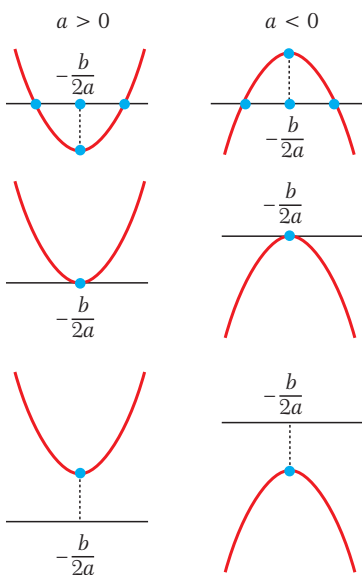
B. GRAPHING $y = ax^2 + bx + c$

1. Vertex Point

Let x_1 and x_2 be zeros of the polynomial $ax^2 + bx + c$.

Then $ax^2 + bx + c = a(x - x_1)(x - x_2)$

$$\begin{aligned} &= a \left(x - \frac{-b + \sqrt{\Delta}}{2a} \right) \left(x - \frac{-b - \sqrt{\Delta}}{2a} \right) \\ &= a \left(x^2 - x \left(\frac{-b + \sqrt{\Delta}}{2a} \right) - x \left(\frac{-b - \sqrt{\Delta}}{2a} \right) + \frac{b^2 - \Delta}{4a^2} \right) \\ &= a \left(x^2 + \frac{xb + x\sqrt{\Delta} + xb - x\sqrt{\Delta}}{2a} + \frac{b^2}{4a^2} - \frac{\Delta}{4a^2} \right) \\ &= a \left(x^2 + \frac{bx}{a} + \frac{b^2}{4a^2} \right) - \frac{\Delta}{4a} \\ &= a \left(x + \frac{b}{2a} \right)^2 - \frac{\Delta}{4a} \\ &= a(x - h)^2 + k \quad \text{where } h = -\frac{b}{2a} \quad \text{and } k = -\frac{\Delta}{4a}. \end{aligned}$$



If a quadratic function has two known x -intercepts, we can find the x -coordinate of the vertex point by averaging them.

$$h = \frac{x_1 + x_2}{2} = -\frac{b}{2a}$$

This is called the **completed square form** of the quadratic function.

COMPLETED SQUARE FORM

The completed square form of a quadratic function is $f(x) = a(x - h)^2 + k$.

The vertex of a parabola of a function in complete square form is the point (h, k) .

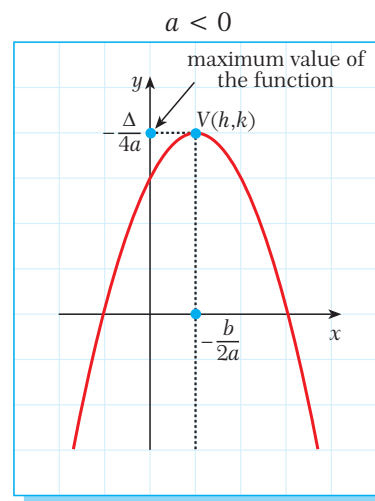
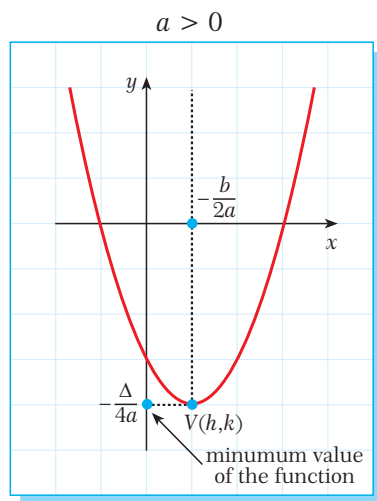
At the vertex point $V(h, k)$, $h = -\frac{b}{2a}$ and $k = f(h)$ or $k = \frac{4ac - b^2}{4a}$.

Since (h, k) is the vertex of a parabola, then the x -coordinate of the vertex point is the equation of the axis of symmetry, $x = h$.

If $a > 0$, then the y -coordinate of the vertex is the minimum value of the function. If $a < 0$, then the y -coordinate of the vertex is the maximum value of the function.



The trajectory of a shell fired from a cannon is a parabola. To reach the maximum value range with a cannon, it is shown in calculus that the barrel of the cannon must be set at 45° .



Example

- 2** Find the coordinates of the vertex point of each function. Write the equation of the axis of symmetry, and determine the maximum or minimum value of the function.

a. $y = x^2 - 2x - 3$

b. $y = -x^2 + 4x + 5$

c. $y = x^2 + 3x + 1$

Solution a. $V(h, k)$ is the vertex point.

$$h = -\frac{b}{2a} = \frac{2}{2} = 1$$

$$k = f(h) = 1^2 - 2 \cdot 1 - 3 = -4$$

So the vertex is $V(1, -4)$.

The equation of the axis of symmetry is $x = 1$.

Since $a = 1 > 0$, the vertex is the minimum value and $k = -4$ is the minimum value of the function.

b. $h = -\frac{b}{2a} = -\frac{4}{2 \cdot (-1)} = 2$

$$k = f(h) = -2^2 + 4 \cdot 2 + 5 = 9$$

So the vertex is $V(2, 9)$.

The equation of the axis of symmetry is $x = 2$.

Since $a = -1 < 0$, the vertex is the maximum value and $k = 9$ is the maximum value of the function.

$$\text{c. } h = -\frac{b}{2a} = -\frac{3}{2 \cdot 1} = -\frac{3}{2}$$

$$k = f(h) = \left(-\frac{3}{2}\right)^2 + 3 \cdot \left(-\frac{3}{2}\right) + 1 = -\frac{5}{4}$$

So the vertex point is $V\left(-\frac{3}{2}, -\frac{5}{4}\right)$.

The equation of the axis of symmetry is $x = -\frac{3}{2}$.

Since $a = 1 > 0$, the vertex is the minimum value and $y = -\frac{5}{4}$ is the minimum value of the function.

Example

3 The line $3x + 1 = 0$ is the axis of symmetry of the parabola $y = -2x^2 + mx - 1$. Find the maximum value of the function.

Solution $3x + 1 = 0$; $x = -\frac{1}{3}$ is the equation of axis of symmetry.

$$h = -\frac{1}{3} = -\frac{b}{2a} = -\frac{m}{2 \cdot (-2)}; \quad m = -\frac{4}{3}$$

Since $a = -2 < 0$, the vertex is the maximum value.

$$k = f(h) = -2\left(-\frac{1}{3}\right)^2 - \frac{4}{3}\left(-\frac{1}{3}\right) - 1 = -\frac{7}{9}$$

So $k = -\frac{7}{9}$ is the maximum value.

Example

4 The minimum value of the function $y = x^2 - 8x + n$ is -4 . What is the value of n ?

Solution Given $k = -4$ and $h = -\frac{b}{2a} = -\frac{-8}{2 \cdot 1} = 4$; $h = 4$.

$$k = f(h); \quad -4 = 4^2 - 8 \cdot 4 + n; \quad n = 12$$

Check Yourself 2

Find the vertex of each parabola.

$$1. \ y = x^2 - 10x + 20 \quad 2. \ y = \frac{1}{2}x^2 + x - 8 \quad 3. \ y = 4 - x^2 \quad 4. \ y = 4x + x^2$$

Answers

$$1. \ (5, -5) \quad 2. \ (-1, -8.5) \quad 3. \ (0, 4) \quad 4. \ (-2, -4)$$

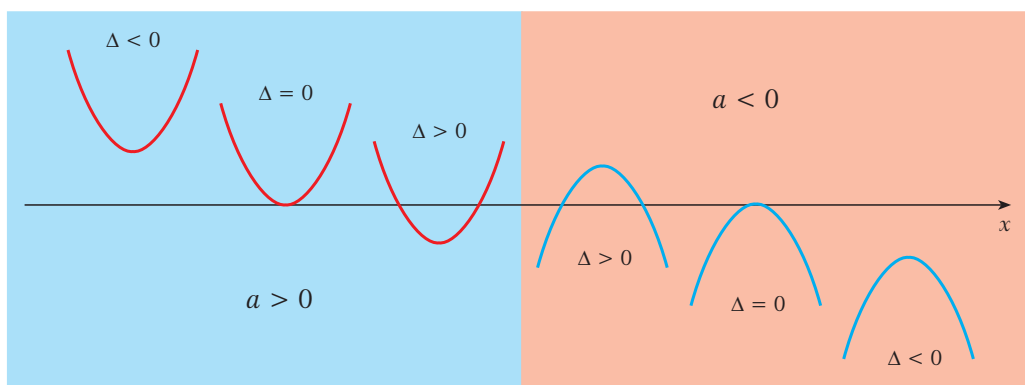
2. Intercepts

Consider the function $y = f(x) = ax^2 + bx + c$, $a \neq 0$.

If $x = 0$, then $y = c$. The point $(0, c)$ is called the y -intercept of the graph. We mean that the parabola intersects the y -axis at this point.

If $y = 0$, then $ax^2 + bx + c = 0$. Let x_1 and x_2 be roots of this equation. The points $(x_1, 0)$ and $(x_2, 0)$ are called the x -intercepts of the graph. We mean that the parabola intersects the x -axis at these points. There are three possibilities for the x -intercepts.

1. If $\Delta > 0$, the parabola intersects the x -axis at two distinct points.
2. If $\Delta = 0$, the parabola is tangent to the x -axis.
3. If $\Delta < 0$, the parabola does not intersect the x -axis.



Example

5 Find the x - and y -intercepts of each function.

a. $y = x^2 - 9$

b. $y = -4x^2 + 5x$

c. $y = 3x^2 + 4x - 2$

d. $y = x^2 + 3x + 7$

e. $y = 4x^2 + 20x + 25$

Solution a. $x = 0$; $y = -9$

So $(0, -9)$ is the y -intercept.

$$y = 0; \quad x^2 - 9 = 0$$

$$x_1 = -3, \quad x_2 = 3$$

So $(-3, 0)$ and $(3, 0)$ are the x -intercepts.

b. $x = 0$; $y = 0$

So $(0, 0)$ is the y -intercept.

$$y = 0; \quad -4x^2 + 5x = 0; \quad x_1 = 0, \quad x_2 = \frac{5}{4}$$

So $(0, 0)$ and $\left(\frac{5}{4}, 0\right)$ are the x -intercepts.

c. $x = 0 ; y = -2$

So $(0, -2)$ is the y -intercept.

$$y = 0 ; 3x^2 + 4x - 2 = 0$$

$$\Delta = 4^2 - 4 \cdot 3 \cdot (-2) = 40$$

$$x_1 = \frac{-4 - 2\sqrt{10}}{6} = \frac{-2 - \sqrt{10}}{3}, \quad x_2 = \frac{-4 + 2\sqrt{10}}{6} = \frac{-2 + \sqrt{10}}{3}$$

So $(\frac{-2 - \sqrt{10}}{3}, 0)$ and $(\frac{-2 + \sqrt{10}}{3}, 0)$ are the x -intercepts.

d. $x = 0 ; y = 7$

So $(0, 7)$ is the y -intercept.

$$y = 0 ; x^2 + 3x + 7 = 0$$

$$\Delta = 3^2 - 4 \cdot 1 \cdot 7 = -19 < 0$$

So the parabola does not cut the x -axis, i.e. there are no x -intercepts.

e. $x = 0 ; y = 25$

So $(0, 25)$ is the y -intercept.

$$y = 0 ; 4x^2 + 20x + 25 = 0$$

$$\Delta = 20^2 - 4 \cdot 4 \cdot 25 = 0$$

$$x_1 = x_2 = -\frac{20}{8} = -\frac{5}{2}$$

$$\Delta = 20^2 - 4 \cdot 4 \cdot 25 = 0$$

So the parabola is tangent to the x -axis at the point $(-\frac{5}{2}, 0)$.

Example



The function $y = x^2 - (m + 2)x + 5 + m$ is given. Find the value(s) of m for which the parabola

a. does not cut the x -axis.

b. is tangent to the x -axis.

c. intersects the x -axis at two distinct points.

Solution $\Delta = (m + 2)^2 - 4(5 + m)$
 $= m^2 - 16$

$$\Delta = m^2 - 16 = 0 ; \quad m = \pm 4$$

Let us consider the sign of Δ .



- a. If $m \in (-4, 4)$, the parabola does not cut the x -axis.
- b. If $m = \pm 4$, the parabola is tangent to the x -axis.
- c. If $m \in (-\infty, -4) \cup (4, \infty)$, the parabola cuts the x -axis at two distinct points.

Check Yourself 3

Find the x - and y -intercepts of each parabola.

1. $y = -x^2 + 3x - 4$

2. $y = 3x^2 - 6x + 3$

3. $y = 3x - x^2$

Answers

1. no x -intercepts, $(0, -4)$ 2. $(1, 0)$, $(0, 3)$ 3. $(0, 0)$, $(3, 0)$

3. Sketching a Graph

To sketch the parabola $y = ax^2 + bx + c$, follow the steps.

1. Check a .

When $a > 0$, the graph has a minimum point and the parabola opens upward.

When $a < 0$, the graph has a maximum point and the parabola opens downward.

- 2. Find the coordinates of the vertex point, $V(h, k)$.
- 3. Find the x - and y -intercepts.
- 4. Sketch the graph of the parabola.

Example

7 Graph the functions.

a. $y = x^2 - 3x - 10$

b. $y = -2x^2 - 8x$

c. $y = x^2 + 4x + 5$

d. $y = -3x^2 + 12$

e. $y = x^2 - 2\sqrt{2}x + 2$

Solution a. $a = 1 > 0$, so the parabola opens upward.

$$h = -\frac{b}{2a} = \frac{3}{2},$$

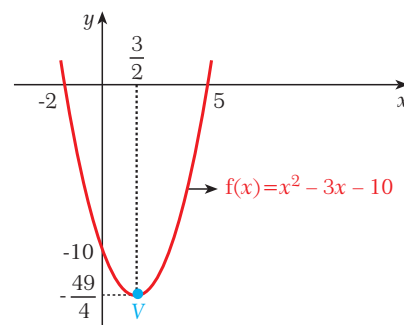
$$k = f(h) = \left(\frac{3}{2}\right)^2 - 3 \cdot \frac{3}{2} - 10 = -\frac{49}{4};$$

$V\left(\frac{3}{2}, -\frac{49}{4}\right)$ is the vertex point.

$x = 0$; $y = -10$, $(0, -10)$ is the y -intercept.

$y = 0$; $x^2 - 3x - 10 = 0$; $x = -2$, $x = 5$;

$(-2, 0)$ and $(5, 0)$ are the x -intercepts.



- b. $a = -2 < 0$, so the parabola opens downward.

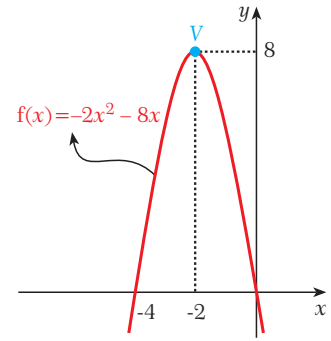
$$h = -\frac{-8}{2 \cdot (-2)} = -2,$$

$$k = f(h) = -2(-2)^2 - 8 \cdot (-2) = 8; \quad V(-2, 8)$$

$x = 0$; $y = 0$, $(0, 0)$ is the y -intercept.

$$y = 0; \quad -2x^2 - 8x = 0;$$

$x = 0$, $x = -4$; $(-4, 0)$ and $(0, 0)$ are the x -intercepts.



- c. $a = 1 > 0$, so the parabola opens upward.

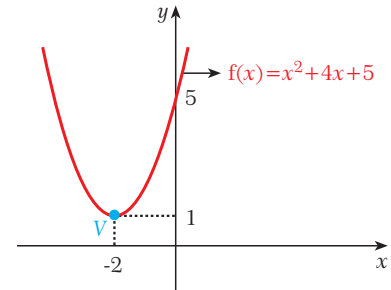
$$h = -\frac{4}{2 \cdot 1} = -2,$$

$$k = f(h) = (-2)^2 + 4 \cdot (-2) + 5 = 1; \quad V(-2, 1)$$

$x = 0$; $y = 5$, $(0, 5)$ is the y -intercept.

$$y = 0; \quad x^2 + 4x + 5 = 0;$$

$\Delta = 4^2 - 4 \cdot 1 \cdot 5 = -4 < 0$; the parabola does not cut the x -axis.



- d. $a = -3 < 0$, so the parabola opens downward.

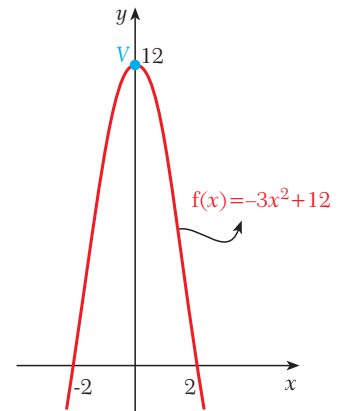
$$h = -\frac{0}{2(-3)} = 0,$$

$$k = f(h) = -3(0)^2 + 12 = 12; \quad V(0, 12)$$

$x = 0$; $y = 12$, $(0, 12)$ is the y -intercept.

$$y = 0; \quad -3x^2 + 12 = 0;$$

$x = \pm 2$; $(-2, 0)$ and $(2, 0)$ are the x -intercepts.



- e. $a = 1 > 0$, so the parabola opens upward.

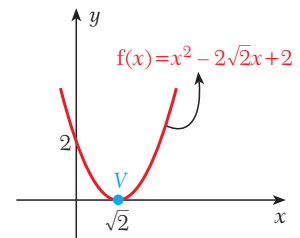
$$h = -\frac{-2\sqrt{2}}{2 \cdot 1} = \sqrt{2}$$

$$k = f(h) = (\sqrt{2})^2 - 2\sqrt{2} \cdot \sqrt{2} + 2 = 0; \quad V(\sqrt{2}, 0)$$

$x = 0$; $y = 2$, $(0, 2)$ is the y -intercept.

$$y = 0; \quad x^2 - 2\sqrt{2}x + 2 = 0;$$

$x_1 = x_2 = \sqrt{2}$; the parabola is tangent to the x -axis at the point $(\sqrt{2}, 0)$.



Example

Sketch the graph of the function $y = -x^2 - 2x + 8$ for $|x| \leq 3$. Find the range of the function.

Solution $a = -1 < 0$, so the parabola opens downward.

$$h = -\frac{-2}{2 \cdot (-1)} = -1,$$

$$k = f(h) = -(-1)^2 - 2 \cdot (-1) + 8 = 9; \quad V(-1, 9)$$

$x = 0$; $y = 8$, $(0, 8)$ is the y -intercept.

$$y = 0; \quad -x^2 - 2x + 8 = 0;$$

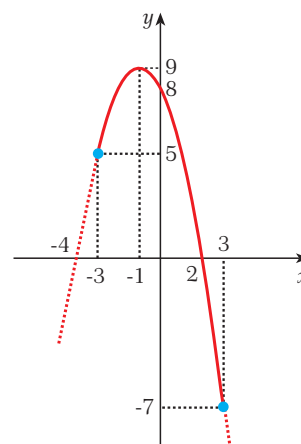
$x_1 = -4$, $x_2 = 2$; $(-4, 0)$ and $(2, 0)$ are the x -intercepts.

$$|x| \leq 3; \quad -3 \leq x \leq 3$$

$x = -3$; $y = -9 + 6 + 8 = 5$, so $(-3, 5)$ is on the parabola.

$x = 3$; $y = -9 - 6 + 8 = -7$, so $(3, -7)$ is on the parabola.

So the range of the function is $[-7, 9]$.

**Example**

Find the values of p for which $px^2 + 4x + p$ is greater than zero for all real values of x .

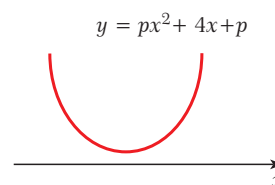
Solution $px^2 + 4x + p > 0$ for all real numbers x .

This is possible only if $p > 0$ and $\Delta < 0$.

$$\begin{cases} p > 0 & (1) \\ 4^2 - 4p^2 < 0 & (2) \end{cases}$$

$$(1) \quad p > 0$$

$$(2) \quad 4^2 - 4p^2 < 0; \quad 4^2 - 4p^2 = 0; \quad p = \pm 2$$



p	$-\infty$	-2	0	2	∞
(1)	$-$	$-$	$\circ$	$+$	$+$
(2)	$-$	$\circ$	$+$	$+$	$\circ$
system		$\circ$	$\circ$	$\circ$	

So $p \in (2, \infty)$.

Check Yourself 4

Sketch the graph of each function.

1. $y = x^2 - 5x + 4$

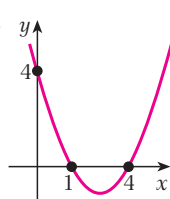
2. $y = -x^2 + 4x - 3$

3. $y = 4x^2 - 20x + 25$

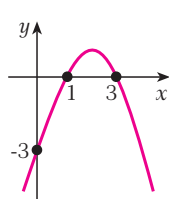
4. $y = 3x^2 - 2x$

Answers

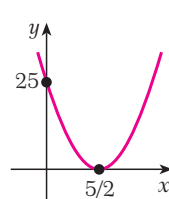
1.



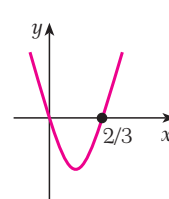
2.



3.



4.



4. Shifting Graphs

Let $y = f(x)$ be a function.

Vertical Shifting

$$y = f(x) + k \quad \begin{cases} k > 0 & \text{shift the graph of } y = f(x) \text{ } k \text{ units upward. (see Figure 1)} \\ k < 0 & \text{shift the graph of } y = f(x) \text{ } |k| \text{ units downward. (see Figure 1)} \end{cases}$$

Horizontal Shifting

$$y = f(x + h) \quad \begin{cases} h > 0 & \text{shift the graph of } y = f(x) \text{ } h \text{ units to the left. (see Figure 2)} \\ h < 0 & \text{shift the graph of } y = f(x) \text{ } |h| \text{ units to the right. (see Figure 2)} \end{cases}$$

Reflection

$$y = -f(x) \quad \text{Reflect the graph of } y = f(x) \text{ in the } x\text{-axis. (see Figure 3)}$$

Horizontal and Vertical Shifting

$y = f(x + h) + k$ Shift the graph of $y = f(x)$ $|k|$ units upward or downward and $|h|$ units to the left or to the right.

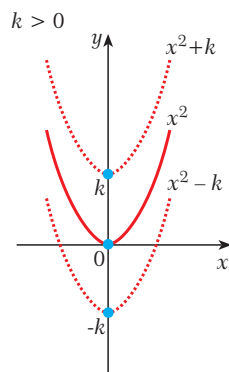


Figure 1

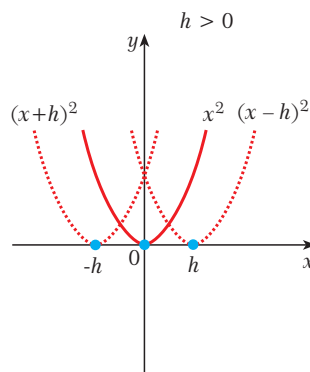


Figure 2

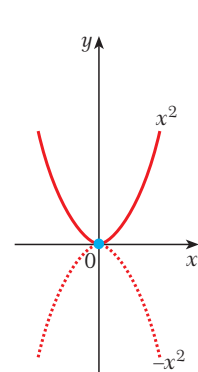


Figure 3

Example

10

Sketch the graph of the each function using the shifting method.

a. $y = x^2 + 3$

b. $y = x^2 - 3$

c. $y = (x - 3)^2$

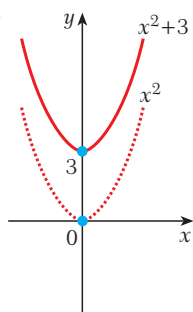
d. $y = (x + 3)^2$

e. $y = -(x - 4)^2$

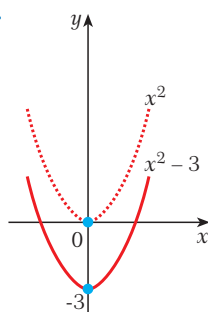
f. $y = (x + 2)^2 - 3$

g. $y = -(x - 2)^2 + 1$

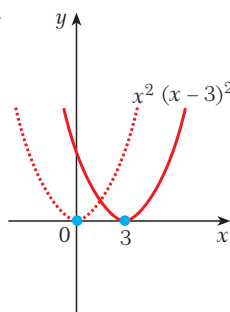
Solution a.



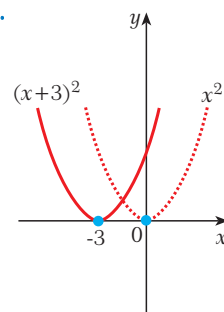
b.



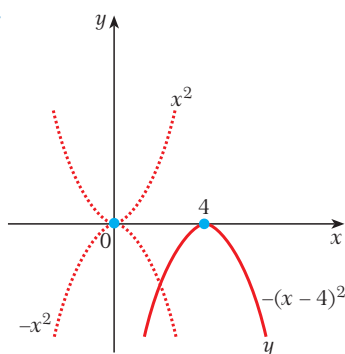
c.



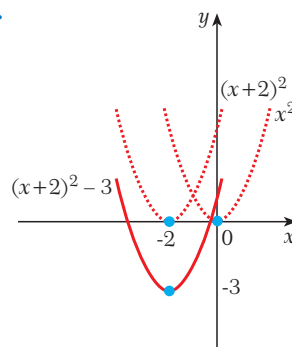
d.



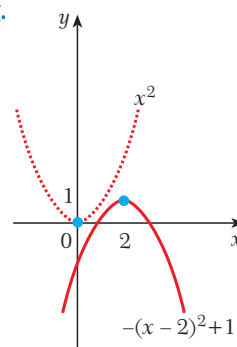
e.



f.



g.



Check Yourself 5

Sketch the graph of the each function by shifting.

1. $y = x^2 + 4$

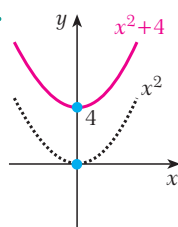
2. $y = -(x + 1)^2$

3. $y = (x - 2)^2 - 2$

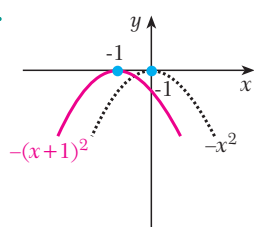
4. $y = -(x + 2)^2 + 1$

Answers

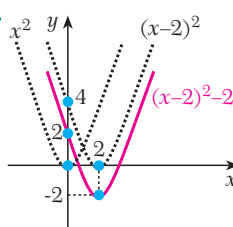
1.



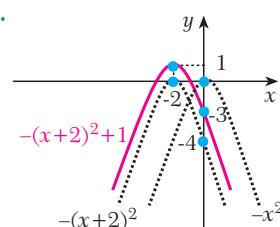
2.



3.



4.



5. Parabolas with Absolute Value (Optional)

Example

11

Sketch the graph of each function.

a. $y = |x^2 - 3x + 2|$

c. $y = x|x + 2|$

b. $y = -x^2 + 2|x| + 3$

d. $y = |2x^2 - 12x + 16| + 2$

Solution a. First graph the function $y = x^2 - 3x + 2$.

Then, take above the part of the graph which is below the x axis, symmetric with respect to x -axis.

$a = 1 > 0$, so the parabola opens upward.

$$h = -\frac{-3}{2 \cdot 1} = \frac{3}{2},$$

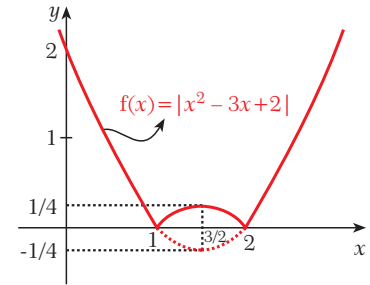
$$k = f(h) = \left(\frac{3}{2}\right)^2 - 3 \cdot \frac{3}{2} + 2 = -\frac{1}{4};$$

$V\left(\frac{3}{2}, -\frac{1}{4}\right)$ is the vertex point.

$x = 0$; $y = 2$, $(0, 2)$ is the y -intercept

$y = 0$; $x^2 - 3x + 2 = 0$;

$x = 1$, $x = 2$; $(1, 0)$ and $(2, 0)$ are the x -intercepts.



b. $y = -x^2 + 2|x| + 3$

Case 1

If $x \geq 0$, $|x| = x$; $y = -x^2 + 2x + 3$

$a = -1 < 0$, so the parabola opens downward.

$$h = -\frac{2}{2 \cdot (-1)} = 1, \quad k = f(1) = -1^2 + 2 \cdot 1 + 3 = 4; \quad V(1, 4) \text{ is the vertex point.}$$

$x = 0$; $y = 3$, $(0, 3)$ is the y -intercept.

$y = 0$; $-x^2 + 2x + 3 = 0$;

$x = -1$, $x = 3$; $(-1, 0)$ and $(3, 0)$ are the x -intercepts.

Case 2

If $x < 0$, $|x| = -x$; $y = -x^2 - 2x + 3$

$a = -1 < 0$, so the parabola opens downward.

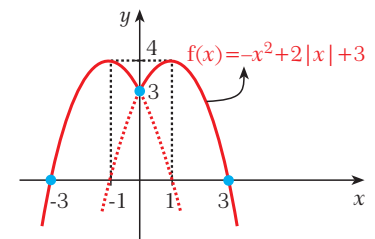
$$h = -\frac{-2}{2 \cdot (-1)} = -1,$$

$$k = f(-1) = -(-1)^2 - 2 \cdot (-1) + 3 = 4;$$

$V(-1, 4)$ is the vertex point.

$x = 0$; $y = 3$, $(0, 3)$ is the y -intercept.

$y = 0$; $-x^2 - 2x + 3 = 0$; $x = -3$, $x = 1$; $(-3, 0)$ and $(1, 0)$ are the x -intercepts.



c. $y = x|x + 2|$

Case 1

If $x + 2 \geq 0$, $x \geq -2$, $|x + 2| = x + 2$; $y = x^2 + 2x$

$a = 1 > 0$, so the parabola opens upward.

$h = -\frac{2}{2 \cdot 1} = -1$, $k = f(-1) = (-1)^2 + 2 \cdot (-1) = -1$; $V(-1, -1)$ is the vertex point.

$x = 0$; $y = 0$, $(0, 0)$ is the y -intercept.

$y = 0$; $x^2 + 2x = 0$; $x = -2$, $x = 0$; $(-2, 0)$ and $(0, 0)$ are the x -intercepts.

Case 2

If $x + 2 < 0$; $x < -2$, $|x + 2| = -(x + 2)$;

$$y = -x^2 - 2x$$

$a = 1 > 0$, so the parabola opens upward.

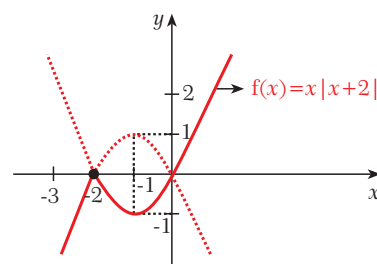
$$h = -\frac{-2}{2 \cdot (-1)} = -1,$$

$$k = f(-1) = -(-1)^2 - 2 \cdot (-1) = 1;$$

$V(-1, 1)$ is the vertex point.

$x = 0$; $y = 0$, $(0, 0)$ is the y -intercept.

$y = 0$; $-x^2 - 2x = 0$; $x = -2$, $x = 0$; $(-2, 0)$ and $(0, 0)$ are the x -intercepts.



d. $y = |2x^2 - 12x + 16| + 2$

First graph the function $y = |2x^2 - 12x + 16|$.

Then shift this graph 2 units up.

$a = 2 > 0$, so the parabola opens upward.

$$h = -\frac{-12}{2 \cdot 2} = 3,$$

$$k = f(h) = 2(3)^2 - 12 \cdot 3 + 16 = -2;$$

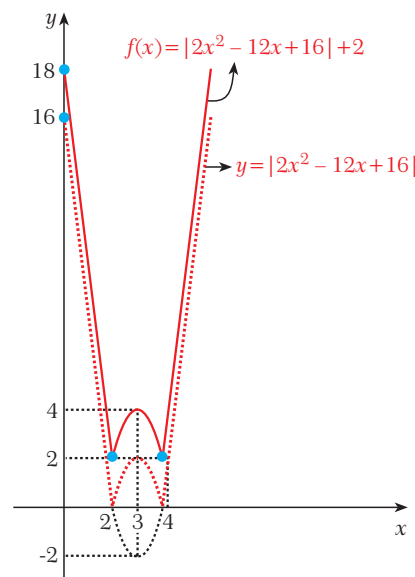
$V(3, -2)$ is the vertex point.

$x = 0$; $y = 16$, $(0, 16)$ is the y -intercept.

$$y = 0; 2x^2 - 12x + 16 = 0;$$

$$x = 2, x = 4;$$

$(2, 0)$ and $(4, 0)$ are the x -intercepts.

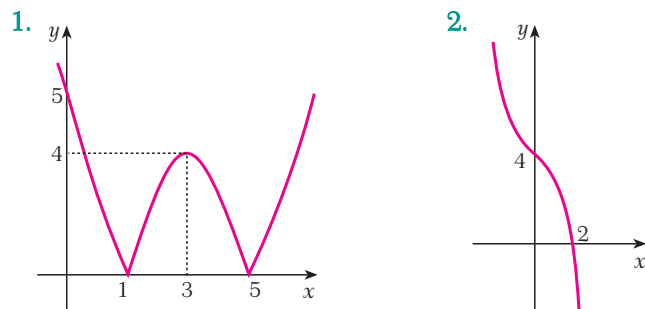


Check Yourself 6

Sketch the graph of each function.

1. $y = |x^2 - 6x + 5|$ 2. $y = -x|x| + 4$

Answers



C. EQUATION OF A PARABOLA

We have learned how to construct a parabola if we are given its equation. But if we are given the graph of a parabola, how can we write its equation?

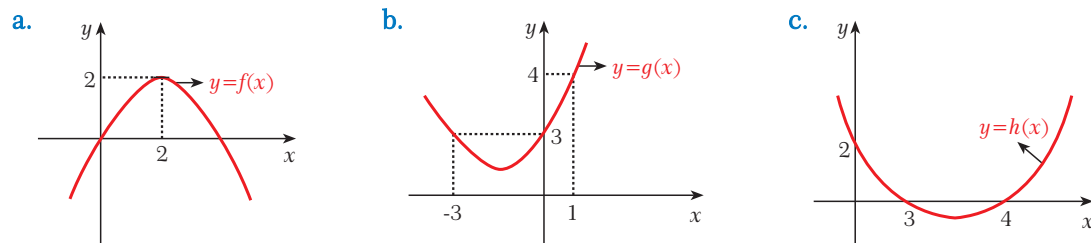
There are three different approaches to finding the equation of a parabola. The approach we use depends on the information we know.

1. If we know the vertex point of the parabola, $V(h, k)$ then we use the formula $y = a(x - h)^2 + k$ to write the equation of the parabola. We need to know another point on the parabola to write the equation.
2. If we know the x -intercepts of the parabola then we use the formula $y = a(x - x_1)(x - x_2)$ to write the equation of the parabola. We need to know another point on the parabola to write the equation.
3. If we know any three points on the parabola, then we use the formula $y = ax^2 + bx + c$ to write the equation of the parabola.

Example

12

Write the equation of each parabola.



Solution a. The vertex point of the parabola is $V(2, 2)$, so $y = a(x - 2)^2 + 2$.

Also, the parabola passes through the origin $(0, 0)$, so this point satisfies the equation of the parabola.

$$0 = a(0 - 2)^2 + 2 ; \quad 0 = 4a + 2 ; \quad a = -\frac{1}{2}$$

$$y = -\frac{1}{2}(x - 2)^2 + 2$$

$$y = -\frac{1}{2}x^2 + 2x$$

This is the equation of the parabola.

b. Let $y = ax^2 + bx + c$.

Since the points $(0, 3)$, $(-3, 3)$ and $(1, 4)$ are on the graph of the parabola, these points satisfy the equation of the parabola.

$$(0, 3), \quad 3 = c \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$(-3, 3), \quad 3 = 9a - 3b + 3 ; \quad 9a - 3b = 0$$

$$(1, 4), \quad 4 = a + b + 3 ; \quad a + b = 1$$

$$\begin{cases} 9a - 3b = 0 & (1) \\ a + b = 1 & (2) \end{cases}$$

$$\text{From (1) and (2),} \quad a = \frac{1}{4}, \quad b = \frac{3}{4}.$$

$$\text{So the equation is } y = \frac{1}{4}x^2 + \frac{3}{4}x + 3.$$

c. Let $y = a(x - 3)(x - 4)$

$(0, 2)$ is on the parabola.

$$2 = a(0 - 3)(0 - 4) ; \quad a = \frac{1}{6}$$

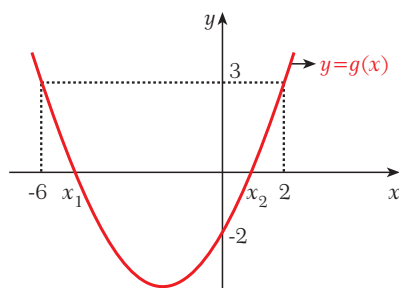
$$y = \frac{1}{6}(x - 3)(x - 4)$$

$$\text{So the equation is } y = \frac{1}{6}x^2 - \frac{7}{6}x + 2.$$

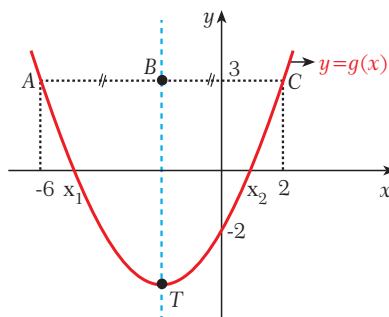
Example

13

x_1 and x_2 are the roots of the parabola below. Find the value of $x_1 + x_2$.



Solution



$$g(2) = g(-6) = 3$$

$AB = BC$, so the x -coordinate of the vertex point is

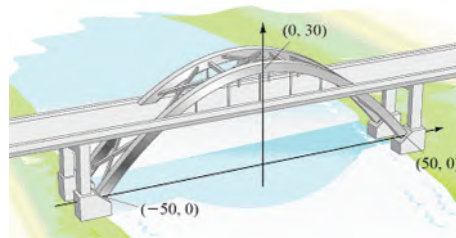
$$h = \frac{-6 + 2}{2} = -2.$$

$$h = -\frac{b}{2a}; \quad \frac{x_1 + x_2}{2} = -2; \quad x_1 + x_2 = -4.$$

Example

14

A parabolic concrete bridge support needs to pass through the points $(-50, 0)$, $(0, 30)$, and $(50, 0)$. Write the equation of the parabola formed by the bridge.



Solution Let $y = ax^2 + bx + c$. Since the points $(-50, 0)$, $(0, 30)$, and $(50, 0)$ are on the graph of the parabola, these points satisfy the equation of the parabola.

$$(0, 30), \quad c = 30$$

$$(-50, 0), \quad 0 = 2500a - 50b + 30; \quad 250a - 5b = -3 \quad (1)$$

$$(50, 0), \quad 0 = 2500a + 50b + 30; \quad 250a + 5b = -3 \quad (2)$$

From (1) and (2),

$$a = \frac{-3}{250}, \quad b = 0.$$

$$\text{So the equation is } y = -\frac{3}{250}x^2 + 30.$$

Example

15

The shape of the famous “Gateway to the West” arch, which is the spirit of the City of St. Lois can be modeled by a parabola.

The equation for the parabola is $y = -\frac{1}{150}x^2 + \frac{21}{5}x$.

- Sketch the graph of the arch's equation on a coordinate axis.
- What is the distance from one side of the arch to the other?



Solution

- $a = -\frac{1}{150} < 0$, so the parabola opens downward.

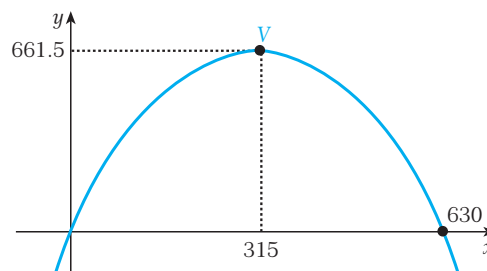
$$h = -\frac{\frac{21}{5}}{2 \cdot (-\frac{1}{150})} = 315,$$

$$k = f(315) = -\frac{1}{150}315^2 + \frac{21}{5} \cdot 315 = 661.5 \quad V(315, 661.5)$$

$x = 0$; $y = 0$ $(0, 0)$ is the y -intercept.

$$y = 0; \quad -\frac{1}{150}x^2 + \frac{21}{5}x = 0;$$

$x = 0$, $x = 630$; $(0, 0)$ and $(630, 0)$ are the x -intercepts.



- From the x -intercepts, we can see that the distance along the bridge is 630 m.

Example

16

A farmer has 120 m of fencing. He wants to put a fence around three sides of a rectangular plot of land, with the side of a barn forming the fourth side. Find the maximum area the farmer can enclose. What dimensions give this area?



Solution

Let x represent the width of the plot. Then, since there are 120 m of fencing,

$$x + x + \text{length} = 120; \quad \text{length} = 120 - 2x.$$

$$\text{The area of the plot is } A = (120 - 2x) \cdot x = 120x - 2x^2.$$

We can see that $a < 0$, so the graph of the function of the area has its maximum value at the vertex point.

To make the area as large as possible, let us therefore find the vertex of the graph of the function $A = -2x^2 + 120x$.

The x -coordinate of the vertex is $\frac{-b}{2a} = -\frac{120}{-4} = 30$. This is the width of the plot at its maximum area. Now we have the solution: $A = -2 \cdot 30^2 + 120 \cdot 30 = 1800 \text{ m}^2$ and the dimensions of the plot are 30 m by $(120 - 2 \cdot 30)\text{m}$, i.e. 30 m by 60 m.

Example**17**Find the maximum vertical distance d between the parabola and the line in the figure.

Solution The distance between the vertex point of the parabola and the midpoint of the AB line segment in the figure will be the maximum distance.

$$\text{So } h = -\frac{b}{2a} = \frac{-4}{-4} = 1$$

$$k = f(1) = 5, \quad V(1, 5).$$

Let us find the coordinates of point A and point B .

$$-2x^2 + 4x + 3 = x - 2$$

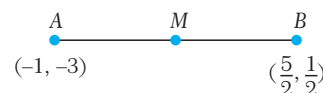
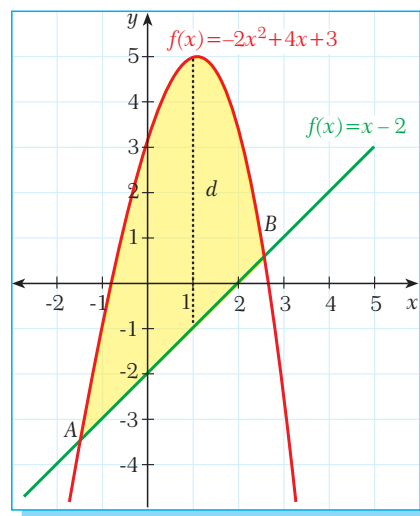
$$2x^2 - 3x - 5 = 0$$

$$(2x - 5)(x + 1) = 0; \quad x = \frac{5}{2}, \quad x = -1$$

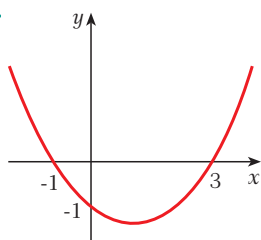
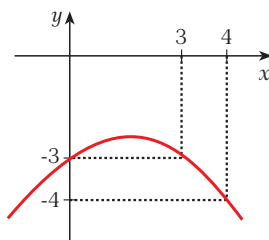
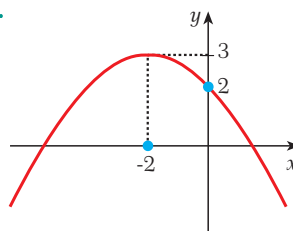
$$A(-1, -3) \text{ and } B\left(\frac{5}{2}, \frac{1}{2}\right)$$

$$M\left(\frac{-1 + \frac{5}{2}}{2}, \frac{-3 + \frac{1}{2}}{2}\right) = \left(\frac{3}{4}, \frac{-5}{4}\right)$$

$$\text{So } d = \sqrt{\left(\frac{3}{4} - 1\right)^2 + \left(\frac{-5}{4} - 5\right)^2} = \sqrt{\frac{1}{16} + \frac{625}{16}} = \sqrt{\frac{626}{16}} = \frac{\sqrt{626}}{4}.$$

**Check Yourself 7**

Write the equation of each parabola.

1.**2.****3.****Answers**

$$1. y = \frac{1}{3}x^2 - \frac{2}{3}x - 1 \quad 2. y = -\frac{1}{4}x^2 + \frac{3}{4}x - 3 \quad 3. y = -\frac{1}{4}x^2 - x + 2$$

The things of this world cannot be made known without a knowledge of mathematics.

PARABOLIC REFLECTORS

Parabolic curves are used in the design of lighting systems, telescopes, and radar antennas, mainly because of the reflective property you can see in the figures below.

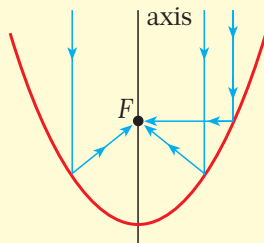


Figure a. A reflecting telescope:
light rays parallel to the
axis are concentrated at
the focus.

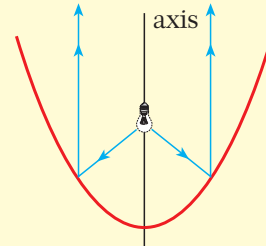


Figure b. A parabolic flashlight:
a light source at the focus
sends out beams of light
parallel to the axis.

Figure a shows the application of the reflective property of a parabola to create a reflecting telescope. The eyepiece of the telescope is placed at the focus F of a parabolic mirror. Light enters the telescope in rays that are parallel to the axis of the parabola. We know from physics that when light is reflected, the angle of incidence equals the angle of reflection. Hence, the parallel rays of light strike the parabolic mirror so that they all reflect through the focus, which means that all the parallel rays are concentrated at the eyepiece. This maximizes the light-gathering ability of the mirror.

Flashlights and automobile headlights (see Figure b) simply reverse this process. A light source is placed at the focus of a parabolic mirror. The light rays strike the mirror with an angle of incidence equal to the angle of reflection, and each ray is reflected along a path parallel to the axis. As a result, the light emits a light beam of parallel rays.

Radar utilizes both of these properties. First, a pulse is transmitted from the focus to a parabolic surface. As with a reflecting telescope, parallel pulses are transmitted in this way. The reflected pulses then strike the parabolic surface and are sent back to be received at the focus.



EXERCISES 3.1

A. Graphing $y = ax^2$

- Graph each set of functions in the same plane.
 - $y = 2x^2$, $y - 4x^2 = 0$, $2x^2 = 3y$
 - $y = -3x^2$, $y + 4x^2 = 0$, $-3x^2 = 2y$
 - $y = 3x^2$, $3y = x^2$, $2y + x^2 = 0$, $y + 2x^2 = 0$
- Determine whether each point lies on the graph of the function $4y + 5x^2 = 0$.
 - $\left(3, -\frac{45}{4}\right)$
 - $\left(\frac{4}{5}, 1\right)$
 - $(-2, -5)$
 - $(4, 10)$
- If the parabola $y = px^2$ passes through the following points, find p .
 - $(-3, 3)$
 - $\left(2, \frac{4}{5}\right)$
 - $(3, -7)$
 - $\left(\frac{4}{5}, -2\right)$
- Graph the function $y = -\frac{x^2}{2}$ if $|x| \leq 4$.

B. Graphing $y = ax^2 + bx + c$

- Find the coordinates of the vertex point of each function. Write the equation of the axis of symmetry and determine the maximum or minimum value of the function.
 - $y = \frac{3}{4}x^2$
 - $y = -3x^2$
 - $y = 3x^2 + 2$
 - $y = -x^2 - 1$
 - $y = -2x^2 + 5x$
 - $y = 0, 7x^2 + 0, 8x$
 - $y = 3x^2 - 4x + 3$
 - $y = -x^2 + x + 1$
 - $y = 3(x + 1)(x - 4)$
 - $y = -2(x + 4)^2$
 - $y = -(x + 3)^2 - 5$
 - $y = (x - 1)^2 + 3$

- The equation of the axis of symmetry of the parabola $y = 2x^2 + (m + 1)x - 4$ is $4x - 3 = 0$. Find the minimum value of the function.
- $-\frac{4}{3}$ is the minimum value of the function $y = 3x^2 + 2x - n$. Find the value of n .
- The vertex point of the parabola $y = -3x^2 - 2(2k + 5)x + 4$ is on the y -axis. Find k .
- Find the x - and y -intercepts of each function.
 - $y = -x^2 + 4x$
 - $y = (x - 3)^2$
 - $y = 8(x + 7)^2 + 4$
 - $y = x^2 - 8$
 - $y = 2x^2 + 5x + 3$
 - $y = -x^2 - 4x - 5$
 - $y = 16x^2 + 24x + 9$
 - $y = -\frac{2}{3}x^2 - 4$
- The function $y = -x^2 + (m - 2)x + m - 3$ is given. Find the values of m for which the parabola
 - does not cut the x -axis.
 - is tangent to the x -axis.
 - cuts the x axis at two distinct points.
- The parabola $y = mx^2 - (m + 1)x + 2m - 1$ passes through the point $A(2, 9)$. Find the value of m .

12. Sketch the graph of each function.

- a. $y = 3x^2 - 9$ b. $y = 2x^2 + 5$
 c. $y = -3(x + 4)^2$ d. $y = 2(x + 7)^2$
 e. $y = x^2 - 2x - 3$ f. $y = -x^2 - 4x$
 g. $y = x^2 - 4x + 1$ h. $y = x^2 - 5$
 i. $y = (x - 3)^2 + 4$ j. $y = (x + 3)(4 - x)$

13. Sketch the graph of the function

$$y = x^2 - 2x - 8 \text{ if } -1 < x < 5.$$

14. Sketch the graph of the function

$$y = -2x^2 + 4x + 1 \text{ if } |x - 1| < 2.$$

15. Find the minimum and the maximum values of the function $y = x^2 + 4x + 7$ if $x \in [-4, 1]$.

16. Find the minimum or the maximum value of the function if $m + n = 16$.

17. Find the minimum or the maximum value of the function $y = m^2 + n$ if $m - 2 = n$.

18. The parabola $y = 2ax^2 + bx - 3b$ passes through the points $(1, -3)$ and $(2, -5)$. Find the values of a and b .

19. Write the equation of the parabola which is symmetric to $y = 3(x + 1)^2$ with respect to the line $x = 2$.

20. Given $f(x) = -3(x - 1)^2 + 2$, show that $f(x) \leq f(1)$.

21. $f(x) = x^2 + 3mx + 4m$ and $f(x) \geq 0$ are given. Find m .

22. Sketch the graph of each function by shifting.

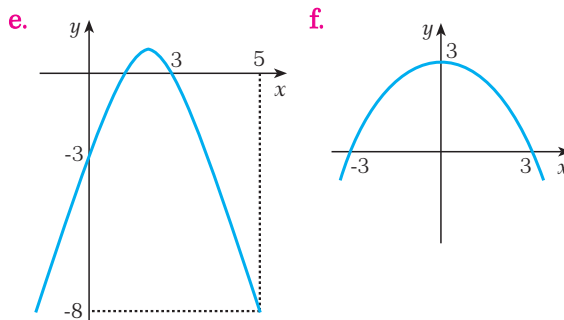
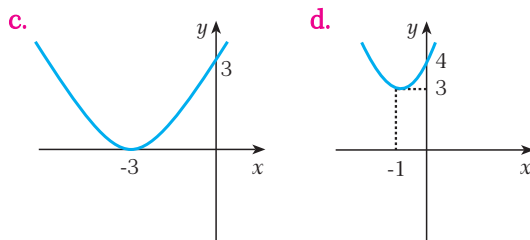
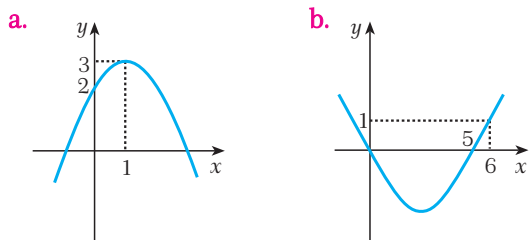
- a. $y = x^2 + 4$ b. $y = x^2 - 2$
 c. $y = -x^2 - 2$ d. $y = -(x + 5)^2$
 e. $y = (x - 4)^2$ f. $y = -(x - 3)^2$
 g. $y = -(x + 1)^2 - 2$ h. $y = (x - 2)^2 + 3$

23. Sketch the graph of each function.

- a. $y = |x^2 - 4x + 3|$ b. $y = -2x^2 + 4|x| + 1$
 c. $y = x|x - 2|$ d. $y = |x^2 - 6x + 8| - 1$

C. Equation of a Parabola

24. Write the equation of each parabola.



25. Write the equation of the parabola through each set of points.

a. $A(-1, 0)$, $B(0, 4)$, $C(1, 2)$

b. $A(0, -4)$, $B(2, 0)$, $C(4, -4)$

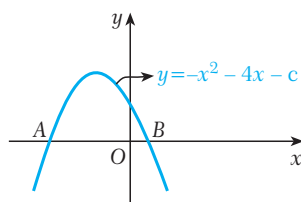
26. Write the equation of the parabola through each vertex point V and point A .

a. $V(0, -4)$, $A(3, 5)$ b. $V(2, 0)$, $A(4, 4)$

c. $V(-1, -4)$, $A(2, 5)$ d. $V\left(-\frac{1}{2}, -\frac{15}{4}\right)$, $A(0, -4)$

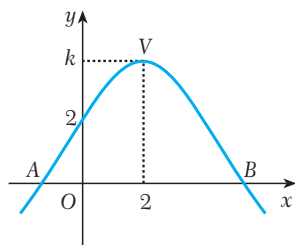
27. In the figure,
 $3 \cdot |OB| = |AO|$.

Find c .



28. In the figure,
✧ $|OB| = 5 \cdot |AO|$ and
 V is the vertex point
of the parabola.

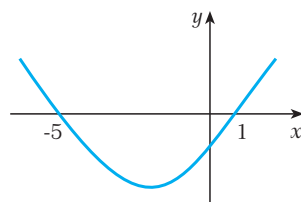
Find k .



29. The figure shows the
✧ graph of the function
 $y = ax^2 + bx + c$.

Find the value of

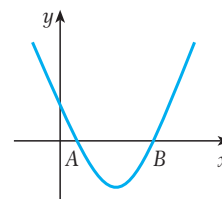
$a + b + c$.



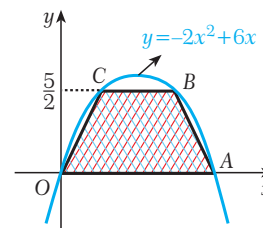
30. The figure shows the graph
✧ of the function

$y = mx^2 - 7mx + 5$.

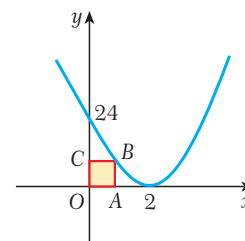
If $|AB| = 3$, find m .



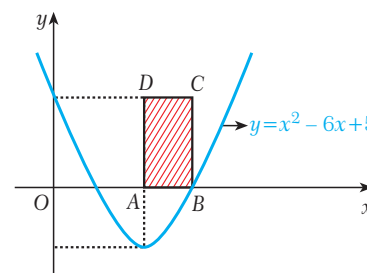
31. Find the area of the
✧ trapezoid $OABC$ in
the figure.



32. Find the area of the
✧ square $OABC$ in the figure.



33. Find the area
✧ of the
rectangle
 $ABCD$ in
the figure.



Mixed Problems

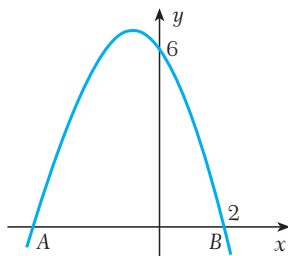
34. x_1 and x_2 are the x -intercepts of the function $y = mx^2 - 4x - 2m - 2$. If $x_1 \cdot x_2 < 0$, find the possible values of m .

35. The perimeter of a rectangle is 32 cm. Find its maximum possible area.

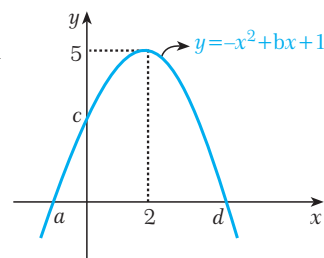
36. The function $y = 5x^2 - 3x + k$ is tangent to the line $y = 2$. Find k .

37. 3 is the minimum value of the function $y = mx^2 - 2mx + 2m + 1$. Find m .

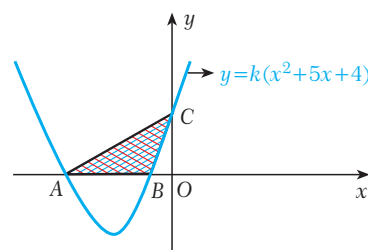
38. The figure shows the graph of the function $y = -x^2 + bx + c$. Find the coordinates of the point A.



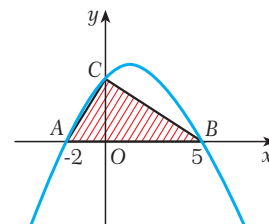
39. The figure shows the graph of the function $y = -x^2 + bx + 1$. Find the value of $a + b + c + d$.



40. In the figure, the area of the triangle ABC is 6 cm^2 . Find k .



41. In the figure, the maximum value of the function is $\frac{49}{4}$. Find the area of the triangle ABC.





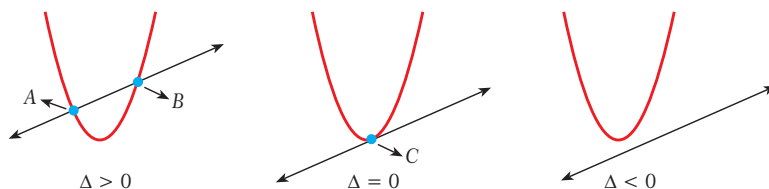
GRAPHICAL SOLUTION OF INEQUALITIES

A. RELATIVE POSITION OF A PARABOLA AND A LINE IN A PLANE

Consider the quadratic function $y = f(x) = ax^2 + bx + c$ and the linear function $y = g(x) = mx + n$. In order to determine whether they have an intersection or not, let us consider the common points of the two functions.

The equation $f(x) = g(x)$ gives us $f(x) - g(x) = 0$ in standard form. Let us examine the discriminant of this equation and its effects on the common points.

1. If $\Delta > 0$, the line intersects the parabola at two distinct points.
2. If $\Delta = 0$, the line is tangent to the parabola.
3. If $\Delta < 0$, the line does not intersect the parabola.



Example

18

The functions $f(x) = x^2 - mx + 3$ and $g(x) = x - 6$ are given. Find the values of m for which

- a. the line does not intersect the parabola.
- b. the line is tangent to the parabola.
- c. the line intersects the parabola at two distinct points.

Solution $f(x) = g(x)$

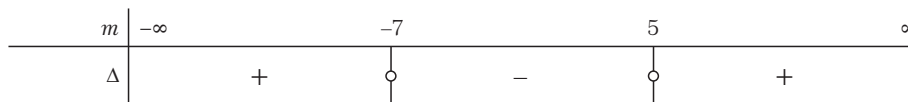
$$x^2 - mx + 3 = x - 6$$

Writing this in standard form gives us $x^2 - (m + 1)x + 9 = 0$.

$$\Delta = (-(m + 1))^2 - 4 \cdot 1 \cdot 9 = m^2 + 2m - 35$$

$$m^2 + 2m - 35 = 0; \quad m_1 = -7, \quad m_2 = 5$$

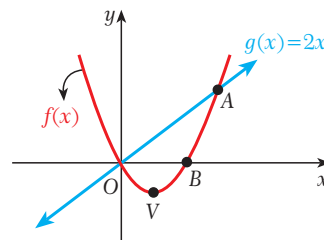
Let us consider the sign of Δ .



- a. If $m \in (-7, 5)$, then the line does not intersect the parabola.
- b. If $m = -7$ or $m = 5$, then the line is tangent to the parabola.
- c. If $m \in (-\infty, -7) \cup (5, \infty)$, then the line intersects the parabola at two distinct points.

Example**19**

$V(2, -4)$ is the vertex of the parabola given in the figure. Find the coordinates of the point A.

**Solution**

$V(2, -4)$ is on the parabola. Therefore, point B is $B(4, 0)$ because of the symmetry of a parabola. Therefore, the x -intercepts of the parabola are 0 and 4.

$$\text{So } f(x) = a(x - 0)(x - 4).$$

$(2, -4)$ is on the parabola, and so $f(2) = -4$.

$$-4 = a(2 - 0)(2 - 4); \quad a = 1$$

$$\text{This gives } f(x) = x^2 - 4x.$$

To find the common solutions, we can now use the equation $f(x) = g(x)$.

$$x^2 - 4x = 2x$$

$$\text{In standard form, } x^2 - 6x = 0; \quad x = 0 \text{ or } x = 6.$$

So, 0 and 6 are the x -coordinates of the points O and A respectively.

$$g(x) = 2x. \text{ If } x = 6, \quad g(x) = 12.$$

Therefore, point A is $(6, 12)$.

Example**20**

The parabola $f(x) = (m - 1)x^2 - (3m - 1)x + 4m - 1$ is tangent to the line $y = 3$. Find m .

Solution

If the parabola is tangent to the line $y = 3$, then the ordinate of the vertex point of the parabola, k , is 3.

$$k = \frac{4ac - b^2}{4a};$$

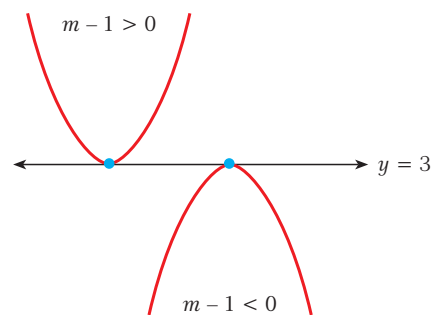
$$3 = \frac{4 \cdot (m - 1) \cdot (4m - 1) - (3m - 1)^2}{4(m - 1)}, \quad m \neq 1$$

$$3 = \frac{16m^2 - 20m + 4 - 9m^2 + 6m - 1}{4m - 4}$$

$$12m - 12 = 7m^2 - 14m + 3$$

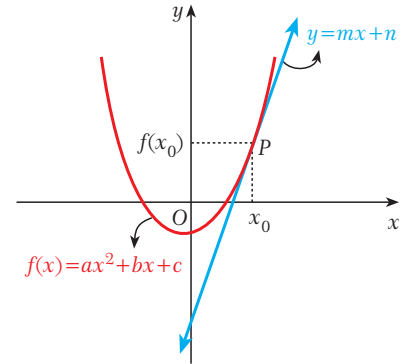
$$7m^2 - 26m + 15 = 0$$

$$\text{So, } m = 3 \text{ or } m = \frac{5}{7}.$$



Note

The slope of the line which is tangent to the parabola $y = f(x) = ax^2 + bx + c$ at the point $P(x_0, f(x_0))$ is $2ax_0 + b$.



Check Yourself 8

Find the intersection point(s) of the functions $y = x^2 + 3x - 3$ and $y = 2x - 1$.

Answer

$(-2, -5), (1, 1)$

B. RELATIVE POSITION OF TWO PARABOLAS IN A PLANE

We have seen how to determine the relative position of a line and a parabola in the same plane. Let us now look at the relative position of two parabolas. Consider the quadratic functions $f(x) = ax^2 + bx + c$ and $g(x) = Ax^2 + Bx + C$. In order to determine whether they have an intersection or not, let us consider the common points of two functions.

$$\begin{aligned} \text{In the equation } f(x) &= g(x), & ax^2 + bx + c &= Ax^2 + Bx + C \\ & & (a - A)x^2 + (b - B)x + (c - C) &= 0. \end{aligned}$$

This is an equation in standard quadratic form.

1. If $a - A \neq 0$, then we examine the discriminant.
 - a. If $\Delta > 0$, the parabolas intersect each other at two distinct points.
 - b. If $\Delta = 0$, the parabolas are tangent to each other.
 - c. If $\Delta < 0$, the parabolas do not intersect each other.
2. If $a - A = 0$ and $b - B \neq 0$, the parabolas intersect each other at one point.
3. If $a - A = 0$, $b - B = 0$, and $c - C = 0$, the parabolas are coincident.

Example

21

The parabolas $f(x) = 3x^2 + 3x + m + 1$ and $g(x) = 2x^2 - x + 2m$ are tangent to each other. Find the coordinates of the point of tangency.

Solution

Let us write the equation $f(x) = g(x)$.

$$3x^2 + 3x + m + 1 = 2x^2 - x + 2m$$

In standard form, $x^2 + 4x + 1 - m = 0$.

Since the parabolas are tangent to each other, $\Delta = 0$.

$$\Delta = 16 - 4 + 4m = 0 ; \quad m = -3$$

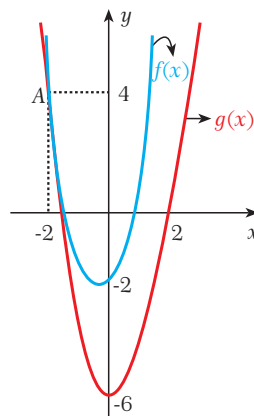
$$x^2 + 4x + 1 - (-3) = 0 ; \quad x^2 + 4x + 4 = 0$$

$$(x + 2)^2 = 0 ; \quad x = -2$$

-2 is therefore the abscissa of the point of tangency.

$$f(-2) = 12 - 6 - 3 + 1 = 4$$

So A(-2, 4) is the point of tangency.



Example

22

The function $f(x) = mx^2 + 2mx + m - 1$ is given. If m is a variable parameter, find the coordinates of the fixed point that all parabolas of the function $f(x)$ pass through.

Solution

Let us look at some values for m .

$$m = 1 ; \quad y_1 = x^2 + 2x$$

$$m = 2 ; \quad y_2 = 2x^2 + 4x + 1$$

At the fixed point, $y_1 = y_2$.

$$x^2 + 2x = 2x^2 + 4x + 1$$

$$x^2 + 2x + 1 = 0$$

$(x + 1)^2 = 0 ; \quad x = -1$ is the x -coordinate of the fixed point.

$$x = -1 ; \quad y_1 = 1 - 2 = -1$$

So the fixed point is $(-1, -1)$.

Check Yourself 9

Find the intersection points of the functions $y = x^2 + 2x - 5$ and $y = 2x^2 - x - 3$.

Answer

$(1, -2), (2, 3)$

C. GRAPHICAL SOLUTION OF INEQUALITIES IN TWO VARIABLES



The graph of an inequality in two variables is the set of all points whose coordinates satisfy the inequality.

We can solve linear and quadratic inequalities in two variables using a graph. To solve such inequalities, we examine the regions which are separated by a line and a parabola in the graph.

1. Linear Inequalities

Definition

linear inequality in two variables

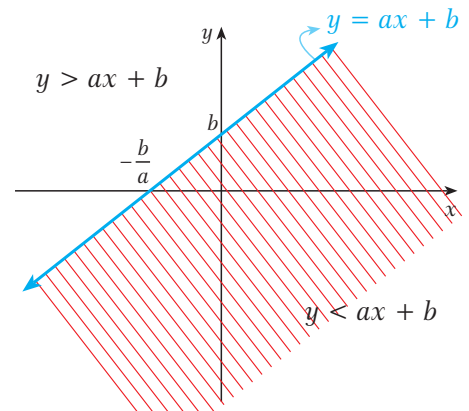
An inequality in one of the forms

$$ax + b > y, \quad ax + b < y$$

$$ax + b \geq y, \quad ax + b \leq y$$

for the real numbers a and b , $a \neq 0$, is called a **linear inequality in two variables**.

A line divides the plane into three regions: the line itself and the two half-planes on either side of the line. The graphs of linear inequalities in two variables are therefore regions in the real number plane and may include a boundary line. The boundary line for the inequality is the graph of the equation $y = ax + b$.



The **boundary** for the graph is found by replacing the inequality symbol with an equality sign and graphing the resulting equation.

To graph a linear inequality in two variables, follow the steps.

1. Graph the linear equation $y = ax + b$. This line becomes the boundary line for the graph. Draw the boundary line using a dashed line if the inequality symbol is $<$ or $>$, or a solid line if the inequality symbol is $\leq$ or $\geq$.
2. Choose a test point which is not on the boundary line and substitute it into the inequality.
3. Shade the region which includes the test point if the resulting inequality is true, and shade the region which does not contain the test point if the resulting inequality is false.

Example

23

Graph the inequalities.

a. $2x - 3y + 6 < 0$

b. $2y + 3x + 4 \leq 0$

Solution

a. $2x - 3y + 6 < 0$

$$2x + 6 < 3y; \quad y > \frac{2}{3}x + 2$$

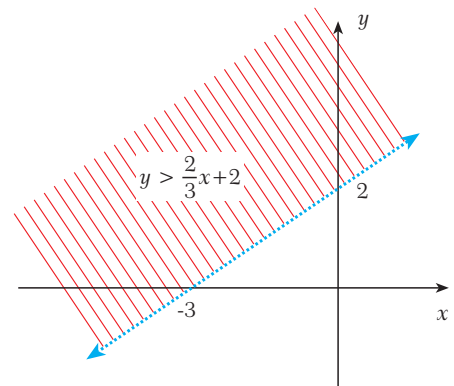
$y = \frac{2}{3}x + 2$ is the boundary line and we draw the line dashed since the inequality symbol is $>$.

The x - and y -intercepts are $(-3, 0)$ and $(0, 2)$.

Test the point $(0, 0)$ in $y > \frac{2}{3}x + 2$:

$$0 > \frac{2}{3} \cdot 0 + 2; \quad 0 > 2 \text{ (false).}$$

So we shade the region which does not contain the test point.



b. $2y + 3x + 4 \leq 0$

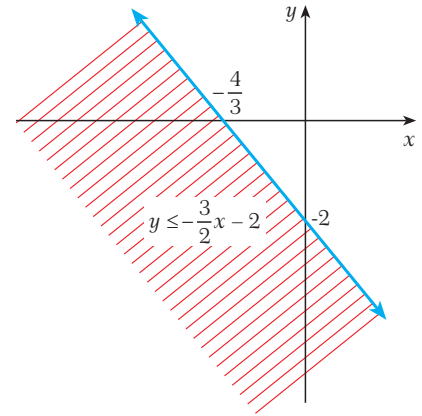
$$2y \leq -3x - 4; \quad y \leq -\frac{3}{2}x - 2$$

$y = -\frac{3}{2}x - 2$ is the boundary line and the line is solid since the inequality symbol is $\leq$. The x - and y -intercepts are $\left(-\frac{4}{3}, 0\right)$ and $(0, -2)$.

Test the point $(0, 0)$ in $y \leq -\frac{3}{2}x - 2$.

$$0 \leq -\frac{3}{2} \cdot 0 - 2; \quad 0 \leq -2 \text{ (false).}$$

So we shade the region which does not contain the test point.



Check Yourself 10

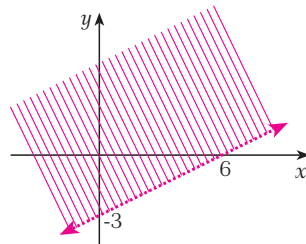
Graph the inequalities.

1. $x - 2y < 6$

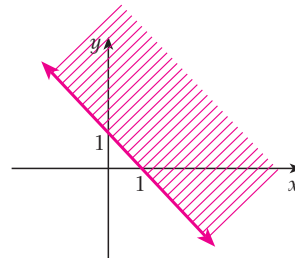
2. $x + y - 1 \geq 0$

Answers

1.



2.



2. Quadratic Inequalities

Definition

quadratic inequality in two variables

An inequality in one of the forms

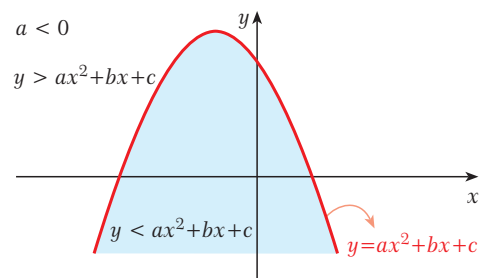
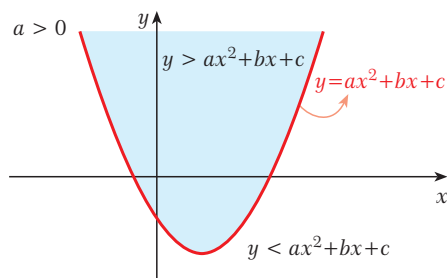
$$ax^2 + bx + c > y, \quad ax^2 + bx + c < y,$$

$$ax^2 + bx + c \geq y, \quad ax^2 + bx + c \leq y,$$

for the real numbers a, b and c , $a \neq 0$ is called a **quadratic inequality in two variables**.

A parabola divides a plane into three regions:

1. The parabola itself.
2. The interior region of the parabola.
3. The exterior region of the parabola.



To graph a quadratic inequality in two variables, follow the steps.

1. Graph the parabola $y = ax^2 + bx + c$. The parabola is the boundary of the graph. Draw the parabola as a dashed line if the inequality symbol is $<$ or $>$, and as a solid line if the inequality symbol is $\leq$ or $\geq$.
2. Choose a test point which is not on the boundary and substitute it into the inequality.
3. Shade the region which includes the test point if the resulting inequality is true and shade the region which does not contain the test point if the resulting inequality is false.

Example

24

Sketch the graph of each inequality.

a. $y \leq (2x + 1)(x - 3)$

b. $y < -(x + 2)^2 - 1$

c. $-x^2 + 4x + 5 \leq 0$

d. $x^2 + 6x + 9 > 0$

Solution

a. $y \leq (2x + 1)(x - 3)$

Graph the parabola $y = (2x + 1)(x - 3)$, $y = 2x^2 - 5x - 3$.

$a = 2 > 0$, so the parabola opens upward.

$$h = -\frac{b}{2a} = \frac{5}{4},$$

$$k = f(h) = 2\left(\frac{5}{4}\right)^2 - 5 \cdot \frac{5}{4} - 3 = -\frac{49}{8};$$

$V\left(\frac{5}{4}, -\frac{49}{8}\right)$ is the vertex point.

$x = 0$; $y = -3$, $(0, -3)$ is the y -intercept.

$$y = 0; \quad 2x^2 - 5x - 3 = 0;$$

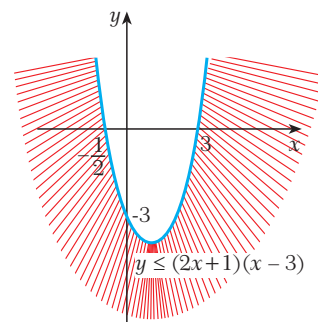
$x = -\frac{1}{2}$, $x = 3$; $\left(-\frac{1}{2}, 0\right)$ and $(3, 0)$ are the x -intercepts.

The parabola will be solid since the inequality symbol is $\leq$.

Test the point $(0, 0)$ in $y \leq (2x + 1)(x - 3)$:

$$0 \leq (2 \cdot 0 + 1)(0 - 3); \quad 0 \leq -3 \text{ (false).}$$

So we shade the exterior region which does not contain the test point.



b. $y < -(x + 2)^2 - 1$

Graph the parabola $y = -(x + 2)^2 - 1$, $y = -x^2 - 4x - 5$.

$a = -1 < 0$, so the parabola opens downward.

$$h = -\frac{-4}{2 \cdot (-1)} = -2, \quad k = f(h) = -(-2)^2 - 4 \cdot (-2) - 5 = -1, \text{ so}$$

$V(-2, -1)$ is the vertex point.

$x = 0$; $y = -5$, so $(0, -5)$ is the y -intercept.

$$y = 0; \quad -x^2 - 4x - 5 = 0;$$

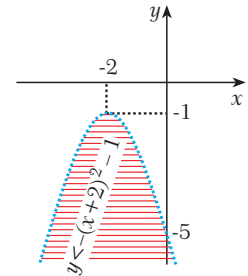
$$\Delta = (-4)^2 - 4(-1)(-5) = -4; \quad \Delta < 0.$$

So the parabola is below the x -axis.

The parabola line will be dashed since the inequality symbol is $<$.

Test the point $(0, 0)$ in $y < -(x + 2)^2 - 1$. $0 \leq -(0 + 2)^2 - 1$; $0 \leq -5$ (false).

So we shade the interior region which does not contain the test point.



c. $-x^2 + 4x + 5 \leq 0$

Graph the parabola $y = -x^2 + 4x + 5$.

$a = -1 < 0$, so the parabola opens downward.

$$h = -\frac{4}{2 \cdot (-1)} = 2, \quad k = f(h) = -(2)^2 + 4 \cdot (2) + 5 = 9, \text{ so}$$

$V(2, 9)$ is the vertex point.

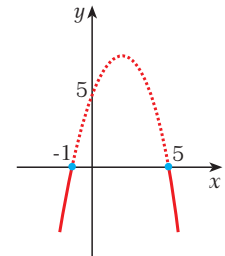
$x = 0$; $y = 5$, so $(0, 5)$ is the y -intercept.

$$y = 0; \quad -x^2 + 4x + 5 = 0;$$

$x = -1$, $x = 5$, so $(-1, 0)$ and $(5, 0)$ are the x -intercepts.

So the part of the parabola below the x -axis is the solution.

$$x \in (-\infty, -1] \cup [5, \infty).$$



d. $x^2 + 6x + 9 > 0$

Graph the parabola $y = x^2 + 6x + 9$.

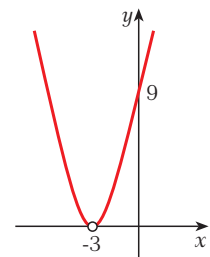
$$y = (x + 3)^2$$

$$h = -3, \quad k = 0, \text{ so}$$

$V(-3, 0)$ is the vertex point.

$(0, 9)$ is the y -intercept.

$$x \in (-\infty, -3) \cup (-3, \infty).$$



Check Yourself 11

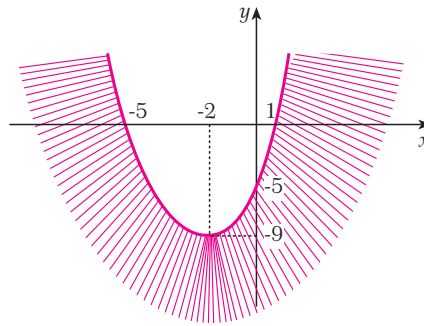
Sketch the graph of each inequality.

1. $y \leq x^2 + 4x - 5$

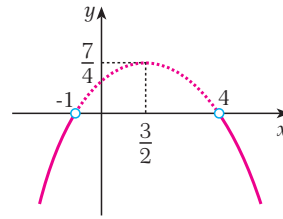
2. $-x^2 + 3x + 4 < 0$

Answers

1.



2.



D. GRAPHICAL SOLUTION OF SYSTEMS OF INEQUALITIES

A system of inequalities in two variables is made up of two or more inequalities. A solution of the system is an ordered pair that satisfies all of the inequalities. The solution set is the set of all ordered pairs that satisfy all of the inequalities. To solve a system of inequalities, we first solve the inequalities individually. The solution to the system is the region where all of the inequalities overlap on a graph.

Example

25

Sketch the graphs of the inequality systems.

a.
$$\begin{cases} x + y + 1 \geq 0 \\ 2x - y + 2 > 0 \end{cases}$$

b.
$$\begin{cases} xy > 0 \\ y \leq (x - 2)(x - 3) \end{cases}$$

c.
$$\begin{cases} x + 2y - 2 < 0 \\ y - x^2 + 4 \geq 0 \end{cases}$$

d.
$$\begin{cases} x < 1 \\ y < 3 \\ y < (x + 1)(3 - x) \end{cases}$$

e.
$$\begin{cases} y > 2x(x + 2) \\ y \leq -2x^2 + 8 \end{cases}$$

Solution

a.
$$\begin{cases} y \geq -x - 1 \\ y < 2x + 2 \end{cases}$$

$$y = -x - 1$$

The x -intercept is $(-1, 0)$.

The y -intercept is $(0, -1)$.

The line is solid.

Test $(0, 0)$ in $y \geq -x - 1$:

$$0 \geq -1 \quad (\text{true}).$$

$$y = 2x + 2$$

The x -intercept is $(-1, 0)$.

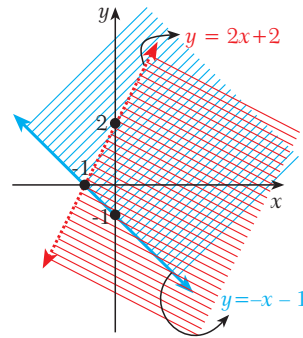
The y -intercept is $(0, 2)$.

The line is dashed.

Test $(0, 0)$ in $y < 2x + 2$:

$$0 < 2 \quad (\text{true}).$$

Shade the regions which include the test points:



The overlapping region is the solution of the system.

b.
$$\begin{cases} y < -\frac{1}{2}x + 1 \\ y \geq x^2 - 4 \end{cases}$$

$$y = -\frac{1}{2}x + 1$$

The x -intercept is $(2, 0)$.

The y -intercept is $(0, 1)$.

The line is dashed.

Test $(0, 0)$ in $y < -\frac{1}{2}x + 1$:

$$0 < 1 \quad (\text{true}).$$

Shade the regions which include the test points:

$$y = x^2 - 4$$

The x -intercept is $(-2, 0)$ and $(2, 0)$.

The y -intercept is $(0, -4)$.

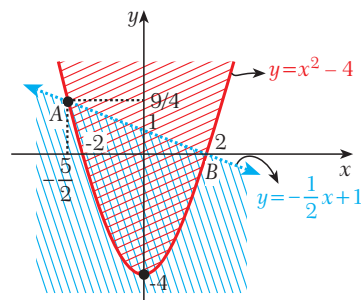
The parabola opens upward.

$V(-4, 0)$ is the vertex point.

The line is solid.

Test $(0, 0)$ in $y \geq x^2 - 4$:

$$0 \geq -4 \quad (\text{true}).$$



The overlapping region is the solution of the system.

c.
$$\begin{cases} y > 2x(x + 2) \\ y \leq -2x^2 + 8 \end{cases}$$

$$y = 2x^2 + 4x$$

The x -intercepts are $(0, 0)$ and $(-2, 0)$.

The y -intercept is $(0, 0)$.

The parabola opens upward.

$V(-1, -2)$ is the vertex point.

The parabola is dashed.

Test $(0, 1)$ in $y > 2x(x + 2)$:

$$1 > 0 \quad (\text{true}).$$

Shade the regions which include the test points:

$$y = -2x^2 + 8$$

The x -intercept are $(-2, 0)$ and $(2, 0)$.

The y -intercept is $(0, 8)$.

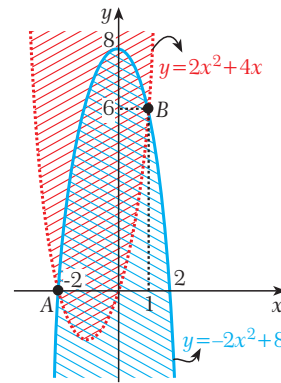
The parabola opens downward.

$V(0, 8)$ is the vertex point.

The parabola is solid.

Test $(0, 1)$ in $y \leq -2x^2 + 8$:

$$1 \leq 8 \quad (\text{true}).$$



The overlapping region is the solution of the system.

d.
$$\begin{cases} xy > 0 \\ y < (x - 2)(x - 3) \end{cases}$$

$xy > 0$ is true if x and y are both positive or negative.

So we can shade the first and the third quadrants as solutions.

The x -axis and y -axis will be dashed since the inequality symbol is $>$.

$$y = (x - 2)(x - 3); \quad y = x^2 - 5x + 6$$

The x -intercepts are $(2, 0)$ and $(3, 0)$.

The y -intercept is $(0, 6)$.

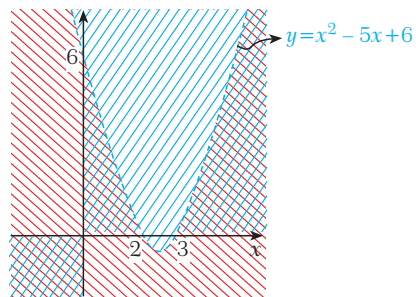
The parabola opens upward.

$V\left(\frac{5}{2}, -\frac{1}{4}\right)$ is the vertex point.

The parabola is solid.

Test $(0, 1)$ in $y < (x - 2)(x - 3)$: $1 < 6$ (true).

Shade the exterior region of the parabola:



The overlapping region is the solution of the system.

e.
$$\begin{cases} x < 1 \\ y < 3 \\ y < (x + 1)(3 - x) \end{cases}$$

Graph the line $x = 1$. The line is dashed. Shade the part on the left of the line.

Graph the line $y = 3$. The line is dashed. Shade the part below the line.

$$y = -x^2 + 2x + 3$$

The x -intercepts are $(-1, 0)$ and $(3, 0)$.

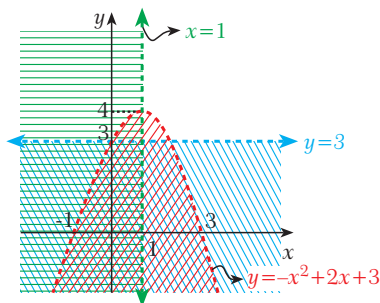
The y -intercept is $(0, -3)$.

The parabola opens downward. $V(1, 4)$ is the vertex point.

The parabola is dashed.

Test $(0, 1)$ in $y < (x + 1)(3 - x)$: $0 < 3$ (true).

Shade the interior region of the parabola.

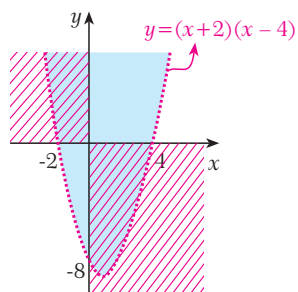


The overlapping region is the solution of the system.

Check Yourself 12

Sketch the graph of the system of inequalities $\begin{cases} x \cdot y < 0 \\ y > (x + 2)(x - 4) \end{cases}$.

Answer



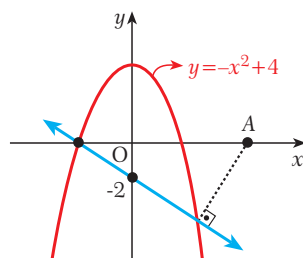
A mathematician, like a painter or a poet, is a maker of patterns. If his patterns are more permanent than theirs, it is because they are made with ideas.

EXERCISES 3.2

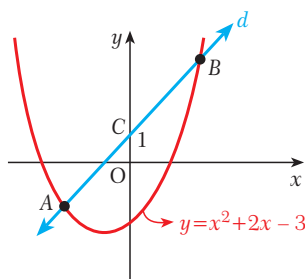
A. Relative Position of a Parabola and a Line in a Plane

- The line $f(x) = 2x + 3$ and the parabola $g(x) = -x^2 + mx + 4$ are given. Find the values of m for which the line
 - does not intersect the parabola.
 - is tangent to the parabola.
 - intersects the parabola at two distinct points.

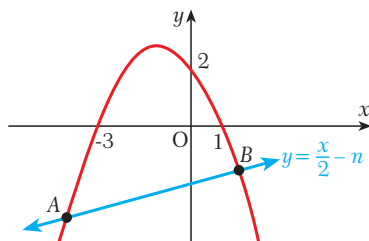
- Find the coordinates of the point A in the figure.



- In the figure, $|AC| = |BC|$. Find the equation of the line d .

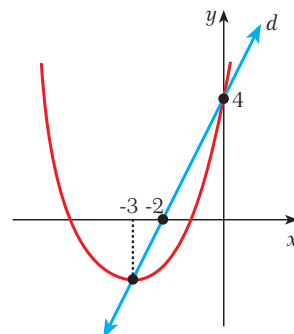


- Find the sum of the x -coordinates of the points A and B in the figure.

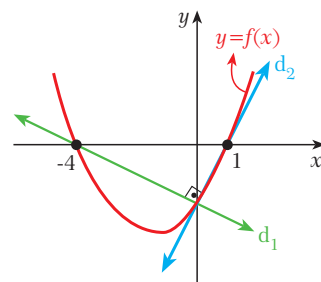


- The line $y + 4b - bx = 1$ is tangent to the parabola $y = x^2 - (b + 2)x + 10$. Find b .

- In the figure, line d passes through the vertex point of the parabola. Find the equation of the parabola.



- In the figure, $d_1 \perp d_2$. Find $f(2)$.

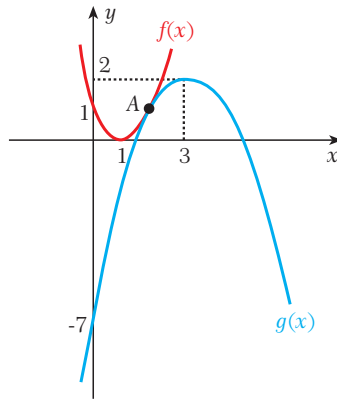


- The parabola $y = (m + 1)x^2 - (3m + 1)x + m - 1$ is tangent to the line $y = -2$. Find m .

- ☆☆ The slope of the line which is tangent to the function $f(x) = ax^2 + bx + c$ at the point $(x_0, f(x_0))$ is $2ax_0 + b$. Find the equation of the line which is tangent to the parabola $y = 3x^2 + 6x + 4$ at the point $x = 1$.

B. Relative Position of Two Parabolas in a Plane

10. In the figure, the parabolas of the functions $f(x)$ and $g(x)$ are tangent to each other at the point A. Find the coordinates of the point A.

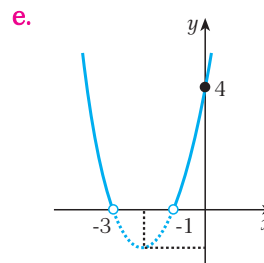
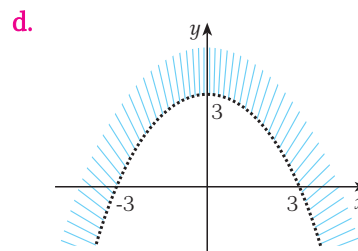
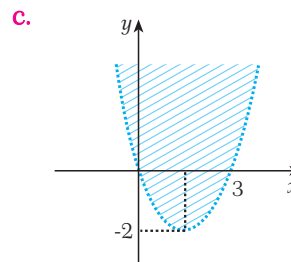
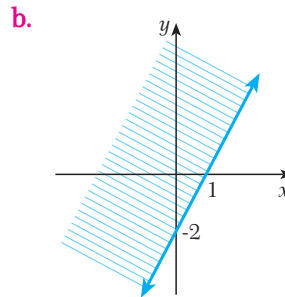
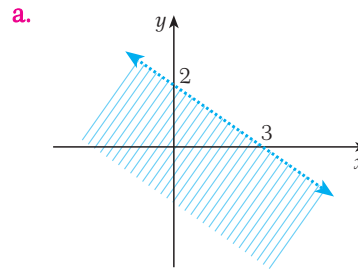


11. The parabolas $y = 3x^2 + 4x + m$ and $y = 2x^2 + 5x + n$ intersect each other at two distinct points. Find the sum of the x -coordinates of the intersection points.
12. The function $f(x) = mx^2 + 2mx + m - 1$ is given. If m is a variable parameter, find the coordinates of the fixed point that all parabolas of the function $f(x)$ pass through.

C. Graphical Solution of Inequalities in Two Variables

13. Show all the points which satisfy each inequality in the analytic plane.
- a. $2x + 3y + 4 < 0$ b. $x - y - 4 \leq 0$
 c. $3x + y - 5 \geq 0$ d. $y < 3x + 4$
14. Sketch the graphs of the solutions of the inequalities in the analytic plane.
- a. $y > -(3 - x)^2 + 4$ b. $y \leq (x + 2)(3x - 4)$
 c. $y > x^2 + x + 1$ d. $y < -x^2 + 8$
 e. $(x + 3)(2x - 5) > 0$ f. $-4x^2 + 20x - 25 \geq 0$

15. Express each shaded region as an inequality.



D. Graphical Solution of Systems of Inequalities

16. Sketch the graph of the solution of each inequality system in the analytic plane.

a.
$$\begin{cases} x + 2y + 3 < 0 \\ 3x - y + 1 \geq 0 \end{cases}$$

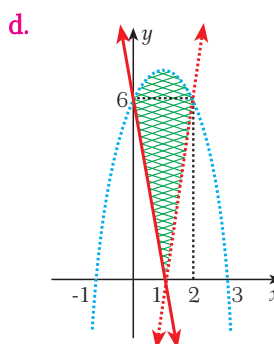
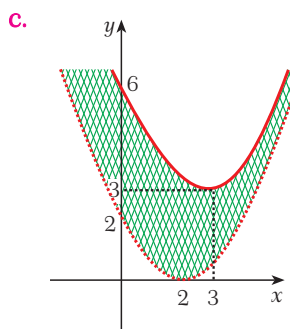
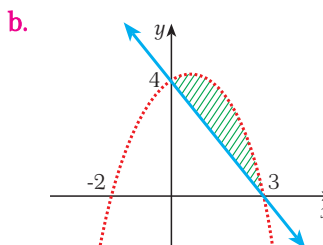
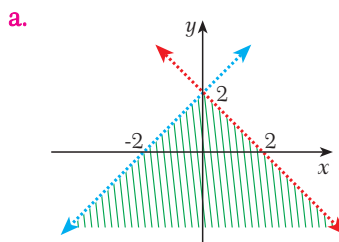
b.
$$\begin{cases} y - x + 1 > 0 \\ y < -(x + 1)^2 + 1 \end{cases}$$

c.
$$\begin{cases} y + x^2 - 4x + 3 \leq 0 \\ y - x^2 + 4 > 0 \end{cases}$$

d.
$$\begin{cases} y < -x + 2 \\ y < x + 2 \\ y > x^2 - 4 \end{cases}$$

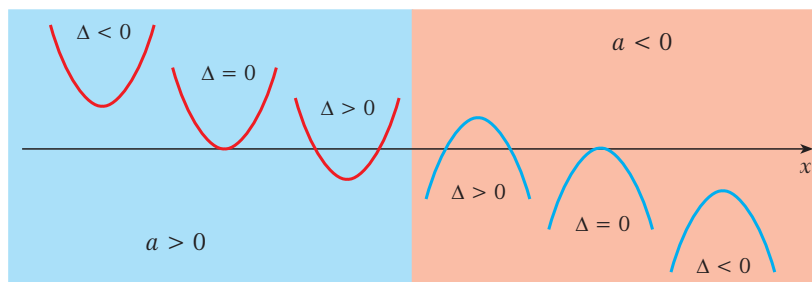


17. Express each shaded region as a system of inequalities.



CHAPTER 3 SUMMARY

- A quadratic function is described by the standard equation $f(x) = ax^2 + bx + c$.
- The graph of a quadratic function is called a parabola. The lowest or highest point of the parabola is called its vertex $V(h, k)$, where $h = -\frac{b}{2a}$ and $k = f(h)$.
- The axis of symmetry is the line that passes through the vertex of a parabola with equation $x = h = -\frac{b}{2a}$.
- When $a > 0$, the parabola opens upward, resulting in a vertex that is a minimum.
When $a < 0$, the parabola opens downward, resulting in a vertex that is a maximum.
- As the absolute value of a increases, the parabola becomes narrower.
When $a = 0$, the graph changes into a line since the x^2 term is eliminated.
- In the equation $y = f(x) = ax^2 + bx + c$, the value c is the y -intercept of the graph. In other words, the graph passes through the point $(0, c)$.
 1. If $\Delta > 0$, the parabola cuts the x -axis at two distinct points, $(x_1, 0)$, $(x_2, 0)$.
 2. If $\Delta = 0$, the parabola is tangent to the x -axis at the point $(x_1, 0)$, $(x_1 = x_2)$.
 3. If $\Delta < 0$, the parabola does not cut the x -axis.



- To sketch the parabola $y = ax^2 + bx + c$, follow the steps.
 1. Check a .
 2. Find the coordinates of the vertex point, $V(h, k)$.
 3. Find the x - and y -intercepts.
 4. Graph the parabola.
- Let $y = f(x)$ be a function.
Vertical shifting:

$$y = f(x) + k \quad \begin{cases} k > 0 & \text{shift graph of } y = f(x) \text{ up } k \text{ units.} \\ k < 0 & \text{shift graph of } y = f(x) \text{ down } |k| \text{ units.} \end{cases}$$

Horizontal shifting:

$$y = f(x + h) \quad \begin{cases} h > 0 & \text{shift the graph of } y = f(x) \text{ left } h \text{ units.} \\ h < 0 & \text{shift the graph of } y = f(x) \text{ right } |h| \text{ units.} \end{cases}$$

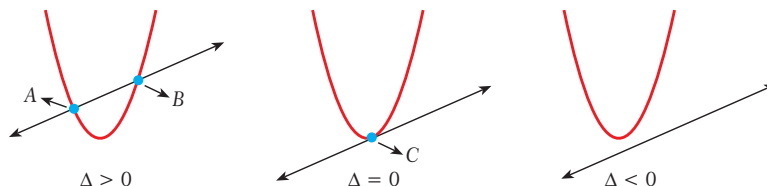
Reflection:

$$y = -f(x) \quad \text{Reflect the graph of } y = f(x) \text{ in the } x \text{ axis.}$$

Horizontal and vertical shifting:

$$y = f(x + h) + k. \text{ Shift the graph of } y = f(x) \text{ up or down } |k| \text{ units and left or right } |h| \text{ units.}$$

- There are three different approaches to writing the equation of a parabola.
 - If we know the vertex point of the parabola, $V(h, k)$, then we use the formula $y = a(x - h)^2$ to write the equation of the parabola. We need to know another point on the parabola to write the equation.
 - If we know the x -intercepts of the parabola, then we use the formula $y = a(x - x_1)(x - x_2)$ to write the equation of the parabola. We need to know another point on the parabola to write the equation.
 - If we know any three points on the parabola, then we use the formula $y = ax^2 + bx + c$ to write the equation of the parabola.
- Let $y = f(x) = ax^2 + bx + c$ and $y = g(x) = mx + n$ be two functions.
In the standard form of the equation $f(x) = g(x)$, there are three possible cases for the discriminant, Δ .
 - If $\Delta > 0$, the line intersects the parabola at two distinct points.
 - If $\Delta = 0$, the line is tangent to the parabola.
 - If $\Delta < 0$, the line does not intersect the parabola.



- Let $f(x) = ax^2 + bx + c$ and $g(x) = Ax^2 + Bx + C$ be two functions.
In the equation $f(x) = g(x)$, $ax^2 + bx + c = Ax^2 + Bx + C$
 $(a - A)x^2 + (b - B)x + (c - C) = 0$.
 - If $a - A \neq 0$, then we examine the discriminant, Δ .
 - If $\Delta > 0$, the parabolas intersect each other at two distinct points.
 - If $\Delta = 0$, the parabolas are tangent to each other.
 - If $\Delta < 0$, the parabolas do not intersect each other.
 - If $a - A = 0$ and $b - B \neq 0$, the parabolas intersect each other at one point.
 - If $a - A = 0$, $b - B = 0$ and $c - C = 0$, the parabolas are coincident.
- An inequality in one of the forms

$$\begin{array}{ll} ax + b > y, & ax + b < y \\ ax + b \geq y, & ax + b \leq y \end{array}$$

for real numbers a and b , $a \neq 0$, is called a linear inequality in two variables.

- To graph a linear inequality in two variables, follow the steps.
 1. Graph the linear equation $y = ax + b$. This line becomes the boundary line for the graph. Draw the boundary line using a dashed line if the inequality symbol is $<$ or $>$, or a solid line if the inequality symbol is $\leq$ or $\geq$.
 2. Choose a test point which is not on the boundary line and substitute it into the inequality.
 3. Shade the region which includes the test point if the resulting inequality is true, and shade the region which does not contain the test point if the resulting inequality is false.
- An inequality in one of the forms

$$ax^2 + bx + c > y, \quad ax^2 + bx + c < y,$$

$$ax^2 + bx + c \geq y, \quad ax^2 + bx + c \leq y,$$
 for real numbers a, b and $c, a \neq 0$ is called a quadratic inequality in two variables.
- To graph a quadratic inequality in two variables, follow the steps.
 1. Graph the parabola $y = ax^2 + bx + c$. The parabola is the boundary of the graph. Draw the parabola as a dashed line if the inequality symbol is $<$ or $>$ and as a solid line if the inequality symbol is $\leq$ or $\geq$.
 2. Choose a test point which is not on the boundary and substitute it into the inequality.
 3. Shade the region which includes the test point if the resulting inequality is true, and shade the region which does not contain the test point if the resulting inequality is false.
- A system of inequalities in two variables is made up of two or more inequalities. A solution of the system is an ordered pair that satisfies all of the inequalities. The solution set is the set of all ordered pairs that satisfy all of the inequalities. To solve a system of inequalities, we first solve the inequalities individually. The solution to the system is the region where all of the inequalities overlap on a graph.

Concept Check

- How many times can a quadratic function cross the x -axis?
- The coefficient of the x^2 -term can be positive or negative. What difference does it make?
- Which of the parabolas $y = -2x^2 - x - 5$ and $y = 4x^2 - 2x + 2$ is wider?
- What is the effect of increasing the *magnitude* of the coefficient of x^2 ?
- Which of the parabolas $y = 2x^2 + x - 5$ and $y = -4x^2 + 2x + 2$ opens upward?
- For what values of a will a parabola's vertex be a maximum? For what values of a will the vertex be a minimum?
- What point on the parabola $y = ax^2 + bx + c$ does the axis of symmetry always pass through?
- In some examples, the x -term is zero and in others it is non-zero. What difference does it make?
- If you write a quadratic function in the form $y = (x + m)^2 + n$, what is the significance on the graph of the values of m and n ?
- If you write a quadratic function in the form $y = (x + b)(x + g)$, what is the significance on the graph of the values of b and g ?
- When graphing an inequality in two variables, how do you find the equation of the boundary?
- What is the significance of a dashed line in the graph of an inequality?
- When graphing an inequality in two variables, how do you know which side of the boundary to shade?

CHAPTER REVIEW TEST 3A

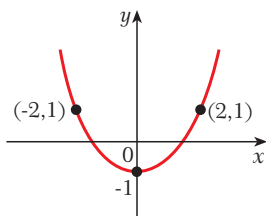
1. Which one of the following points is the vertex of the parabola $y = 2x^2 - 4x + 3$?

A) $(-1, 9)$ B) $(-2, 19)$ C) $(1, 1)$
D) $(-4, 1)$ E) $(2, 3)$

2. What is the minimum value of the expression $x^2 + 8x$ for any real number x ?

A) $-16,25$ B) -16 C) -15 D) -8 E) -7

3. Which one of the following functions represents the graph in the figure?



A) $x^2 = 2y + 2$ B) $x^2 = 4y + 4$
C) $y = \frac{1}{2}x^2 + 1$ D) $y = 2x^2 - 1$
E) $x^2 = 4y - 1$

4. The points $(1, y_1)$ and $(-1, y_2)$ are on the parabola $y = ax^2 + bx + c$ and $y_1 - y_2 = -6$. Find b .

A) 3 B) 0 C) -3 D) $\sqrt{ac}$ E) $\frac{a+c}{2}$

5. The parabola $y = x^2 - mx + m - 1$ is tangent to the x -axis. Find the value of m .

A) 4 B) 2 C) -2 D) -1 E) 0

6. The parabolas $y = x^2 + 2x - 5$ and $y = 2x^2 - x - 3$ intersect each other at two distinct points. What is the abscissa of one of these points?

A) 1 B) 3 C) -3 D) 5 E) -5

7. A particle on the parabola $y = x^2 - 2x - 8$ is moved from a point P with ordinate 7 to another point Q with ordinate -9 . What is the horizontal distance covered by this particle?

A) 5 B) 4 C) 3 D) 2 E) 1

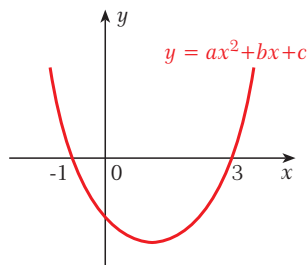
8. The parabola $y = x^2 - 6x + 1$ is given. The axis of symmetry of the parabola cuts the x -axis at one point. What is the abscissa of this point?

A) 6 B) 5 C) 4 D) 3 E) 2

9. $x^2 - 5x + 6 < 0$ and $p = x^2 + 5x + 6$ is given.
Which one of the following expressions is true?

A) $p > 30$ B) $20 < p < 30$
C) $0 < p < 20$ D) $p < 0$
E) p is any real number

10. Which one of the following statements is false about the parabola given in the figure?

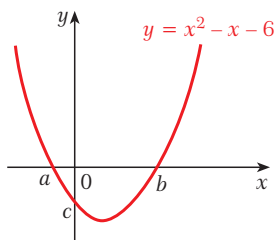


A) a is positive.
B) b is less than zero.
C) The abscissa of the vertex point is positive.
D) There are two distinct real roots.
E) The minimum value of the function is positive.

11. For which value of x does the function $f(x) = (3 - x)(2x - 1)$ have its maximum value?

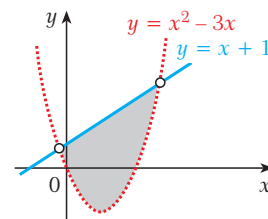
A) 0 B) $\frac{7}{4}$ C) 2 D) 3 E) 6

12. The figure shows the graph of the function $y = x^2 - x - 6$. What is the value of $a + b + c$?



A) -7 B) -6 C) -5 D) 3 E) 6

13. Which system of inequalities expresses the shaded region in the figure?

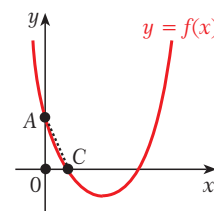


A) $\begin{cases} x^2 - 3x > y \\ x + 1 \geq 0 \end{cases}$ B) $\begin{cases} x^2 - 3x < y \\ x + 1 \leq y \end{cases}$
C) $\begin{cases} x^2 - 3x > y \\ x + 1 \leq y \end{cases}$ D) $\begin{cases} x^2 - 3x < y \\ x + 1 \geq y \end{cases}$
E) $\begin{cases} x^2 - 3x < y + 1 \\ x + 1 > 0 \end{cases}$

14. The sum of two numbers is 5. What is the greatest possible value of the product of these numbers?

A) 5 B) -5 C) 25 D) $\frac{25}{2}$ E) $\frac{25}{4}$

15. The figure shows the graph of the parabola $f(x) = x^2 - 5x + 4$. What is the area of the triangle AOC?



A) 1 B) 2 C) $\frac{5}{2}$ D) $\frac{7}{3}$ E) 3

16. The point $V(1, 2)$ is the vertex of the parabola $y = x^2 + bx + c$. What is the product of b and c ?

A) -6 B) -3 C) -2 D) -3 E) 6

CHAPTER REVIEW TEST 3B

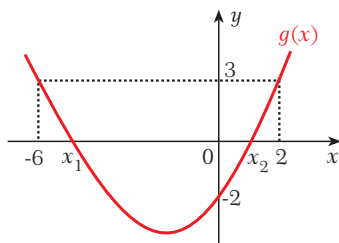
1. The parabola $y = 2x^2 + bx - 2$ passes through the point $K(1, -1)$. What is the value of b ?

A) -3 B) -2 C) -1 D) 0 E) 1

2. $s(t) = 1600t - 1600t^2$ is the displacement function of a particle starting to move from the origin in the analytic plane, in terms of time t . What is the maximum height that this particle can reach?

A) 800 B) 640 C) 400 D) 320 E) 160

3. The figure shows the graph of the function $g(x)$. What is the product of the values of x_1 and x_2 ?



A) $-\frac{24}{5}$ B) -4 C) $-\frac{1}{3}$ D) -3 E) $-\frac{10}{3}$

4. The parabolas $y = x^2 + 2x - 5$ and $y = x - 1$ intersect each other at two distinct points. What is the sum of the x -coordinates of these points?

A) -1 B) 3 C) 4 D) 5 E) 12

5. The vertex point of the parabola $y = x^2 - 8x + c$ is tangent to the x -axis. What is the value of c ?

A) -16 B) -4 C) 4 D) 8 E) 16

6. The parabolas $y = ax^2 + bx + c$ and $y = ax^2 - bx + c$ are given. The coefficients a , b , and c are real numbers. Which one of the following statements is true about the intersection of these parabolas?

A) There are two intersections. One intersection is on the x -axis and the other is on the y -axis.
 B) The intersection point is not on the axes.
 C) The parabolas only intersect at the origin.
 D) The parabolas only intersect on the y -axis.
 E) The parabolas only intersect on the x -axis.

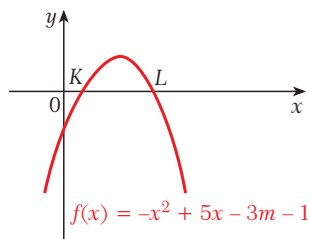
7. The minimum value of the function $y = x^2 + px + q$ is zero. What is the value of q ?

A) $\frac{p^2}{4}$ B) $-\frac{p^2}{4}$ C) $\frac{p}{2}$ D) $-\frac{p}{2}$ E) p

8. The parabola $y = ax^2 - ax - 1$ does not cut the x -axis. How many possible integer values of a are there?

A) 2 B) 3 C) 4 D) 5 E) 6

9.



The figure shows the graph of the parabola $f(x) = -x^2 + 5x - 3m - 1$. $|OL| = 4 \cdot |OK|$ is given. What is the value of m ?

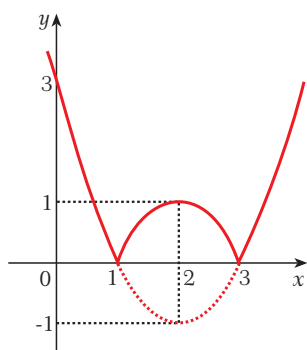
- A) 1 B) -1 C) -2 D) 2 E) 3

10. $f(x) = x^2 - 6x + 9$ and $g(x) = |f(x)|$ are given. Which one of the following is the solution of the inequality $g(x) > f(x)$?

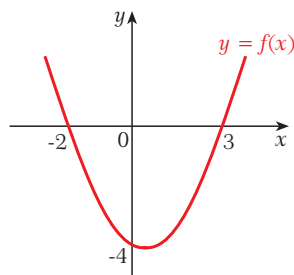
- A) $3 < x < 5$ B) $\emptyset$ C) $3 < x < 9$
D) $x > 9$ E) $x < 5$

11. Which one of the following functions represents the graph in the figure?

- A) $y = |x^2 - 4x - 3|$
B) $y = x^2 + 4|x| + 3$
C) $y = x^2 - 4|x| + 3$
D) $y = |x^2 - 4x + 3|$
E) $y = |x^2 + 4x - 3|$



12. The figure shows the parabola $y = f(x)$. What is $f(1)$?

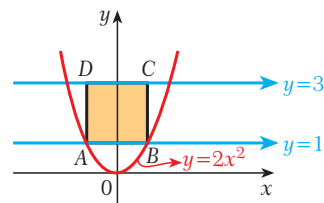


- A) -4 B) -2 C) -3 D) 1 E) 2

13. What is the minimum value of the function $f(x) = 4 - x^2$ on the closed interval $[-1, 3]$?

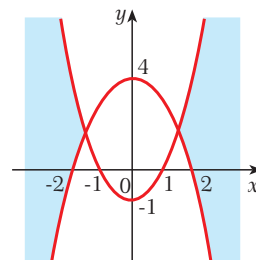
- A) -6 B) -5 C) -4 D) 2 E) 3

14. What is the area of the rectangle ABCD in the figure?



- A) $\frac{1}{2}$ B) $\frac{\sqrt{2}}{2}$ C) $\sqrt{2}$ D) 2 E) $2\sqrt{2}$

15. Which system of inequalities expresses the shaded region in the figure?



- A) $\begin{cases} y \geq x^2 - 1 \\ y \geq -x^2 + 4 \end{cases}$ B) $\begin{cases} y \leq x^2 - 1 \\ y \leq -x^2 + 4 \end{cases}$
C) $\begin{cases} y \geq x^2 - 1 \\ y \leq -x^2 + 4 \end{cases}$ D) $\begin{cases} y \leq x^2 - 1 \\ y \geq -x^2 + 4 \end{cases}$
E) $\begin{cases} y \leq x^2 - 1 \\ y \geq x^2 + 4 \end{cases}$

16. The parabola $y = x^2 + 4x - 2$ is given. What is the distance from the point $A(1, -2)$ to the vertex of the parabola?

- A) 5 B) $5\sqrt{2}$ C) 6 D) $6\sqrt{2}$ E) 2

ANSWERS TO EXERCISES

EXERCISES 1.1

1. a. 0 b. 0 c. 0 d. 0 2. a. $0, -\frac{5}{2}$ b. $0, \frac{3}{7}$ c. $0, \frac{8}{3}$ d. $0, \frac{1}{9}$ 3. a. 0, -2 4. $0, -\frac{5}{2}$ 5. -1, 2 6. a. ± 4 b. no real solution
c. $\pm \frac{5}{3}$ d. ± 5 e. $\pm \frac{2\sqrt{5}}{5}$ f. $\pm \frac{4}{5}$ 7. a. -1, 5 b. no real solution c. $(-1 - \sqrt{5}), (-1 + \sqrt{5})$ d. -2, 4 8. $\pm \sqrt{3}$ 9. a. 0, 1
b. $0, \frac{1}{3}$ c. ± 7 d. ± 5 e. -1, 2 f. -1, -2 g. 1 h. $-\frac{5}{3}, \frac{3}{2}$ i. $\frac{2}{5}, \frac{3}{2}$ j. $-\frac{1}{4}, \frac{2}{3}$ 10. a. -a, -1 b. no real solution c. $-\frac{5b}{2a}, 1$
11. a. $(2 + \sqrt{5}), (2 - \sqrt{5})$, b. $(\sqrt{7} - 2), (-\sqrt{7} - 2)$ c. $(3 \pm \sqrt{22})$ d. no real solution e. no real solution f. no real solution
g. $3 \pm 2\sqrt{2}$ h. $\frac{-1 \pm \sqrt{7}}{6}$ 12. a. $2 \pm \sqrt{2}$ b. -1, $\frac{1}{2}$ c. $x = -\frac{3}{2}$ d. $\frac{5 \pm \sqrt{13}}{6}$ e. no real root f. $\frac{-1 \pm 2\sqrt{7}}{3}$ g. no real root
h. $x = -\frac{4}{5}$ i. $\frac{7}{4}, 4$ j. $-\frac{2}{3}, 1$ k. $\frac{1}{26}, 1$ l. -1, 23 13. a. $\frac{1 \pm \sqrt{33}}{8}$ b. $\frac{3 \pm \sqrt{73}}{4}$ c. 1, 5 d. $\frac{8}{5}, 5$ 14. a. two real roots
b. two real roots c. one double root d. one double root e. no real solution f. no real solution 15. a. $a > \frac{9}{40}$
b. $a = \frac{9}{40}$ c. $a > \frac{9}{40}$ 16. a. $m > -\frac{7}{2}$ b. $m > -\frac{7}{2}$ c. $m > -\frac{7}{2}$ 17. $-\frac{1}{4}, 1$ 18. 8 cm, 10 cm
19. 12 cm, 20 cm 20. 4, 6, 8, 10 21. $256 \text{ cm}^2, 441 \text{ cm}^2$ 22. 7, 49

EXERCISES 1.2

1. a. $\frac{11}{10}, -\frac{6}{5}$ b. $-\frac{72}{49}, -\frac{40}{21}$ c. $\frac{10}{3}, -\frac{5}{3}$ d. $\frac{4}{3}, -\frac{5}{6}$ e. $-\frac{18}{7}, \frac{8}{7}$ f. $(3\sqrt{2} - 2), 0$ 2. $\frac{24}{7}$ 3. $-\frac{2}{5}$ 4. a. $\frac{21}{4}$ b. 53 c. $-\frac{17}{9}$
d. $\frac{153}{256}$ 5. $m_1 = -5, m_2 = 4, x_2 = -\frac{5}{4}$ 6. $n_1 = -4, n_2 = 2, x_2 = -\frac{2}{3}$ 7. 0 8. $m = 2n$ 9. $k = 3$
10. $\frac{14}{3}, 4$ 11. 5 12. $-\frac{7}{3}, \frac{5}{3}$ 13. -6, 2 14. 1 15. $k > 6$ 16. 6, 7 17. $a \cdot b$ 18. 0
19. 7 20. $c = -2, 1, \frac{1 \pm \sqrt{5}}{2}$ 21. -13 22. $-\frac{1}{2}, \frac{1 \pm \sqrt{13}}{2}$ 23. $4 + 2\sqrt{3} \text{ cm}$

EXERCISES 1.3

1. a. $x^2 - 1 = 0$ b. $4x^2 - 8x + 3 = 0$ c. $x^2 - 4x = 0$ d. $x^2 - 4x + 2 = 0$ e. $x^2 - 2\sqrt{3}x + 1 = 0$ f. $p^2q^2x^2 - (p^2 + q^2)x + 1 = 0$
 2. $x^2 + 2x + 5 = 0$ 3. $x^2 - 8x - 12 = 0$ 4. $x^2 + x + 27 = 0$ 5. a. $x^2 - 4 = 0$ b. $x^2 + 8x + 7 = 0$
 c. $4x^2 - 25 = 0$ d. $x^2 - 2x + 1 = 0$ e. $x^2 + 2mx - 8m^2 + 27m - 18 = 0$ f. $mx^2 - 2mx - 8m - 9 = 0$ 6. 5
 7. -7, -6, -5 or 5, 6, 7 8. 9 9. 15 hours, 10 hours 10. 3 hours, 2 hours 11. 6 hours 12. $\frac{4}{7}$ 13. $18(\sqrt{2} + 1)$
 14. from A to B 60 km/h, from B to A 80 km/h 15. 160 km/h 16. 16 km/h 17. 7 women 18. 80 km/h
 19. 24 hours, 48 hours 20. 20 km/h 21. height = 6 m, base = 12 m 22. 40 kg 25%, 20 kg 40%
 23. 60 hours, 84 hours 24. 4 hours, 8 hours 25. 6 m, 8 m

EXERCISES 1.4

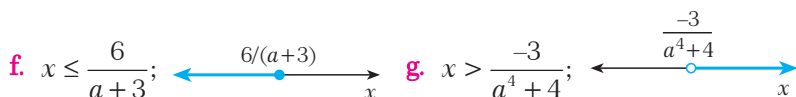
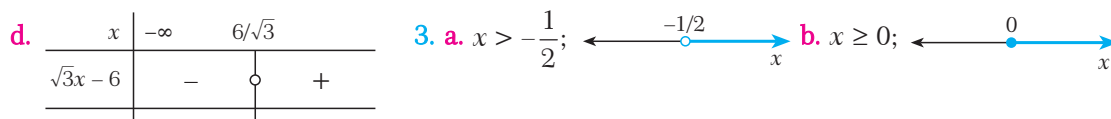
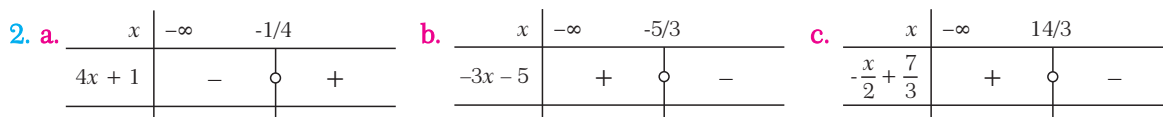
1. a. $\pm 2, \pm 3$ b. $\pm\sqrt{3}$ c. $\pm\sqrt{10}, \pm 2\sqrt{3}$ d. $-5 \pm \sqrt{7}$ e. $-5\frac{1}{2}$ f. 79 g. 4 h. $2 \pm \sqrt{3}$ i. $2 \pm \sqrt{2}$ 2. a. $\frac{3 \pm \sqrt{5}}{2}$ b. $(1 \pm \sqrt{10}), -1, 3$
 c. -2, -1, 0, 1 d. 1, 2 e. -6, 1 3. a. 0, $\pm\frac{1}{4}, 1, 5$ b. -10, $(-4 \pm \sqrt{2}), 2$ c. 2 d. 5 e. $\frac{6 \pm \sqrt{42}}{3}$ f. $\pm\sqrt{5}, \frac{7}{3}, 1$ g. no solution
 h. $\pm 1, 2, 4$ i. 0 j. $-3 \pm \sqrt{15}$ 4. a. 2 b. $\frac{11 - \sqrt{21}}{2}$ c. -7 d. 1, 2 e. 12 f. 3 g. -1, 2 h. 0 i. -1, 7 j. no solution
 5. a. 7, 8 b. 2 c. 0 d. 0, 2 e. 0 f. no solution g. $-\frac{8}{3}, 1$ h. 2 6. a. -1 b. $\frac{7}{3}, 3$ c. 2 d. -1, 5 e. -1, 3 f. no solution
 g. $3 - \sqrt{7}, 2$ h. 2 i. no solution 7. a. 0 b. $\pm\frac{1}{\sqrt{2}}$ c. -2, 0 d. 2 e. $-\frac{2}{3}, \frac{1}{2}, 2$ f. $x \in (-\infty, \frac{3}{2}]$ 8. a. 1 b. 0, 3 c. $\frac{3}{4}$
 d. -1, 4

EXERCISES 1.5

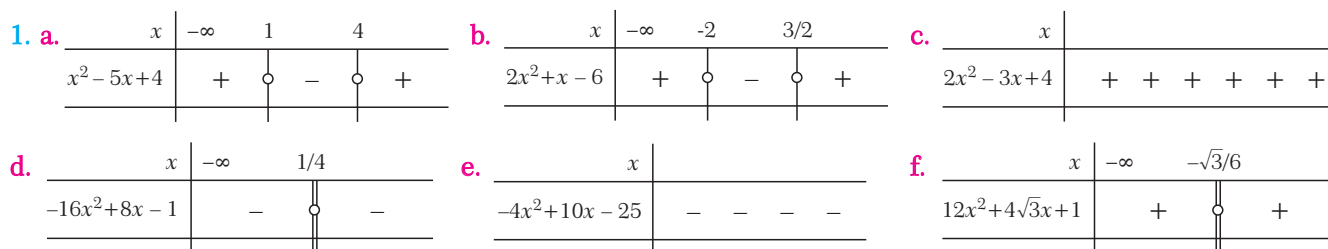
1. a. (-6, -18), (18, 6) b. $(\frac{5}{2}, -\frac{7}{2})$ c. (-3, 6), (10, -7) d. (10, 15), (15, 10) e. (4, -1), (-4, 1) f. (3, 3), (-3, -3)
 g. (-3, -4) h. (0, -5), (1, -4)
 2. a. (-4, -4), (-6, -2) b. (0, 0), $(\frac{3}{2}, 3)$ c. (0, 0) d. (1, 0) e. (2, 4), (4, 2), $(\sqrt{22} - 4, -\sqrt{22} - 4), (-\sqrt{22} - 4, \sqrt{22} - 4)$
 f. (0, -2), (2, 0) g. (0, $\sqrt{2}$), (0, $-\sqrt{2}$) h. (2, 1), (-2, -1), $(\frac{\sqrt{7}}{2}, -\frac{\sqrt{7}}{2})(-\frac{\sqrt{7}}{2}, \frac{\sqrt{7}}{2})$

EXERCISES 2.1

1. a. $(-2, \infty)$ b. $(-\infty, 6]$ c. $(-\infty, -7)$ d. $[4, \infty)$ e. $[2, 4)$ f. $(-1, 0]$ g. $(-5, \frac{1}{2})$ h. $[0.5, \frac{3}{2}]$



EXERCISES 2.2



2. a. $(4, 5)$ b. $(0, \frac{4}{7})$ c. $(-\infty, -\frac{7}{6}] \cup [0, \infty)$ d. $\square$ e. $x = -\frac{4}{3}$ f. $(\frac{4}{5}, 1)$ g. $(-\infty, -4) \cup (2, \infty)$ 3. a. $(0, 2)$ b. $0, 2$

- c. $(-\infty, 0) \cup (2, \infty)$ 4. $(-\infty, -2\sqrt{3}-1) \cup (2\sqrt{3}-1, \infty)$ 5. $(3-2\sqrt{2}, 3+2\sqrt{2})$ 6. $\{-6, -5, -4, -3, -2, -1, 0, 1, 2\}$

7. a. $(0, \infty), x \neq 1$ b. $(-\infty, -\frac{1}{3}) \cup (\frac{3}{2}, 2)$ c. $x \neq -2, (-\infty, -1) \cup (\frac{2}{3}, 3)$ d. $(-\infty, \frac{2}{3}) \cup (1, \frac{5}{2})$

- e. $(-4, -3) \cup (-2, -1) \cup (\frac{1}{2}, 3)$ f. $(-1, 5)$ g. $(-\infty, \frac{3}{2}) \cup (\frac{7}{3}, \infty)$ h. $(-8, -1)$ i. $\square - \{2\}$ j. $(-\infty, -7] \cup (-1, 0) \cup (0, 1] \cup (3, \infty)$

- k. $(-\infty, \frac{-\sqrt{6}}{2}] \cup [-1, 0) \cup [1, \frac{\sqrt{6}}{2}]$ l. $(-\infty, 1) \cup (\frac{4}{3}, 2)$ m. $(-\infty, 2) - \{-4, -2\}$ n. $[0, \frac{1}{3}]$ o. $(-\infty, 1) \cup (\frac{11}{5}, \infty)$

- p. $(-\infty, -2) \cup (-2, -1) \cup (-1, 0)$ q. $[-\frac{1}{2}, 0] \cup [1, \infty)$ 8. a. $(0, 1)$ b. $(0, 1)$ c. $\emptyset$ d. $[\frac{3}{2}, 2)$ 9. a. $\square - \{7, -7\}$

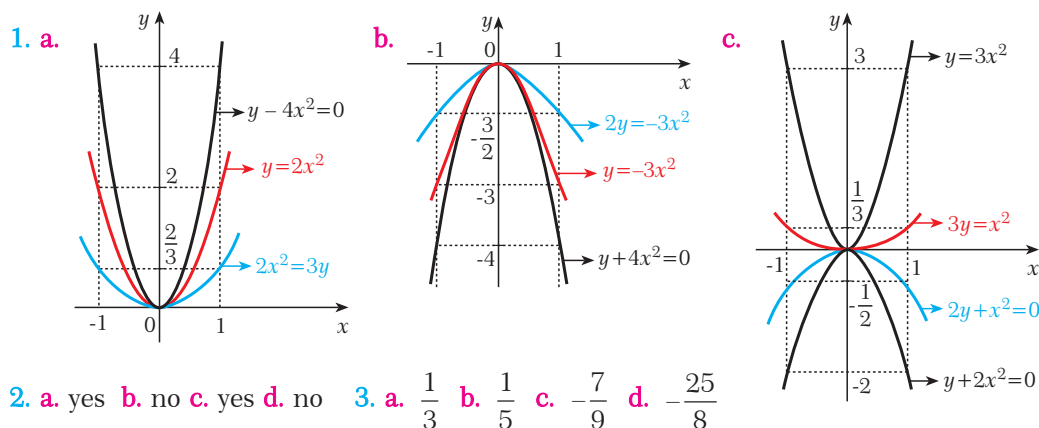
- b. $(0, 16) \cup (16, \infty)$ c. $(-\infty, -2) \cup [2, \infty)$ d. $[2, 3] \cup [-1, 1]$ 10. a. $(0, \sqrt{3}-1) \cup (-4, -3) \cup (-\sqrt{3}-1, -2) \cup (1, 2)$

11. $x \neq 0, (-2, 1)$

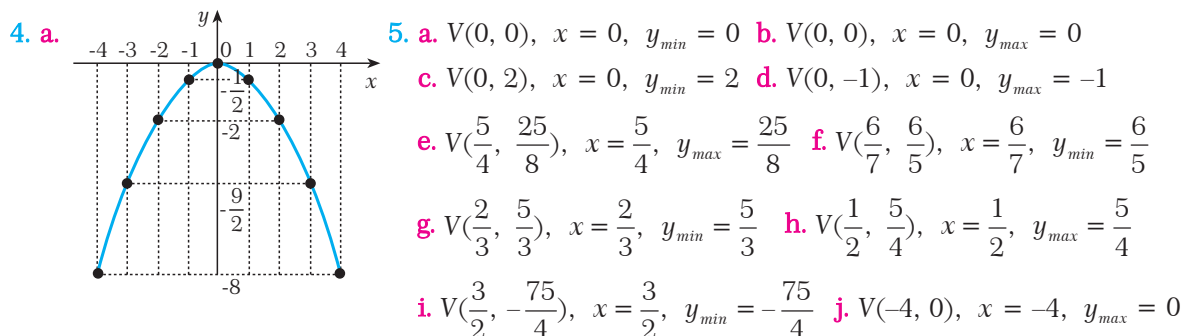
EXERCISES 2.3

1. a. $(-2, \frac{3}{2})$ b. $(1, 3)$ c. $(-6, -5)$ d. $(-1, 1)$ e. $(1, 2)$ f. $[1, 2)$ g. $(2, 4)$ h. $(0, 1)$ 2. a. $(0, 1)$ b. $(-1, 1) \cup (3, 5)$
 c. $(-8, -\frac{13}{2}) \cup (0, 5)$ 3. a. $(-\infty, 1] \cup [\frac{3}{2}, \infty)$ b. $(-\infty, -\frac{5}{3}) \cup (3, \infty)$ c. $(-\infty, -5) \cup (-1, \infty)$ d. $(-\infty, -4) \cup (0, \infty)$
 e. $[\frac{3}{2}, 2)$ f. $[\frac{2}{7}, \frac{2}{3}]$ g. $(-\infty, 3)$ 4. a. $(3, \infty)$ b. $[\frac{13}{5}, 4)$ c. $[4, \infty)$ d. $(-3, 1)$ e. $[3, \infty)$ f. $(9, \infty)$ 5. a. $[3, \infty)$ b. $[-2, \frac{\sqrt{5}}{10} - \frac{3}{2}]$
 c. $[2, 3]$ d. $[-15, \frac{\sqrt{13}-5}{2})$ e. $[3, \frac{417}{64})$ 6. a. $[\frac{-\sqrt{5}-1}{2}, \frac{\sqrt{13}+3}{2}]$ b. $[-1, 3]$ c. $[-\sqrt{6}, \frac{-\sqrt{11}-1}{2}) \cup (\frac{\sqrt{11}+1}{2}, \sqrt{6}]$
 7. $[1, 5) \cup (10, \infty)$ 8. a. $(2, \infty)$ b. $(1, \infty)$ c. $(-\infty, -\frac{5}{2})$ d. $(-\infty, -2) \cup (2, \infty)$ 9. $(-5, -1)$

EXERCISES 3.1



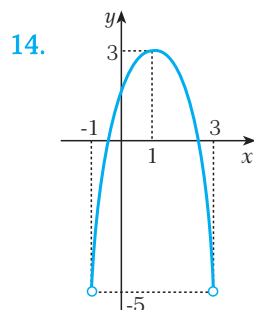
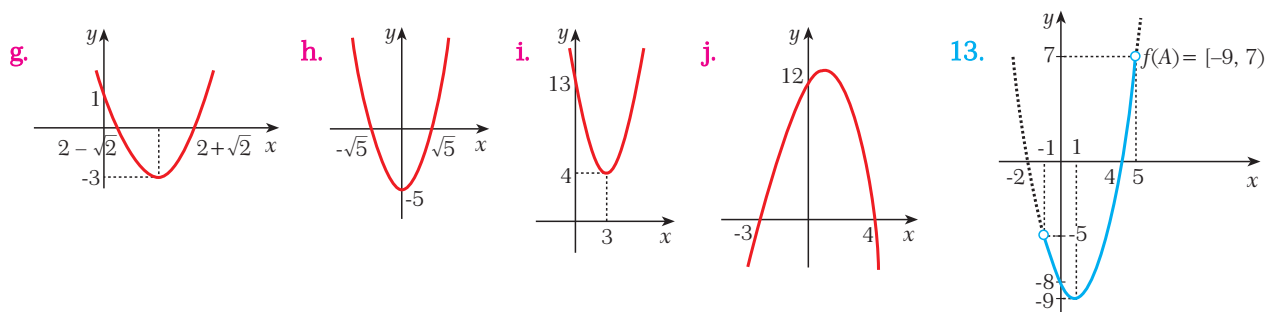
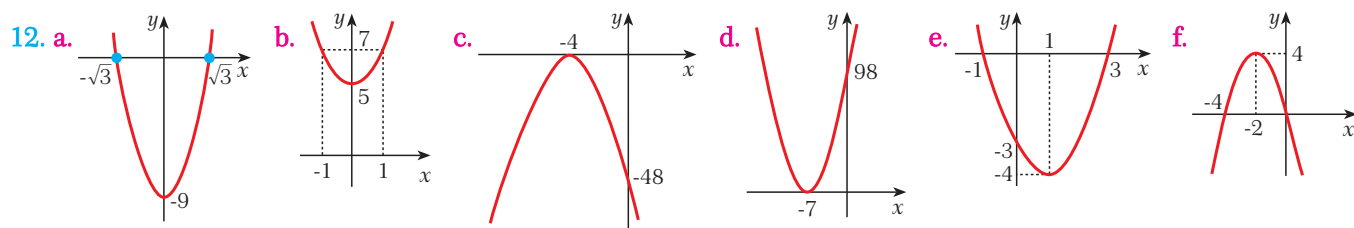
2. a. yes b. no c. yes d. no 3. a. $\frac{1}{3}$ b. $\frac{1}{5}$ c. $-\frac{7}{9}$ d. $-\frac{25}{8}$



- k. $V(-3, -5)$, $x = -3$, $y_{\max} = -5$ l. $V(1, 3)$, $x = 1$, $y_{\min} = 3$ 6. $-\frac{41}{8}$ 7. 1 8. $-\frac{5}{2}$ 9. a. $(0, 0)$, $(4, 0)$

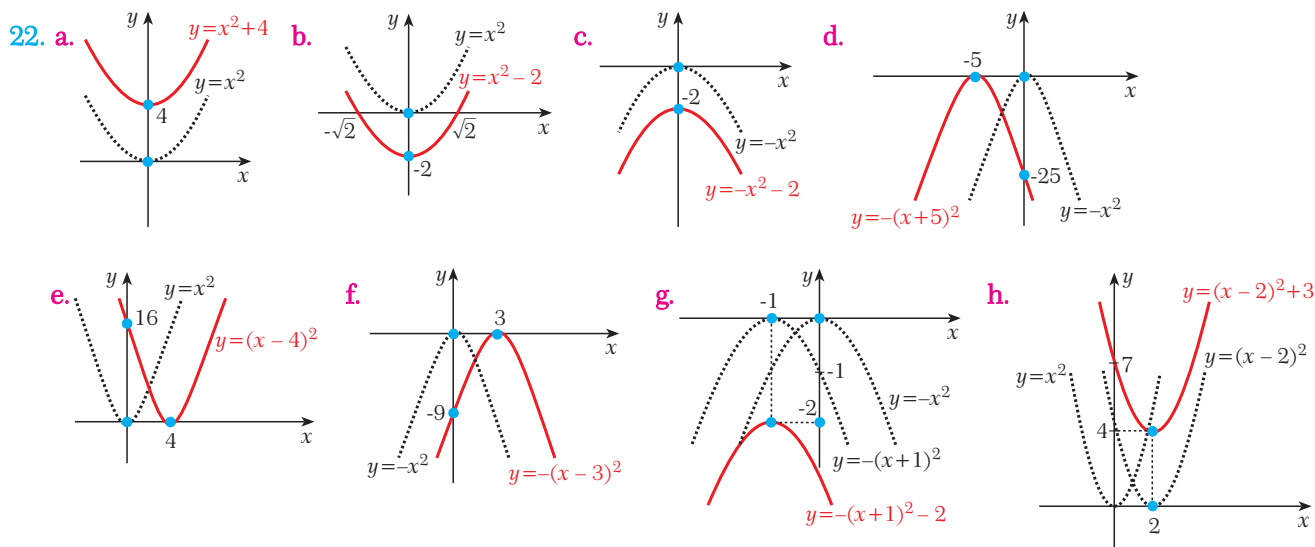
- b. $(3, 0)$, $(0, 9)$ c. $(0, 396)$ d. $(-2\sqrt{2}, 0)$, $(2\sqrt{2}, 0)$, $(0, -8)$ e. $(-\frac{3}{2}, 0)$, $(-1, 0)$, $(0, 3)$ f. $(0, -5)$ g. $(-\frac{3}{4}, 0)$, $(0, 9)$

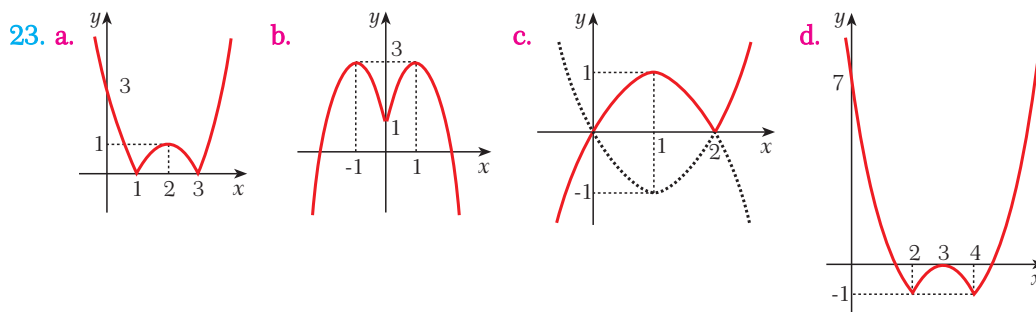
h. $(0, -4)$ 10. a. $m \in (-2\sqrt{2}, 2\sqrt{2})$ b. $m = \pm 2\sqrt{2}$ c. $m \in (-\infty, -2\sqrt{2}) \cup (2\sqrt{2}, \infty)$ 11. 3



15. a. $y_{\max} = 12$, $y_{\min} = 3$ 16. $y_{\max} = 64$ 17. $y_{\min} = -\frac{9}{4}$ 18. $a = -\frac{1}{2}$, $b = 1$

19. $y = 3(x - 5)^2$ 20. $-3(x - 1)^2 + 2 \leq 2$; $-3(x - 1)^2 \leq 0$; $3(x - 1)^2 \geq 0$ 21. $m \in [0, \frac{16}{9}]$





24. a. $y = -(x - 1)^2 + 3$ b. $y = \frac{1}{6}x(x - 5)$ c. $y = \frac{1}{3}(x + 3)^2$ d. $y = (x + 1)^2 + 3$ e. $y = -x^2 + 4x - 3$

f. $y = -\frac{1}{3}(x - 3)(x + 3)$ 25. a. $y = -3x^2 + x + 4$ b. $y = -x^2 + 4x - 4$ 26. a. $y = x^2 - 4$ b. $y = (x - 2)^2$

c. $y = (x + 1)^2 - 4$ d. $y = -x^2 - x - 4$ 27. -12 28. $\frac{18}{5}$ 29. 0 30. $\frac{1}{2}$ 31. $\frac{25}{4}$ 32. $\frac{9}{4}$ square units

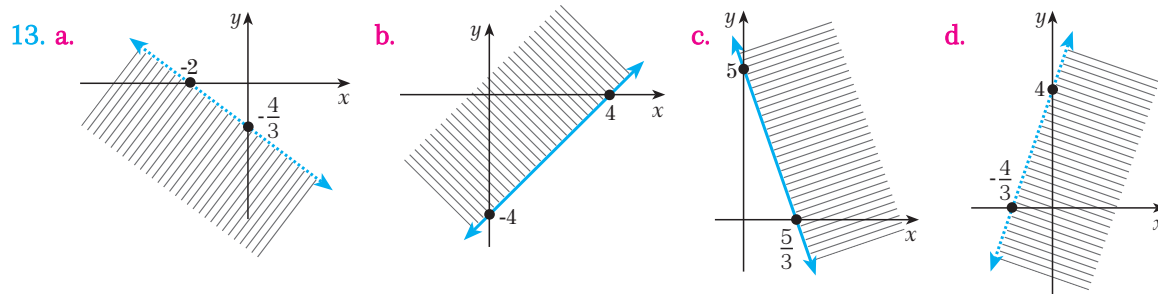
33. 10 square units 34. $m \in (-\infty, -1) \cup (0, \infty)$ 35. 64 square units 36. $\frac{49}{20}$ 37. 2 38. $(-3, 0)$ 39. 9 40. 1

41. 35 square units

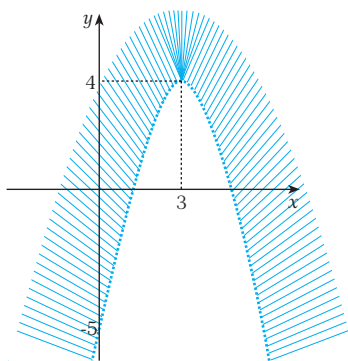
EXERCISES 3.2

1. a. $\emptyset$ b. $\emptyset$ c. $m \in \square$ 2. $(8, 0)$ 3. $y = 2x + 1$ 4. $-\frac{11}{4}$ 5. -2 or 4 6. $y = \frac{2}{3}(x + 3)^2 - 2$ 7. 3

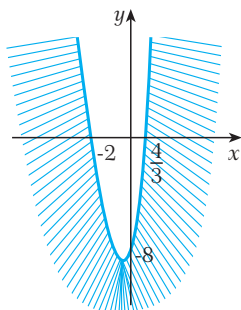
8. 1 or $-\frac{3}{5}$ 9. $y = 12x + 1$ 10. $(2, 1)$ 11. 1 12. $(-1, -1)$



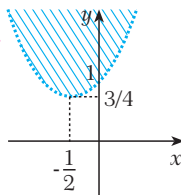
14. a.



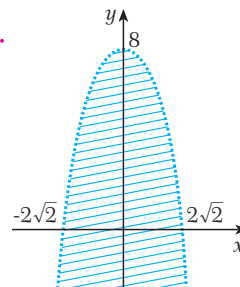
b.



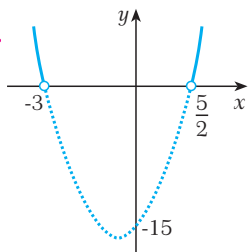
c.



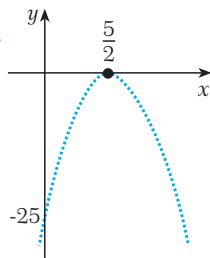
d.



e.



f.

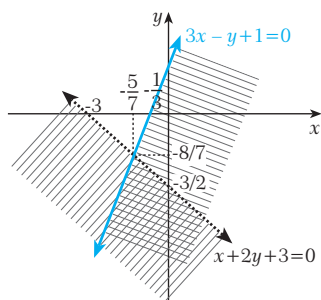


15. a. $3y + 2x - 6 < 0$ b. $y - 2x + 2 \geq 0$

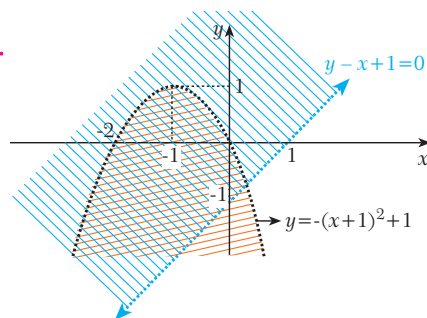
c. $y > \frac{8}{9}x^2 - \frac{8}{3}x$ d. $y > -\frac{1}{3}x^2 + 3$

e. $\frac{4}{3}(x+3)(x+1) > 0$

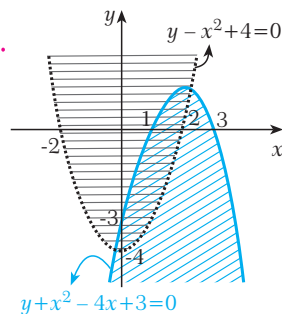
16. a.



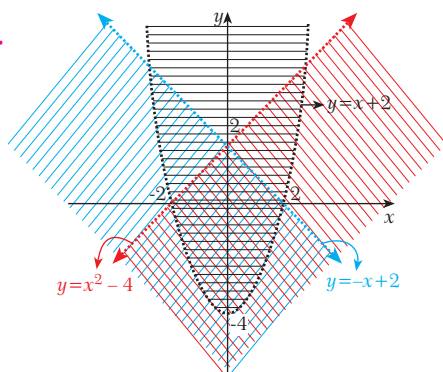
b.



c.



d.



17. a. $\begin{cases} y < x + 2 \\ y < -x + 2 \end{cases}$ b. $\begin{cases} y < -\frac{2}{3}(x+2)(x-3) \\ 3y + 4x - 12 \geq 0 \end{cases}$

c. $\begin{cases} y \leq \frac{1}{3}(x-3)^2 + 3 \\ y > \frac{1}{2}(x-2)^2 \end{cases}$ d. $\begin{cases} y < -2(x+1)(x-3) \\ y + 6x - 6 \geq 0 \\ y > 6x - 6 \end{cases}$

ANSWERS TO REVIEW TESTS

Chapter 1

TEST 1A

1. C
2. E
3. C
4. B
5. E
6. D
7. C
8. C
9. B
10. B
11. D
12. A
13. E
14. C
15. B
16. D

TEST 1B

1. B
2. D
3. E
4. A
5. D
6. B
7. A
8. C
9. A
10. E
11. B
12. D
13. B
14. A
15. B
16. B

Chapter 2

TEST 2A

1. E
2. B
3. E
4. A
5. C
6. D
7. A
8. D
9. E
10. B
11. D
12. D
13. C
14. B
15. A
16. B

TEST 2B

1. D
2. B
3. A
4. C
5. E
6. B
7. E
8. C
9. C
10. D
11. A
12. B
13. E
14. C
15. D
16. D

Chapter 3

TEST 3A

1. C
2. B
3. A
4. C
5. B
6. A
7. B
8. D
9. B
10. E
11. B
12. C
13. D
14. E
15. B
16. A

TEST 3B

1. C
2. C
3. A
4. A
5. E
6. D
7. A
8. B
9. A
10. B
11. D
12. A
13. B
14. E
15. D
16. A

GLOSSARY

A

absolute value: the distance between a number and zero on the number line.

altitude of a triangle: a perpendicular line drawn from any vertex of a triangle to the opposite side or to an extension of the side.

area: the number of unit squares that fit inside a figure.

B

base of a triangle: a side of a triangle with a perpendicular altitude drawn to it.

C

coefficient: the numerical factor of a term.

common factor: a number that is a factor of two or more numbers.

consecutive: following one after another without interruption.

constant: a symbol that presents a static quantity. The value of a constant never changes.

coordinates: a number or set of numbers that identify the location of a point on a line, in a plane, or in a space.

coordinate plane: a plane on which the locations of points are given by x - and y -coordinates that represent horizontal and vertical distances along the x -axis and y -axis lines, respectively.

D

denominator: the number below the bar in a fraction.

dependent variable: a variable, usually y , that depends on an input value of an independent variable, usually x . The set of all possible values of the dependent variable forms the range of the function.

distance: the positive difference of the values of two points on a number line.

domain: a list of the x -values that are possible in a specific function $f(x)$. The domain of a function excludes values that result in imaginary numbers or division by zero, and values that make no sense in the problem.

E

equation: a mathematical statement that establishes the equality of two expressions.

F

factor: a is a factor of b if it divides b exactly, with no remainder.

factoring: the process of rewriting an algebraic expression as a product of two or more factors.

formula: an algebraic equation that explains how two or more variables are related.

fraction: a number of the form $\frac{a}{b}$ (or a decimal) that shows the relationship of a part to a whole.

function: a rule that associates one and only one output value with each input value.

H

hypotenuse: the side opposite the right angle in a right triangle.

I

independent variable: a variable, usually x , whose value can change freely in a function $f(x)$.

inequality: an algebraic expression which shows that a quantity is greater than or less than another quantity.

L

line: a line consists of infinitely many points in a plane, extending infinitely in opposite directions.

linear equation: an equation in the form $ax + by + c = 0$. The graph of a linear equation is a straight line.

N

numerator: the number above the bar in a fraction.

nonlinear equation: an equation that has at least one variable raised to a power greater than one.

O

ordered pair: a pair of numbers that represents a point in the rectangular coordinate system.

P

parabola: the curved graph of a quadratic equation of the form $y = ax^2 + bx + c$.

parameter: a parameter is a variable that can be varied or changed in an expression.

perfect square: a number that is the square of another number.

perimeter: the distance around a closed figure.

point: a location on a line, in a plane, or in space. A point has no length, width, or depth.

polynomial: an algebraic expression that consists of one or more terms.

Pythagorean Theorem: the theorem stating that for a right triangle, the sum of the squares of the sides equals the square of the hypotenuse.

Q

quadratic equation: a second degree polynomial equation in a single variable x , which we can write as $ax^2 + bx + c = 0$ with $a \neq 0$.

R

range: the set of y -values that can be produced by a specific function $y = f(x)$.

ratio: a fraction that compares quantities by division.

right triangle: a triangle that contains one right angle.

S

simplified form: an algebraic expression in simplified form is in its most basic form.

substitute: to put or use (a number or an expression) in place of another in a formula or equation.

system of equations: a group of linear or nonlinear equations that use the same variables.

T

term: any number or variable that is added or subtracted.

tangent line: a line that touches a curve or a circle at exactly one point and lies in the same plane as the curve or circle.

theorem: a statement that can be proved through deductive reasoning by using undefined terms, definitions, postulates, axioms, or assumptions.

triangle: a polygon with three sides.

V

variable: not having a fixed numerical value.

vertex of a parabola: the highest or the lowest point of a parabola.

X

x -intercept: the point where a graph crosses the x -axis.

Y

y -intercept: the point where a graph crosses the y -axis.

